A Bias-Aware Multi-Modal Deep Learning Framework for Pancreatic Cancer Detection Using CT Imaging and Urinary Biomarkers

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DEDICATION

This dissertation is dedicated to my family for their unwavering support, encouragement, and patience throughout my academic journey and to my friend Siddharth, whose encouragement was pivotal in pushing me to see this through to completion.

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ABSTRACT

Pancreatic ductal adenocarcinoma (PDAC) is among the most lethal malignancies, with a five-year survival rate of approximately 12% because diagnosis is often late. Computed tomography (CT) and urinary biomarker panels offer diagnostic potential, but each modality alone has practical limitations for reliable early detection. Cross-dataset CT studies are additionally vulnerable to scanner-related shortcut learning, which can inflate apparent model performance.

This study developed and evaluated a bias-aware multimodal deep learning framework that integrates CT imaging and urinary biomarker data for pancreatic cancer detection. Three objectives guided the work: establish a bias-aware data foundation with patient-level leakage control; develop and evaluate unimodal CT and biomarker models and test whether multimodal fusion adds complementary value under synthetic pairing constraints.

The CT pipeline processed 122 patients from two public repositories. Bias detection showed that a logistic regression model achieved an AUC of 0.866 using pixel mean alone. A body-region-specific extreme intensity standardization approach reduced bias AUC to 0.569, near random. A ResNet50 CT classifier then achieved patient-level AUC of 1.000, although unsupervised domain analysis still indicated residual domain structure in deep features. A multilayer perceptron trained on seven urinary biomarker features achieved an AUC of 0.944, closely reproducing published benchmarks. Decision-level and feature-level fusion experiments, validated across five random seeds with label-mismatch controls, showed CT-dominated behavior rather than true multimodal synergy, indicating that synthetic pairing cannot replace matched patient data.

The study contributes a transparent bias detection and mitigation pipeline for cross-institutional imaging, independent reproducibility evidence for the urinary biomarker panel, and a rigorous framework for fusion evaluation under data-alignment constraints. The findings emphasize that transparent reporting of limitations and negative results is essential for responsible clinical AI development.

TABLE OF CONTENTS

DEDICATION i

ACKNOWLEDGEMENTS ii

ABSTRACT iii

LIST OF TABLES x

LIST OF FIGURES xi

LIST OF ABBREVIATIONS xii

CHAPTER 1 1

INTRODUCTION 1

1.1 Background and Motivation 1

1.2 Artificial Intelligence in Medical Diagnosis 2

1.3 Challenges in Multimodal Learning and Dataset Bias 3

1.4 Methodological Perspective and Research Focus 4

1.5 Research Aim and Objectives 5

1.5.1 Research Aim 5

1.5.2 Research Objectives 5

1.6 Research Questions 5

1.7 Scope of the Study 6

1.8 Significance of the Study 7

1.9 Structure of the Dissertation 7

1.10 Chapter Summary 8

CHAPTER 2 9

LITERATURE REVIEW 9

2.1 Introduction 9

2.2 Clinical Background of Pancreatic Cancer 9

2.3 Medical Imaging for Pancreatic Cancer Detection 10

2.3.1 Computed Tomography in Pancreatic Cancer 10
2.3.2 Deep Learning in Medical Imaging 11
2.3.3 Deep Learning for Pancreatic Cancer Detection 13
2.4 Biomarkers for Pancreatic Cancer Detection 14
2.4.1 Serum Biomarkers and Their Limitations 14
2.4.2 Urinary Biomarkers: The Debernardi et al. Panel 14
2.5 Machine Learning for Biomarker Analysis 16
2.6 Multimodal Learning in Healthcare 16
2.6.1 Rationale for Multimodal Integration 16
2.6.2 Fusion Strategies 17
2.6.3 The Synthetic Pairing Challenge 17
2.7 Dataset Bias and Shortcut Learning 18
2.7.1 The Problem of Spurious Correlations 18
2.7.2 Types and Sources of Dataset Bias 19
2.7.3 Detection and Mitigation Strategies 19
2.7.4 Implications for the Present Study 20
2.8 Evaluation Protocols and Reproducibility 21
2.9 Limitations of Existing Multimodal Studies 21
2.10 Positioning of the Present Study 22
2.11 Summary 23
CHAPTER 3 24
METHODOLOGY 24
3.1 Introduction 24
3.2 Datasets 25
3.2.1 CT Imaging Data 25
3.2.2 Urinary Biomarker Data 25

3.2.3 Dataset Challenges and Limitations 26
3.3 CT Imaging Preprocessing 26
3.3.1 Initial Processing Pipeline 26
3.4 Dataset Bias Detection and Mitigation 27
3.4.1 Bias Detection Procedures 27
3.4.2 Extreme Intensity Standardization 27
3.4.3 Spatial Bias Detection and Mitigation 28
3.4.4 Unsupervised Domain Analysis 29
3.4.5 Final Preprocessing Pipeline 29
3.5 Biomarker Data Preprocessing 30
3.6 Model Architectures 30
3.6.1 CT Classification Model 30
3.6.2 Biomarker Classification Model 31
3.7 Multimodal Fusion Methodology 31
3.7.1 The Synthetic Pairing Challenge 31
3.7.2 Decision-Level Fusion 32
3.7.3 Feature-Level Fusion 32
3.7.4 Bias-Aware Controls 33
3.8 Evaluation Protocol 33
3.8.1 CT Model Evaluation 33
3.8.2 Biomarker Model Evaluation 33
3.8.3 Fusion Model Evaluation 34
3.8.4 Statistical Robustness 34
3.9 Implementation Details 34
3.10 Ethical Considerations 35
3.11 Alignment with Research Objectives 35

3.12 Summary 36
CHAPTER 4 37
IMPLEMENTATION 37
4.1 Introduction 37
4.2 Development Environment 37
4.3 Data Preparation and Preprocessing 39
4.3.1 CT Image Loading and Initial Data Organization 39
4.3.2 Exploratory Data Analysis and Bias Detection 39
4.3.3 Bias Mitigation: Iterative Development of Intensity Standardization 40
4.3.4 Orientation Correction and Body Segmentation 42
4.3.5 Unsupervised Domain Analysis 43
4.3.6 Biomarker Data Preprocessing 45
4.4 Model Implementation 46
4.4.1 CT Classification Model (ResNet50) 47
4.4.2 Biomarker Classification Model (MLP) 49
4.4.3 Multimodal Fusion Implementation 51
4.5 Training and Evaluation Pipeline 53
4.6 Implementation Challenges and Solutions 55
4.7 Summary 56
CHAPTER 5 57
RESULTS AND DISCUSSION 57
5.1 Introduction 57
5.2 Dataset Bias Analysis 57
5.2.1 Initial Bias Detection 57
5.2.2 Iterative Mitigation Results 58
5.3 CT Classification Model Results 60

5.3.1 Slice-Level Performance 60
5.3.2 Patient-Level Performance 61
5.3.3 Unsupervised Domain Analysis and Cross-Cluster Generalization 61
5.3.4 Grad-CAM Explainability Analysis 63
5.3.5 Training Dynamics 64
5.4 Biomarker Classification Model Results 65
5.4.1 Test Set Performance 65
5.4.2 Comparison with Original Study 66
5.4.3 Feature Importance and Training Dynamics 66
5.5 Multimodal Fusion Results 68
5.5.1 Decision-Level Fusion 68
5.5.2 Feature-Level Fusion 69
5.6 Comparison with Existing Literature 71
5.7 Limitations 73
5.8 Summary 74
CHAPTER 6 75
CONCLUSION AND RECOMMENDATIONS 75
6.1 Introduction 75
6.2 Achievement of Research Objectives 75
6.2.1 Objective 1: Bias-Aware Data Foundation 75
6.2.2 Objective 2: Unimodal Model Development and Evaluation 76
6.2.3 Objective 3: Multimodal Fusion Investigation 77
6.3 Addressing the Research Questions 78
6.4 Contributions to Knowledge 79
6.5 Limitations 80
6.6 Recommendations for Future Work 81

6.6.1 Addressing the Domain Bias Problem 81
6.6.2 Enabling Genuine Multimodal Evaluation 82
6.6.3 Extending Clinical Scope 83
6.6.4 Prospective Validation and Clinical Integration 83
6.6.5 Technical Improvements 83
6.7 Final Remarks 84
REFERENCES 86
APPENDIX A: RESEARCH PROPOSAL 90
APPENDIX B: Key Code Listings 104
B.1: Reproducibility Seed Initialization 104
B.2: Extreme Intensity Standardization Function 104
B.3: ResNet50 Model Construction 105
B.4: Biomarker MLP Architecture 105
B.5: CT Embedding Extraction via Forward Hook 106

LIST OF TABLES

Table 4.1: Software dependencies used in the implementation. 38

Table 4.2: Summary of bias mitigation attempts. 42

Table 4.3: Biomarker dataset characteristics after preprocessing. 46

Table 4.4: CT model hyperparameter configuration. 48

Table 4.5: Biomarker model hyperparameter configuration. 50

Table 4.6: Computational requirements of major experiments. 55

Table 5.1: Bias mitigation results summary. 58

Table 5.2: CT model slice-level test performance. 60

Table 5.3: CT model patient-level test performance. 61

Table 5.4: Biomarker model test performance. 65

Table 5.5: Decision-level fusion performance across CT weights. 68

Table 5.6: Feature-level fusion sanity-check summary. 69

Table 5.7: Comparison with selected prior studies. 71

LIST OF FIGURES

Figure 4.1: Dataset bias detection results. 40

Figure 4.2: Body segmentation and cropping pipeline examples. 43

Figure 4.3: K-selection analysis (elbow and silhouette). 45

Figure 4.4: K-Means cluster distribution (k = 2) on CT embeddings. 45

Figure 4.5: CT model training history. 49

Figure 4.6: Biomarker model training curves. 51

Figure 4.7: Biomarker test performance (ROC and confusion matrix). 51

Figure 4.8: Decision-level fusion AUC versus CT weight. 53

Figure 4.9: Fusion sanity-check performance under label perturbation. 53

Figure 5.1: CT cohort composition and class distribution. 59

Figure 5.2: CT test-set performance. 62

Figure 5.3: Comparative model performance summary. 63

Figure 5.4: Grad-CAM visualisation examples. 64

Figure 5.5: Biomarker model calibration plot. 64

Figure 5.6: Biomarker permutation feature importance. 67

Figure 5.7: Biomarker score distribution by class. 67

Figure 5.8: Decision-level fusion sanity-check AUC. 70

Figure 5.9: Feature-level fusion sanity-check AUC. 71

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AMP	Automatic Mixed Precision
AUC	Area Under the Curve
AUC-ROC	Area Under the ROC Curve
BRCA1	Breast Cancer Gene 1
CA19-9	Carbohydrate Antigen 19-9
CA242	Carbohydrate Antigen 242
CBCT	Cone Beam Computed Tomography
CDKN2A	Cyclin-Dependent Kinase Inhibitor 2A
CEA	Carcinoembryonic Antigen
CECT	Contrast-Enhanced Computed Tomography
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
COVID-19	Coronavirus Disease 2019
CPU	Central Processing Unit
CSV	Comma-Separated Values
CT	Computed Tomography
CT-CB	Pancreas-CT-CB dataset (TCIA)
CUDA	Compute Unified Device Architecture
DICOM	Digital Imaging and Communications in Medicine
DL	Deep Learning
EDA	Exploratory Data Analysis
ELISA	Enzyme-Linked Immunosorbent Assay
F1	F1 Score
FPR	False Positive Rate

GANTT_____Generalized Activity Normalization Timetable

GB_____Gigabyte

GEO_____Gene Expression Omnibus

GLOBOCAN_____Global Cancer Observatory

GPU_____Graphics Processing Unit

Grad-CAM_____Gradient-weighted Class Activation Mapping

HU_____Hounsfield Units

I-IV_____Stage I-IV

JSON_____JavaScript Object Notation

KNN_____k-Nearest Neighbors

KRAS_____Kirsten rat sarcoma viral oncogene homolog

LSTM_____Long Short-Term Memory

LYVE1_____Lymphatic Vessel Endothelial Hyaluronan Receptor 1

miRNA_____microRNA

MLP_____Multi-Layer Perceptron

MRI_____Magnetic Resonance Imaging

NBIA_____National Biomedical Imaging Archive

PDAC_____Pancreatic Ductal Adenocarcinoma

PET-CT_____Positron Emission Tomography – Computed Tomography

PNG_____Portable Network Graphics

REG1A_____Regenerating Islet-Derived 1A

REG1B_____Regenerating Islet-Derived 1B

ROC_____Receiver Operating Characteristic

SEED_____Random Seed

SEER_____Surveillance, Epidemiology, and End Results

SMAD4_____SMAD Family Member 4

SVM-RFE	Support Vector Machine – Recursive Feature Elimination
TCIA	The Cancer Imaging Archive
TFF1	Trefoil Factor 1
TP53	Tumor Protein p53
TPR	True Positive Rate
UID	Unique Identifier
VGG16	Visual Geometry Group 16-layer Model
VRAM	Video RAM

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal cancers, characterized by silent onset, rapid progression, and high treatment resistance. GLOBOCAN 2020 reports pancreatic cancer as the seventh leading cause of cancer death worldwide, with about 466,000 deaths annually. The mortality-to-incidence ratio is close to one, meaning that most patients diagnosed with PDAC will ultimately die from the disease. This near-parallel pattern between incidence and mortality across regions reflects the aggressive biology of PDAC and the limited impact of current diagnostic and treatment pathways. These features have made PDAC a priority target for early-detection research in translational oncology.

Prognosis remains poor. Five-year survival hovers around 10% in many developed settings and is often worse in resource-limited systems. What makes this particularly frustrating is the lack of meaningful progress: unlike breast, colorectal, or even lung cancers, where survival gains over the past few decades have been substantial, pancreatic cancer outcomes have barely budged despite advances in surgery, systemic therapy, and supportive care. The survival gap between PDAC and other common malignancies reinforces just how urgently earlier diagnosis is needed.

Poor outcomes are linked to both anatomy and biology. The pancreas sits deep in the retroperitoneum, hidden behind other organs and practically inaccessible to routine examination. Its proximity to major vessels and lymphatic pathways means that even small tumors can spread early. By the time a patient presents with symptoms, local invasion or occult metastatic spread has often already occurred, and curative options have narrowed considerably.

PDAC is also characterized by dense desmoplastic stroma, hypovascular tumor microenvironments, and recurrent genomic alterations, including frequent KRAS activation and inactivation of TP53, CDKN2A, and SMAD4. Together, these factors promote therapeutic resistance and contribute to the aggressive clinical behavior associated with the disease.

The stage at which PDAC is diagnosed largely determines a patient's survival prospects. When the disease is still localized, five-year survival can reach about 39%. But only a small minority of cases are caught that early. Once the cancer has spread regionally, survival drops to roughly 7-12%, and in metastatic cases, where more than half of the patients present it plummets to

around 3%. The gradient is steep, and the message is clear, diagnostic delay carries a disproportionately large clinical cost in PDAC.

Late presentation is common because early PDAC is usually asymptomatic or causes vague symptoms such as dyspepsia, mild back pain, fatigue, or new-onset glycemic changes. By the time classic features such as jaundice, severe abdominal or back pain, and marked weight loss appear, disease is often advanced.

Computed tomography (CT) remains the primary imaging modality for pancreatic cancer diagnosis and staging. Contrast-enhanced multiphasic CT can detect lesions, assess local extension, and map vascular anatomy for surgical planning. However, CT sensitivity drops substantially for small lesions, particularly those below two centimeters, which are precisely the tumors where early detection would matter most. Adding to the difficulty, early pancreatic lesions can look a lot like benign inflammatory change, chronic pancreatitis, or normal tissue variation.

Serum biomarkers offer a complementary angle: they are minimally invasive, cheap to run at scale, and suitable for serial monitoring. CA19-9 is the most widely used marker in pancreatic cancer, but it has well-known limitations. Elevations appear in a range of benign hepatobiliary and inflammatory conditions, and in individuals who lack Lewis antigen expression, the test is uninformative altogether. More recently, urinary biomarker panels combining LYVE1, REG1B, and TFF1 have shown encouraging discriminative performance, though they still need robust external validation before they could be deployed routinely.

Taken together, these limitations create a clear need for early detection strategies that integrate complementary sources of diagnostic information. The present study takes on this challenge by developing and critically evaluating a bias-aware multimodal learning framework that combines CT imaging with biomarker data for pancreatic cancer detection.

1.2 Artificial Intelligence in Medical Diagnosis

Deep learning, and convolutional neural networks (CNNs) in particular, has substantially reshaped medical image analysis. Rather than relying on hand-crafted feature pipelines, CNNs learn hierarchical representations directly from pixel data, picking up complex non-linear visual patterns that would be difficult or impossible to specify manually. Transfer learning adapting models pretrained on large general-purpose datasets has made it practical to apply these methods even when annotated medical data is scarce.

In controlled evaluation settings, deep learning models have reached expert-level performance on a range of clinical imaging tasks. For pancreatic cancer specifically, the potential benefits are clear: if a model can reliably flag suspicious lesions before symptoms develop, this could substantially improve survival outcomes. The key question, though, is whether such models hold up when faced with realistic data heterogeneity rather than curated benchmarks.

Pretrained architectures such as ResNet variants are commonly fine-tuned for medical classification tasks. This transfer-learning strategy generally improves optimization stability and convergence in small-to-moderate datasets, and it provides a practical foundation for many clinically oriented imaging pipelines.

Similarly, machine learning models have shown effectiveness in modelling complex relationships within biomarker and clinical datasets. Ensemble methods, neural networks, and probabilistic models have been widely explored for disease risk prediction using combinations of clinical variables, laboratory measurements, and molecular markers. Multi-layer perceptrons (MLPs) offer a flexible framework for learning non-linear decision boundaries in tabular biomarker data, enabling integration of multiple clinical and molecular features into unified risk scores.

Despite this progress, clinical deployment of AI-based diagnostic systems remains limited. Many reported systems have been trained and evaluated on narrow datasets that fail to capture real-world variability. Models may end up learning trivial correlations, scanner quirks, institutional protocols, acquisition artifacts, rather than actual pathological features. A model that appears to work brilliantly on one hospital's data can fall apart when applied to another's. This is why rigorous evaluation, focused on reliability and robustness rather than headline accuracy, matters as much as the models themselves.

1.3 Challenges in Multimodal Learning and Dataset Bias

Conceptually, multimodal learning leverages the fact that different data sources capture different aspects of disease, so combining them can improve diagnostic reliability relative to any single modality. Imaging reveals structural and anatomical patterns; biomarkers reflect molecular and systemic signals that may appear before any visible anatomical change.

In practice, however, multimodal systems introduce several challenges: the modalities may be structurally very different, fusion design involves unavoidable trade-offs, aligning data from different sources is rarely straightforward, and the risk of confounding increases. Without

careful validation, it is easy to mistake artificial performance gains for genuine complementary value.

Most existing multimodal approaches assume ideal patient-level alignment between modalities. In practice, publicly available medical datasets are rarely aligned, and multimodal cohorts are limited in size. Consequently, researchers frequently rely on synthetic pairing or approximate alignment strategies, which introduce additional sources of bias and uncertainty. Furthermore, when one modality exhibits substantially stronger discriminative power, fusion models may become dominated by that modality, reducing the effective contribution of complementary information. Understanding and characterizing this modality dominance effect is essential for interpreting fusion results and designing balanced integration strategies.

A major challenge in medical AI is dataset bias (domain shift), where models learn acquisition-specific signatures instead of pathology. Differences in scanner hardware, reconstruction settings, site protocols, image processing pipelines, and cohort composition can create strong non-clinical cues that inflate internal performance metrics.

This risk is particularly high in cross-dataset settings where cancer and control cohorts originate from different institutions. In such scenarios, apparent classification success may reflect source discrimination rather than disease detection, with substantial degradation during external use. Detecting and mitigating this behavior is therefore a central focus of this dissertation.

1.4 Methodological Perspective and Research Focus

Although multimodal AI holds genuine promise for early detection, the literature includes studies whose reported gains may be inflated by unaddressed bias. Models may exploit scanner or protocol differences that correlate with class labels, producing strong metrics that do not necessarily reflect disease biology.

Fusion results can also be misleading when assumptions of patient alignment and data homogeneity are violated. With independently sourced public datasets, observed gains may reflect modality dominance instead of genuine cross-modal complementation.

This study is designed with these risks in view. The approach prioritizes transparent methodology, negative controls, multi-seed validation, and cautious interpretation. The concern is not only whether a model performs well on a test set, but whether it learns clinically meaningful patterns that can generalize beyond the specific datasets used for development..

1.5 Research Aim and Objectives

1.5.1 Research Aim

The primary aim of this research is to develop and rigorously evaluate a bias-aware multimodal deep learning framework for pancreatic cancer detection that integrates CT imaging and urinary biomarker data, while systematically analyzing the impact of dataset bias, modality dominance, and alignment constraints on model performance.

1.5.2 Research Objectives

To achieve this aim, three specific research objectives were defined:

- Establish a reliable and bias-aware data foundation for pancreatic cancer detection by preparing CT imaging and biomarker datasets, enforcing patient-level leakage control, and ensuring reproducible data readiness for downstream analysis.
- Develop and evaluate unimodal diagnostic models for CT imaging and urinary biomarkers to quantify the independent predictive value and limitations of each modality under transparent evaluation.
- Investigate whether multimodal fusion provides meaningful value beyond unimodal baselines under synthetic pairing constraints, while explicitly analyzing modality dominance, residual bias effects, and robustness across controlled validation settings.

Together, these objectives emphasize clinical relevance, methodological validity, and transparent reporting rather than metric-driven optimization alone. Detailed procedures used to carry out these objectives are provided in Chapter 3, with implementation specific details presented in Chapter 4.

1.6 Research Questions

How does dataset bias influence CT-based pancreatic cancer detection models, and what is the magnitude of bias in cross-dataset studies as quantified by the ability of simple classifiers to discriminate between datasets based solely on low-level image statistics such as mean pixel intensity?

To what extent can preprocessing and normalization strategies, including histogram matching, forced standardization, and novel body-region-specific intensity standardization, mitigate bias-related effects while preserving clinically relevant features necessary for disease classification?

Do multimodal fusion strategies provide genuine complementary diagnostic value beyond unimodal CT performance under synthetic pairing conditions, and which fusion strategy, decision-level versus feature-level demonstrates superior performance for pancreatic cancer detection?

How are fusion outcomes affected by residual CT bias and modality dominance, and what procedural safeguards are necessary for reliable multimodal evaluation?

Can machine learning models trained on the validated urinary biomarker panel achieve area under the curve values comparable to the 0.94 reported in the original (Debernardi et al., 2020) study, thereby confirming reproducibility and robustness using an entirely independent computational implementation?

1.7 Scope of the Study

This dissertation focuses deliberately on binary PDAC versus control classification. This focus is intended to produce interpretable findings within the constraints of the available data and resources. Staging, grading, and broader pancreatic pathology differentiation are left for future work, once a robust binary classification foundation has been established.

All analyses use publicly available datasets. For CT, the processed cohort used in this project contains 122 patients (40 PDAC, 82 controls). Biomarker modelling uses the Debernardi et al. (2020) dataset with 590 samples and seven selected features, with data governance aligned to original usage terms.

One important limitation to flag upfront, there is no patient-level alignment between the CT and biomarker datasets. The multimodal experiments rely on class-consistent synthetic pairing, which is enough to test fusion methodology but cannot demonstrate true individual patient level complementation. This constraint shaped how the fusion results are interpreted throughout the dissertation and is addressed directly in the discussion of future work.

Evaluation uses strict patient-level splitting to prevent leakage: an initial stratified 70/30 split, followed by equal validation and test partitioning of the temporary set. Experiments were executed in Google Colab and compatible local notebook environments using Python, PyTorch, Albumentations, and scikit-learn.

1.8 Significance of the Study

The contributions of this research cut across several areas: pancreatic cancer detection, medical imaging analysis, machine learning methodology, and translational clinical research.

Methodological Contribution in Bias Detection and Mitigation: The study documents a practical, stepwise pipeline for detecting and reducing cross-dataset CT bias, including negative findings from initial mitigation attempts and improvement through body-region-focused intensity standardization. This contributes a transparent template for handling shortcut learning risk in cross-cohort medical imaging studies.

Multimodal Fusion Contribution: The work implements and compares decision-level and feature-level fusion under synthetic pairing constraints, with explicit controls for modality dominance and cautious interpretation of outcomes. The contribution is therefore methodological as well as empirical, highlighting both capability and boundary conditions for unpaired multimodal experiments.

Independent Reproducibility Assessment: This study provides an independent computational assessment of the (Debernardi et al., 2020) urinary biomarker panel using the same core feature set and transparent reporting practices. By evaluating whether comparable discriminative performance can be obtained in an independent implementation, the work contributes evidence on reproducibility in pancreatic cancer biomarker modelling.

Commitment to Open Science and Transparency: The dissertation uses public datasets, reproducible pipelines, and explicit reporting of both successful and unsuccessful approaches to support independent verification and realistic interpretation.

Educational Value: The study provides a transparent case example of building and auditing a medical AI pipeline under real constraints, including bias risks, alignment limitations, and reproducibility considerations.

1.9 Structure of the Dissertation

The remainder of this dissertation is organized into five chapters covering literature context, methodology, implementation and final conclusions.

Chapter 2 reviews related work in pancreatic cancer diagnosis, medical imaging AI, biomarker based detection, multimodal fusion, and dataset bias, and then positions the research gap addressed in this dissertation.

Chapter 3 presents the research methodology, including datasets, preprocessing design, bias-mitigation strategy, model development plan, fusion formulation, and evaluation protocol.

Chapter 4 describes implementation details of the full experimental pipeline, including environment configuration, preprocessing execution, model training workflow, and experiment orchestration.

Chapter 5 reports results by research question, covering unimodal and fusion outcomes, robustness checks, and bias-aware interpretation of performance behavior.

Chapter 6 summarizes key conclusions, practical and methodological contributions, study limitations, and future directions toward clinically grounded multimodal validation.

1.10 Chapter Summary

This chapter established the clinical motivation and research foundation for the study. Pancreatic ductal adenocarcinoma was identified as a high-priority early detection challenge due to its late presentation, aggressive biology, and poor survival outcomes that are strongly dependent on stage at diagnosis. The limitations of current diagnostic approaches were outlined, including the reduced sensitivity of CT imaging for small lesions and the insufficient specificity of existing serum biomarkers, motivating the integration of complementary data sources through multimodal learning. Key methodological challenges were introduced, particularly dataset bias arising from cross institutional imaging differences and the constraints of synthetic pairing in the absence of patient aligned multimodal data. The research aim, three specific objectives, and five research questions were defined, centering on bias aware data preparation, transparent unimodal evaluation, and rigorous assessment of fusion value under realistic constraints.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter surveys the research landscape that the present study sits within, pancreatic cancer diagnosis, medical imaging, biomarker analysis, deep learning in healthcare, multimodal fusion, and the problem of dataset bias in AI systems. The goal is not simply to catalogue prior work but to look critically at the assumptions, evaluation protocols, and unacknowledged confounders that run through much of the existing literature. Many published studies treat classification accuracy as the primary measure of success. In contrast, this dissertation places equal emphasis on methodological validity and transparent reporting, particularly given the clinical implications of the models under study.

2.2 Clinical Background of Pancreatic Cancer

Pancreatic ductal adenocarcinoma (PDAC) accounts for the majority of pancreatic cancer cases ranks among the deadliest malignancies in oncology. According to GLOBOCAN 2022 estimates, pancreatic cancer accounted for 510,566 new cases globally, representing 2.6% of all cancer diagnoses, yet contributed disproportionately to cancer mortality with 467,005 deaths, comprising 4.8% of all cancer-related fatalities (Bray et al., 2024). The gap between those two numbers tells the story, with a mortality-to-incidence ratio approaching 94% (Rawla et al., 2019). In other words, the disease is fatal to nearly everyone who gets this diagnosis. The five-year relative survival rate for pancreatic cancer remains low at approximately 12–13%, as opposed to survival rates observed in other common malignancies such as breast cancer (90%), prostate cancer (98%), or even lung cancer (20%) (Park et al., 2021) and (Stoffel et al., 2023).

The global burden of pancreatic cancer varies substantially with geography, with the highest age-standardized incidence rates observed in Western Europe, Eastern Europe, and Northern America (Bray et al., 2024). a pattern that tracks closely with the Human Development Index and the lifestyle factors that come with it: obesity, diabetes, and western dietary patterns (Hesami et al., 2024). Additional established risk factors include cigarette smoking, which accounts for approximately 25% of cases, chronic pancreatitis, family history of pancreatic cancer, and hereditary cancer syndromes including BRCA1/2 mutations, Lynch syndrome. Projections forecast a substantial increase in pancreatic cancer prevalence through 2040, with

anticipated global prevalence rates representing a 31.45% increase since 1990 (Hesami et al., 2024). The trend is moving in the wrong direction, making the case for better detection tools all the more urgent.

The core of the problem is a cruel timing paradox. Survival depends heavily on stage at diagnosis, but the vast majority of patients are diagnosed too late for curative treatment. According to SEER data, the five-year relative survival rate for localized pancreatic cancer reaches 43.6%, yet only 14.6% of cases are diagnosed at this early, potentially curable stage (National Cancer Institute, 2024). Conversely, approximately 52% of patients present with distant metastatic disease, for which the five-year survival rate plummets to 3.0% (Siegel et al., 2024), essentially a death sentence for nearly half the patients at diagnosis. This nearly 15 fold difference in survival between localized and metastatic cases highlights the importance of early detection.

Several anatomical and biological characteristics contribute to late-stage presentation. The pancreas occupies a retroperitoneal location deep within the abdomen, rendering it anatomically inaccessible to routine physical examination (Park et al., 2021). The organ's close proximity to major vascular structures lead to early vascular invasion and metastasis even from relatively small primary tumors. Also, pancreatic tumors typically remain clinically silent during early stages, with the classic symptoms of jaundice, abdominal pain, and weight loss manifesting only after substantial spread of the tumor (Stoffel et al., 2023). At the molecular level, PDAC is characterized by near-universal KRAS mutations (present in 90–95% of cases), frequent inactivation of tumor suppressor genes TP53, CDKN2A, and SMAD4, and extensive desmoplastic stroma that constitutes up to 90% of tumor volume (Khalaf et al., 2021). This dense fibrous stroma contributes to the characteristic imaging appearance of pancreatic tumors, which often appear isodense or only subtly hypodense relative to adjacent normal tissue on CT imaging, particularly for lesions smaller than 2 centimeters (Chen et al., 2023).

2.3 Medical Imaging for Pancreatic Cancer Detection

2.3.1 Computed Tomography in Pancreatic Cancer

Computed tomography has emerged as the cornerstone imaging modality for pancreatic cancer evaluation, offering superior spatial resolution, rapid acquisition times, and comprehensive visualization of both the pancreatic parenchyma and surrounding vascular structures critical for surgical planning. Modern multidetector CT protocols typically employ thin-section imaging

(1–3 millimeters) with contrast enhancement, acquiring images during pancreatic parenchymal phase (40–50 seconds post-contrast) and portal venous phase (60–70 seconds post-contrast) to optimize tumor visibility (Park et al., 2021).

Despite these technical refinements, CT imaging demonstrates significant limitations in detecting early-stage pancreatic cancer, particularly for lesions measuring less than 2 centimeters. Reported sensitivity values range from 70% to 95% overall, but decline substantially for small tumors, with sensitivity dropping below 60% for lesions under 2 centimeters (Chen et al., 2023). Several factors contribute to this diminished performance, small pancreatic tumors may be truly isodense with surrounding parenchyma, background heterogeneity from chronic pancreatitis, fatty replacement, or age-related decline can obscure subtle mass lesions and technical quality varies considerably across institutions with suboptimal bolus timing, inadequate contrast dose degrading diagnostic performance.

Inter-observer variability represents an additional substantial limitation. Studies examining agreement between radiologists in detecting and characterizing pancreatic lesions report moderate inter observer agreement with kappa values typically ranging from 0.4 to 0.6(Liu et al., 2020). Alternative imaging modalities including MRI, endoscopic ultrasound, and PET-CT offer distinct advantages but remain limited by practical constraints including cost, availability, invasiveness, and insufficient spatial resolution, hindering their application in population level screening programmes.

These imaging limitations are directly relevant to the present study. The CT datasets used in this dissertation Pancreas-CT-CB and PANCREAS from the Cancer Imaging Archive originate from different institutions and scanning protocols. This heterogeneity, while representative of real-world clinical diversity, simultaneously introduces the risk that models may learn to distinguish scanner characteristics rather than genuine disease features. This concern motivates the rigorous bias detection methodology described in Chapter 3.

2.3.2 Deep Learning in Medical Imaging

The advent of deep learning, particularly convolutional neural networks (CNNs), has led to a dramatic shift in medical image analysis, demonstrating capabilities that approach or exceed human expert performance across diverse diagnostic tasks. CNNs comprise hierarchical layers that progressively extract increasingly abstract feature representations from raw pixel intensities (LeCun et al., 2015) and (Litjens et al., 2017). This hierarchical feature learning circumvents the need for manual feature engineering, instead allowing networks to discover

optimal feature representations directly from training data through iterative optimization via backpropagation.

Transfer learning has emerged as a particularly powerful strategy for medical imaging applications, where annotated training data remains scarce. This approach leverages networks pretrained on massive datasets such as ImageNet to initialize model weights before fine-tuning on medical images (Kim et al., 2022). Architectures such as ResNet, DenseNet, and EfficientNet have demonstrated particular success when adapted to medical imaging tasks. ResNet architectures, introduced by (He et al., 2016), incorporate skip connections that enable training of networks exceeding 100 layers without performance degradation from vanishing gradients. In medical imaging contexts, ResNet variants have achieved remarkable performance across diverse applications including breast cancer detection, lung nodule classification, skin lesion classification, and diabetic retinopathy grading (Litjens et al., 2017), (Kim et al., 2022), (Esteva et al., 2017) and (Gulshan et al., 2016).

However, a critical observation motivating the present study is that many of these headline performance claims require careful scrutiny. Deep learning models exhibit well-documented sensitivity to domain shifts, scanner variations, and preprocessing pipelines (Pooch et al., 2020). A model achieving 95% accuracy on data from one scanner manufacturer may perform at near-chance levels on data from another. This sensitivity is not merely a theoretical concern but a practical barrier to clinical deployment, and it directly informs the methodological caution adopted throughout this dissertation.

The clinical translation of deep learning for cancer detection has progressed beyond academic research into real-world deployment, with multiple systems receiving regulatory approval. For breast cancer screening, commercial AI systems have demonstrated ability to reduce radiologist workload while maintaining or improving diagnostic performance (Leibig et al., 2022). In lung cancer screening, AI algorithms have achieved sensitivity comparable to experienced thoracic radiologists while substantially reducing false positive rates (Kim et al., 2022). For pathology applications, deep learning systems analyzing whole slide histopathologic images have achieved expert-level performance in tumor grading and molecular subtype classification (Litjens et al., 2017). These real-world implementations validate deep learning's potential while simultaneously revealing challenges related to generalization across institutions, integration with clinical workflows, and maintenance of performance in heterogeneous patient populations.

2.3.3 Deep Learning for Pancreatic Cancer Detection

Application of deep learning to pancreatic cancer detection from CT imaging has yielded promising yet heterogeneous results. (Chen et al., 2023) conducted a nationwide population-based study in Taiwan, training a deep learning model on contrast enhanced CT scans from over 400 institutions, achieving an AUC of 0.92 for detecting pancreatic cancer within 180 days before clinical diagnosis. Their model demonstrated particular strength in detecting small tumors measuring less than 2 centimeters, the population most likely to benefit from early intervention. (Liu et al., 2020) developed and externally validated a system across institutions in Taiwan, Japan, and the United States, achieving AUCs ranging from 0.86 to 0.94 across external validation cohorts. Importantly, the authors demonstrated that ensemble approaches combining predictions from multiple architectures outperformed single model approaches, and that incorporating clinical variables improved classification performance, leading to the multimodal integration strategies central to the present investigation.

(Ma et al., 2020) employed 3D CNNs to exploit volumetric information across multiple consecutive CT slices, achieving accuracy of 93.7% on an independent test set. (Zavalsız et al., 2023) reported that ResNet-50 achieved the highest classification accuracy at 94.8% among pretrained architectures compared. (Dinesh et al., 2023) proposed a hybrid approach combining CNNs with attention mechanisms, achieving a Dice coefficient of 0.87 for tumor segmentation and classification accuracy of 96.2%, demonstrating that attention mechanisms can guide the network to focus on diagnostically relevant regions.

Despite these encouraging results, critical limitations curb enthusiasm regarding immediate clinical implementation. First, most studies employ retrospective designs using curated datasets of confirmed cases and healthy controls, a case-control paradigm that likely inflates apparent performance. Second, external validation cohorts typically originate from a limited number of institutions with similar protocols, raising questions about generalizability. Third, and most critically for the present investigation, few studies have explicitly addressed or even acknowledged the potential for dataset bias arising from systematic differences in imaging characteristics between cancer and control cohorts. This concern is not merely theoretical (Zech et al., 2018) and (DeGrave et al., 2021) have demonstrated that deep learning models can achieve spuriously high performance by learning dataset specific characteristics rather than relevant disease features.

The present study deliberately adopts a more cautious stance than much of the existing literature. Rather than treating strong CT classifier performance as direct evidence of disease learning, the focus is on whether a ResNet50-based pipeline can learn pathology relevant features after systematic bias detection and mitigation. This framing prioritizes methodological validity over optimistic performance claims.

2.4 Biomarkers for Pancreatic Cancer Detection

2.4.1 Serum Biomarkers and Their Limitations

Serum biomarkers have long served as tools in pancreatic cancer management, with carbohydrate antigen 19-9 (CA19-9) representing the most extensively validated and clinically utilized marker. Meta analyses report sensitivity values ranging from 75% to 81% and specificity from 82% to 90% when distinguishing pancreatic cancer from benign pancreatic disease using the standard cutoff of 37 U/mL. However, these seemingly reasonable performance metrics mask several critical limitations that affect CA19-9 utilization as a screening tool.

First, sensitivity for early-stage disease remains suboptimal, with detection rates falling below 60% for stage I cancers. Second, CA19-9 exhibits false negative results in approximately 5–10% of the population who possess the Lewis-negative phenotype, completely abolishing biomarker utility in this subset. Third, false positive elevations occur frequently in benign conditions including obstructive jaundice (affecting 10–60% of cases), cholangitis, and chronic pancreatitis, substantially limiting discriminatory power. These factors contribute to an extremely low positive predictive value of 0.5–0.9% when applied to asymptomatic populations. Additional serum biomarkers including carcinoembryonic antigen (CEA) and carbohydrate antigen 242 (CA242) have been explored but present similar fundamental limitations for early detection.

2.4.2 Urinary Biomarkers: The Debernardi et al. Panel

A shift in pancreatic cancer biomarker research emerged with the identification of urinary protein panels capable of distinguishing PDAC patients from controls with substantially superior performance to traditional serum markers. The seminal work by (Debernardi et al., 2020) established a three-protein urine biomarker panel comprising lymphatic vessel

endothelial hyaluronan receptor 1 (LYVE1), regenerating islet-derived protein 1B (REG1B), and trefoil factor 1 (TFF1).

The biological rationale for these biomarkers reflects distinct aspects of pancreatic cancer pathophysiology. LYVE1, a transmembrane glycoprotein primarily expressed in lymphatic endothelium, demonstrates elevated urinary concentration in PDAC patients, potentially reflecting lymph angiogenesis associated with tumor progression. REG1B belongs to the regenerating islet-derived protein family implicated in pancreatic cell proliferation and differentiation. TFF1, a member of the trefoil factor family involved in mucosal repair, exhibits altered expression in multiple adenocarcinomas including pancreatic cancer.

The validation study by (Debernardi et al., 2020) examined 590 urine specimens comprising 183 controls, 208 benign hepatobiliary disease cases (including 119 chronic pancreatitis patients), and 199 PDAC samples (102 stage I–II and 97 stage III–IV). Utilizing commercially available ELISA assays, the three-biomarker panel combined with age and sex demonstrated AUC values of 0.936 in both training and validation datasets for discriminating stage I–II PDAC from controls, with sensitivity and specificity both exceeding 85%. When combined with plasma CA19-9, the urinary panel achieved AUC of 0.992 with sensitivity 96.3% and specificity 96.7%, substantially surpassing either modality alone.

The clinical utility extends beyond diagnostic accuracy. The biomarker panel demonstrated stability in urine samples stored at room temperature for up to five days and exhibited minimal intra individual daily variation characteristics essential for practical clinical implementation. The associated PancRISK scoring algorithm provides clinically interpretable binary risk stratification optimized for different surveillance scenarios. Subsequent prospective cohort analyses demonstrated that the urinary biomarker panel could detect PDAC up to two years prior to clinical diagnosis, with urinary REG1B levels showing statistically significant elevation 12–24 months before diagnosis. This window represents a transformative opportunity for early detection.

The present study utilizes the publicly available (Debernardi et al., 2020) dataset (590 samples, 7 features) as the biomarker modality. Biomarker modelling in this dissertation produced performance around AUC approximately 0.94, broadly consistent with the original study and supportive of reproducibility under an independent implementation.

2.5 Machine Learning for Biomarker Analysis

Traditional machine learning approaches including support vector machines, random forests, and logistic regression have been applied to biomarker-based pancreatic cancer diagnosis with varying success. More recent work has explored neural networks and ensemble methods. Feature selection, normalization, and handling of missing values are critical preprocessing steps that substantially influence model performance. The relatively small sample sizes typical of biomarker studies (often fewer than 500 samples) constrain the complexity of models that can be reliably trained without overfitting.

The present study employs a multi-layer perceptron (MLP) architecture with a deliberately simple design (7 → 64 → 32 → 2) to avoid overfitting on the 590-sample biomarker dataset. This architectural simplicity is an intentional methodological choice, in a small-sample data set, model simplicity is preferable to architectural complexity, even at the potential cost of marginally lower performance on training data.

2.6 Multimodal Learning in Healthcare

2.6.1 Rationale for Multimodal Integration

The fundamental premise underlying multimodal approaches in cancer detection rests on the recognition that different data modalities capture complementary information about disease state. Imaging modalities excel at capturing spatial and structural information regarding tumor morphology, location, and anatomical relationships but provide limited insight into molecular activity. Conversely, biomarker panels reflect systemic and local biochemical changes associated with malignancy but lack anatomical specificity. The integration of these orthogonal information streams theoretically enables more robust disease detection by leveraging the strengths of each modality while compensating for individual weaknesses.

In oncology, multimodal systems have combined imaging, genomics, proteomics, and clinical records with demonstrable improvements in diagnostic performance. In breast cancer, integration of mammography with gene expression profiles has improved risk stratification and subtype classification beyond unimodal approaches. Lung cancer studies combining CT imaging with genomic and proteomic data have enhanced predictions and treatment response assessment. Prostate cancer studies fusing MRI with molecular biomarkers have refined risk stratification and biopsy decision support. However, application of systematic multimodal

fusion specifically to pancreatic cancer detection remains notably limited, partly reflecting practical challenges including data scarcity from prospectively matched imaging and biomarker acquisitions, patient cohort heterogeneity, and technical complexities of integrating varied data types with vastly different dimensionalities and statistical properties.

2.6.2 Fusion Strategies

The deep learning literature categorizes multimodal fusion approaches into three primary baskets. Early fusion (feature-level) integrates information by concatenating features from different modalities before feeding them into a unified model for joint processing. The principal advantage lies in its capacity to learn complex cross modal interactions, though it presents challenges including sensitivity to differences in feature scales and dimensionalities. Late fusion (decision-level) trains independent models for each modality and combines their outputs through ensemble techniques or learned weighting schemes. Late fusion offers practical advantages including independent model optimization, graceful handling of missing modalities, and improved interpretability. Intermediate fusion attempts to balance the advantages of both through hierarchical architectures with cross-modal interactions at intermediate feature levels.

The present study implements both feature-level and decision-level fusion strategies. However, a critical methodological caveat must be emphasized: the CT imaging data and biomarker data in this study originate from entirely different patient cohorts, requiring synthetic pairing through random matching. This means that no genuine patient-level correspondence exists between the two modalities. Any observed fusion benefit must therefore be interpreted with extreme caution, it may reflect statistical relevance between modality level predictions rather than genuine cross modal information about individual patients. This limitation is candidly acknowledged throughout the dissertation and distinguishes the present work from studies that may overstate fusion benefits without discussing data alignment assumptions.

2.6.3 The Synthetic Pairing Challenge

Real-world implementation of multimodal models is hindered by the difficulty of obtaining matched imaging and biomarker data from the same patients. Most available datasets contain either imaging or biomarkers, but not both. The present study addresses this constraint through synthetic pairing, where CT imaging predictions and biomarker predictions are randomly matched to simulate multimodal input. This approach has precedent in the literature but introduces fundamental limitations.

Synthetic pairing severs any genuine patient-level correspondence between modalities. In a clinically deployed system, a patient's CT scan and their biomarker profile would reflect the same underlying disease state, creating meaningful cross-modal correlations. With synthetic pairing, these correlations do not exist. Consequently, feature-level fusion which attempts to learn joint representations operates on concatenated embeddings that share no underlying patient identity. Decision-level fusion, which merely averages independent predictions, is somewhat less affected by this limitation, as it does not assume cross-modal feature interactions.

This study explicitly tests whether synthetic pairing produces genuine fusion benefit or merely reflects the dominance of the stronger individual modality. This question is rarely addressed in the multimodal medical imaging literature, where fusion improvements are often reported without rigorous controls for modality dominance.

2.7 Dataset Bias and Shortcut Learning

2.7.1 The Problem of Spurious Correlations

Deep learning models in medical imaging have demonstrated remarkable performance on internal validation datasets. However, numerous studies have revealed that these models frequently learn "shortcuts" spurious correlations between easily detectable non-pathological features and disease labels rather than genuine disease specific patterns (Geirhos et al., 2020). This phenomenon poses a critical threat to model generalizability and clinical deployment, as models exploiting dataset specific features often fail when applied to new hospitals, scanners, or patient populations.

The work of (Zech et al., 2018) demonstrated that pneumonia detection models trained on 158,323 chest radiographs across three hospital systems achieved near-perfect accuracy (AUC > 0.99) in identifying the source hospital system, revealing that subtle acquisition protocol differences were being exploited for disease detection. Gradient-based visualization revealed that models focused heavily on image corners and edges, where radiology technicians placed metal tokens and marker features completely irrelevant to pulmonary pathology. Most alarmingly, artificially increasing the imbalance in pneumonia prevalence between training institutions further encouraged shortcut learning, illustrating how models can achieve deceptively high accuracy by learning to associate hospital specific features with disease status.

The COVID-19 pandemic intensified these concerns when (DeGrave et al., 2021) conducted a rigorous audit of deep learning models for COVID-19 detection from chest radiographs, revealing widespread reliance on misleading features. By applying explainable AI techniques, they demonstrated that models consistently prioritized medically irrelevant features such as text markers, patient orientation labels, and dataset specific image characteristics over genuine pathology. When these characteristics were swapped between COVID-positive and COVID-negative images, models frequently reversed their predictions, confirming that decisions were based on irrelevant factors rather than clinical findings. These shortcuts emerged from the common practice of pooling images from disparate sources where positive and negative samples originated from different data collection pipelines, creating a scenario where the disease label became perfectly aligned with dataset origin.

2.7.2 Types and Sources of Dataset Bias

Dataset bias in medical imaging can be categorized into several distinct types (Rabanser et al., 2019). Data Acquisition Bias arises from differences in imaging hardware, scanner manufacturers, acquisition protocols, and institutional practices. Different hospital systems employ distinct scanner models, reconstruction algorithms, and image compression methods, creating subtle "fingerprints" that neural networks readily exploit (Pooch et al., 2020). Selection and Sampling Bias occurs when training datasets do not represent the target population. Label Bias emerges from inconsistencies in annotation processes. Confounding by Protected Attributes occurs when models inadvertently learn to use demographic information as shortcuts (Glocker et al., 2023). Intrinsic shortcuts such as contrast, sharpness, and noise characteristics are particularly challenging because they are entangled with genuine anatomical features.

2.7.3 Detection and Mitigation Strategies

Identifying dataset bias requires systematic approaches beyond standard performance metrics. Cross-dataset generalization testing remains the gold standard. Dataset origin classification provides a direct measure of bias severity: if a model can accurately predict which dataset produced a given image (AUC > 0.85), this strongly suggests exploitable institutional signatures (Zech et al., 2018). Explainable AI techniques including saliency maps and Grad-CAM can visualize which image regions most influence predictions.

Mitigation strategies span data-centric approaches and model-centric approaches. Data-centric strategies include careful dataset curation that mirrors the target deployment environment, data

preprocessing and augmentation to remove extrinsic shortcuts and normalize image characteristics across sources, and employ learning that exposes models to greater diversity. Intensity normalization, histogram equalization, and contrast standardization attempt to harmonize images from different scanners, though studies have shown that even after aggressive preprocessing, intrinsic shortcuts often remain (Teixeira et al., 2021). Model-centric strategies include domain adversarial training that explicitly penalizes learning of dataset-specific features (Trivedi et al., 2022), robust training objectives such as group distributionally robust optimization that focus on worst-case subgroup performance, and knowledge extraction from curated bias-free datasets (Boland et al., 2024). Despite these strategies, completely eliminating dataset bias remains an open challenge, and most successful approaches combine multiple interventions with transparent reporting of residual limitations.

2.7.4 Implications for the Present Study

The dataset bias literature has direct and profound relevance to this dissertation. For the implemented project cohort, CT data used in this work comprise 122 patients (40 cancer, 82 control) drawn from Pancreas-CT-CB and PANCREAS, with known cross-institutional acquisition differences. Preliminary analysis revealed that a simple logistic regression classifier trained on pixel intensity statistics alone could distinguish between the datasets with an AUC of 0.866, confirming substantial acquisition bias.

This finding fundamentally shaped the methodological approach of the present study. Rather than proceeding with standard model training and reporting potentially inflated accuracy, an extreme intensity standardization pipeline was developed to remove scanner-specific signatures while preserving anatomical structure. The standardization approach applies body-region-specific z-score normalization, computing statistics only from foreground tissue pixels and mapping the resulting z-scores to a fixed target distribution (mean 128, standard deviation 40). This approach was developed iteratively after initial attempts using histogram matching and simple global normalization proved insufficient, an important negative result that is documented in Chapter 4.

The effectiveness of this mitigation was validated by demonstrating that the bias classifier's AUC dropped from 0.866 to 0.569 (near random) after standardization, confirming that the scanner-specific fingerprints had been substantially attenuated. However, it must be acknowledged that even this reduced bias AUC exceeds the theoretical 0.500 of a perfectly unbiased dataset, suggesting that some residual bias may persist. This residual bias is

considered when interpreting CT model performance in subsequent chapters. The bias detection and mitigation process constitutes a central methodological contribution of the dissertation.

2.8 Evaluation Protocols and Reproducibility

Robust evaluation of medical AI systems requires external validation, cross-institutional testing, and transparent reporting. However, many studies rely solely on internal splits from single-institution datasets, creating an optimistic picture of model capabilities. The common practice of reporting only the best result from multiple experimental configurations a form of implicit p-hacking, further inflates apparent performance. Studies rarely report performance variability across different random seeds, data splits, or hyperparameter configurations, making it impossible to assess whether reported results are robust or represent lucky outcome.

Reproducibility is further hindered by incomplete documentation of preprocessing pipelines, training procedures, hyperparameter choices, and random seed selection. A recent systematic review found that fewer than 30% of medical imaging AI studies provided sufficient methodological detail for independent replication, and that studies with higher reported accuracy were less likely to provide complete code or data (Roberts et al., 2021). This reproducibility crisis is particularly acute in pancreatic cancer research, where small sample sizes and heterogeneous datasets make results highly sensitive to methodological choices.

Recent initiatives advocate standardized benchmarks, open-source code, and comprehensive reporting guidelines. The present study contributes to this movement by using exclusively public datasets (Cancer Imaging Archive and (Debernardi et al., 2020), documenting all preprocessing decisions including failed approaches, performing multi-seed analysis to assess result stability, and providing complete code for replication. Negative results and limitations are reported alongside successes, reflecting the view that transparent documentation of what does not work is as valuable as reporting what does.

2.9 Limitations of Existing Multimodal Studies

A critical assessment of existing multimodal medical imaging studies reveals several recurring weaknesses. Many suffer from small sample sizes, lack of data alignment between modalities, limited or absent bias analysis, and optimistic interpretation of results. Negative controls and sensitivity analyses are rarely performed, increasing the risk of misleading conclusions. Few studies explicitly test whether observed fusion improvements exceed what would be expected

from modality dominance alone, whether the fusion model genuinely integrates information from both modalities or simply inherits the performance of the stronger individual model.

The tendency to report only best case results, often from a single random seed or a single hyperparameter configuration, further inflates the apparent effectiveness of multimodal approaches. Publication bias compounds this problem: studies reporting null or negative fusion results are less likely to be published, creating a distorted literature in which multimodal fusion appears universally beneficial. Furthermore, few studies address the fundamental question of data alignment: when imaging and molecular data originate from different patient cohorts, any reported fusion benefit must be interpreted as arising from statistical complementarity rather than genuine patient-level multimodal integration.

The present study aims to address these limitations through systematic bias testing, explicit acknowledgment of synthetic pairing constraints, multi-seed evaluation, and critical interpretation of fusion results. The goal is not to achieve the highest possible accuracy metric but to understand what each modality and fusion strategy genuinely contributes under realistic conditions.

2.10 Positioning of the Present Study

In contrast to prior work that emphasizes maximizing classification accuracy, this study prioritizes methodological rigor and transparent evaluation. The key distinctions from existing literature are as follows. First, dataset bias is treated as a primary concern rather than an afterthought, with systematic detection and mitigation preceding model development. Second, the study explicitly acknowledges the synthetic pairing limitation and interprets fusion results accordingly, rather than presenting them as evidence of genuine multimodal synergy. Third, negative results, including near-perfect internal performance in biased CT settings that can be explained by scanner-related shortcuts, are reported as contributions rather than failures. Fourth, multi-seed analysis and careful evaluation protocols are employed to provide honest estimates of model performance variability.

This positioning reflects a broader argument that the medical AI field would benefit from a shift in emphasis: from headline accuracy metrics toward methodological validity, reproducible evaluation, and honest reporting of limitations. The following chapter details the specific methodological choices made to operationalize this perspective.

2.11 Summary

This chapter reviewed clinical, technical, and methodological literature related to pancreatic cancer detection and multimodal AI. PDAC remains among the deadliest malignancies, with a five-year survival rate of approximately 12–13% and survival critically dependent on early detection that current diagnostic methods inadequately provide. CT imaging, while serving as the primary diagnostic modality, exhibits limited sensitivity for early-stage tumors and is subject to significant inter-observer variability. Deep learning has shown promise for CT-based detection but faces serious challenges from dataset bias, limited external validation, and the tendency to learn scanner-specific features rather than disease features.

Urinary biomarkers, particularly the (Debernardi et al., 2020) panel of LYVE1, REG1B, and TFF1, offer a promising complementary modality with demonstrated ability to detect PDAC up to two years before clinical diagnosis. Multimodal fusion holds theoretical appeal but requires careful validation, especially when modalities originate from different patient cohorts and synthetic pairing must substitute for genuine patient-level correspondence.

The limitations of existing approaches, uncritical performance reporting, undetected dataset bias, unacknowledged synthetic pairing constraints, and insufficient reproducibility practices collectively motivate the bias-aware, reliability-oriented framework presented in the subsequent methodology chapter. The present study addresses these gaps by treating bias detection as a prerequisite for model training, acknowledging synthetic pairing as a fundamental constraint on fusion interpretation, and reporting negative results alongside successes as part of a transparent and scientifically rigorous approach to medical AI development.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter describes the methodology employed to develop and evaluate a bias-aware multimodal deep learning framework for early pancreatic cancer detection. The proposed approach integrates two complementary data modalities, CT imaging for spatial and structural information and urinary biomarkers for molecular and functional insights within an experimental design that prioritizes transparency and dataset bias analysis

The research follows a systematic pipeline comprising six stages:

- Data acquisition from publicly available repositories.
- Rigorous bias detection to identify dataset-specific artifacts that could enable shortcut learning,
- Comprehensive preprocessing including intensity standardization, spatial normalization, and artifact removal for CT images alongside biomarker cleaning and normalization.
- Unimodal model development using transfer learning for CT classification and a multi-layer perceptron for biomarker classification.
- Multimodal fusion through both decision-level and feature-level strategies under synthetic pairing conditions.
- Evaluation using slice-level, patient-level, per-cluster, cross-cluster, and multi-seed robustness analyses with negative controls.

As discussed in Chapter 2, medical imaging models frequently learn spurious correlations based on scanner characteristics and acquisition protocols rather than disease-specific features. Methodologically, implementation was executed in two stages: an offline script-based preprocessing stage (ct_preprocess.py and biomarker_preprocess.py) that prepared cleaned inputs, followed by the notebook stage for bias analysis, model training, and fusion experiments.

3.2 Datasets

3.2.1 CT Imaging Data

CT imaging data were acquired from two publicly available TCIA collections representing cancer and non-cancer cohorts. Raw inputs were DICOM abdominal CT series from different institutions and acquisition settings, which introduced substantial heterogeneity before modelling. A dedicated preprocessing script converted selected CT series to 8-bit grayscale PNG slices using soft-tissue windowing (window level = 50 HU, window width = 400 HU) and resizing to 224x224 for downstream analysis.

For this project run, the processed CT cohort comprised 122 patients (40 cancer, 82 control). The preprocessing script indexed 24,627 slices and retained 23,319 slices after script-level quality control, with 1,308 slices dropped. These retained slices formed the baseline input to the notebook pipeline.

The core challenge is that the cancer and control datasets come from different institutions, each with its own scanners, reconstruction algorithms, and acquisition protocols. This creates a real risk that models will learn to tell the datasets apart based on how the images were acquired, rather than picking up on anything related to actual pancreatic pathology, a concern that motivated the extensive bias detection and mitigation procedures described in Section 3.4.

Patient-Level Data Splitting: To prevent data leakage, splitting was enforced at the patient level so that all slices from a given patient appear in only one subset. In the notebook implementation, labels were stratified for the initial train versus temporary partition (70/30), and the temporary partition was then shuffled and divided into validation and test subsets of equal size.

3.2.2 Urinary Biomarker Data

Biomarker data were obtained from the publicly available dataset accompanying (Debernardi et al., 2020), collected at the Barts Cancer Institute, Queen Mary University of London. The dataset comprises 590 samples across three clinical categories: 199 PDAC patients (33.7%), 208 benign controls (35.3%), and 183 patients with chronic pancreatitis or benign pancreatic conditions (31.0%). For binary classification, benign and chronic pancreatitis samples were merged into a single control class, yielding 199 positive and 391 negative samples.

The raw dataset contains demographic and biomarker variables including sex, age, plasma CA19-9, creatinine, LYVE1, REG1B, TFF1, and REG1A. In the notebook modelling stage, the

biomarker classifier used a seven-feature subset: age, plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, and REG1A.

The dataset contains minimal missing values (less than 2% of entries), primarily in the CA19-9 field. PDAC patients were older (median age 65 years) compared to controls (median age 57 years), consistent with known epidemiology, and sex distribution was relatively balanced (52% male, 48% female overall).

3.2.3 Dataset Challenges and Limitations

Several challenges constrain the experimental design and are worth flagging upfront, since they directly shape how the results in Chapter 4 should be read. For CT imaging, the primary concerns are dataset heterogeneity arising from different source institutions and scanner types, systematic batch effects in image intensity and noise patterns, the limited sample size of 122 patients, and the availability of only patient-level labels without individual slice-level tumor annotations.

For multimodal integration, the most significant constraint is that CT and biomarker datasets contain entirely different patients with no overlap, warranting synthetic pairing strategies rather than genuine patient-matched data. The two modalities also differ in geographic origin (biomarker data from UK institutions; CT data from multiple countries), temporal collection periods, and information scale (hundreds of samples per CT patient versus a single measurement vector per biomarker sample). These challenges, particularly the absence of paired patient data and the dataset bias risk, define the boundaries within which all subsequent methodological choices and interpretations must be understood.

3.3 CT Imaging Preprocessing

3.3.1 Initial Processing Pipeline

Raw DICOM data were preprocessed using an offline CT script prior to notebook experiments. CT volumes were converted to 8-bit grayscale PNG slices using soft-tissue windowing (window level = 50 HU, window width = 400 HU) and resized to 224x224. Series selection used NBIA digest Series UID mappings when available; otherwise, the largest CT series with at least 20 slices was selected.

Script-level quality control removed technically invalid or low-value slices using file-size and near-blank pixel-variance checks, producing ct_index_clean.csv and drop logs. Anatomical

filtering and body-region-focused operations (orientation correction, segmentation, body-ratio filtering, and cropping refinements) were then executed in the notebook pipeline. All downstream bias analysis and model development used these processed PNG-based inputs.

3.4 Dataset Bias Detection and Mitigation

Before training any classification models, the first step is to measure how severe the dataset bias actually was. This section covers the detection procedures and mitigation strategies in detail, including two approaches that failed which are documented here.

3.4.1 Bias Detection Procedures

Three complementary approaches were employed to detect systematic differences between the cancer and control datasets. First, pixel-level statistical analysis was performed on sampled images using metrics such as mean intensity, standard deviation, median, and percentile-based summaries. These analyses showed clear distributional differences between cohorts, indicating that simple low-level intensity cues could act as shortcut signals.

Second, a logistic-regression classifier was used to test whether simple pixel statistics could predict dataset origin. High separability before mitigation confirmed shortcut-learning risk. After extreme intensity standardization, this separability dropped toward near-chance behavior (reported around AUC approximately 0.57), indicating substantial bias reduction.

Third, visual inspection of sample images revealed additional systematic differences including overall contrast patterns (cancer images appearing darker), distinct noise textures between datasets, scanner table and gantry artifacts at different vertical positions, and a 180-degree rotation difference in patient orientation (spine position inverted between datasets). These spatial biases provide further trivial shortcuts independent of intensity-based features.

3.4.2 Extreme Intensity Standardization

The extreme intensity standardization procedure was developed after two earlier attempts: histogram matching and forced global standardization, both failed to solve the problem. The histogram matching approach attempted to align the cumulative distribution function of each image to a reference distribution computed from the control dataset, but failed to eliminate bias because it matched entire distributions including background and air regions. Forced global standardization applied a fixed target mean and standard deviation across all pixels, which similarly failed because background regions dominated the statistics.

The extreme standardization method addresses these failures by operating exclusively on body tissue pixels. For each image, a body mask is computed by thresholding at intensity value 5 to exclude air and background. Body pixels are then z-score normalized independently per image, and the resulting z-scores are rescaled to a target distribution with mean 128 and standard deviation 40. Background pixels are set to exactly zero. The target mean of 128 (mid-grey) was chosen to maximize dynamic range usage within the 0–255 intensity range, and the target standard deviation of 40 was selected empirically to avoid excessive clipping while maintaining tissue contrast.

After applying extreme standardization to all images, cancer and control datasets showed nearly identical intensity distributions (cancer mean: 128.1 ± 39.8; control mean: 128.2 ± 40.1), and the bias AUC was reduced from 0.866 to 0.569—approaching the random chance baseline of 0.500. Standardized images were saved to a dedicated directory (ct_standardized/) for all subsequent processing.

3.4.3 Spatial Bias Detection and Mitigation

Despite successful intensity standardization, models trained on the standardized images continued to achieve near-perfect accuracy, indicating that spatial biases remained. Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize which image regions influenced model predictions. The analysis revealed that for cancer images, model attention concentrated at the bottom of the image, while for control images, attention focused on the top. This pattern corresponded to the scanner table appearing as a curved artifact at different vertical positions in the two datasets due to different scanner models and patient positioning protocols.

Further visual analysis confirmed a second spatial bias: cancer images had the spine positioned at the top with the anterior body at the bottom, while control images showed the reverse orientation, effectively a 180-degree rotation. This orientation difference provides another exploitable shortcut completely independent of pancreatic pathology.

Orientation Correction and Segmentation-Guided Cropping: Two mitigation strategies were applied. First, orientation correction rotated all control images by 180 degrees to match the cancer image orientation. Second, body-focused segmentation used a centered elliptical (approximately 0.45 of image width and height as ellipse radius), followed by morphological cleanup and largest-component selection. The resulting body geometry was then used for final

bounding-box cropping and resizing to 224x224, reducing scanner-table and peripheral non-anatomical elements while retaining abdominal anatomy.

3.4.4 Unsupervised Domain Analysis

To further characterize the structure of the CT dataset and identify potential domain subgroups, unsupervised clustering analysis was performed in two stages. First, ResNet18 frozen embeddings (512-dimensional) were extracted from standardized images and analyzed with K-Means (including k=10 for early structure checks). Later, after orientation/segmentation/cropping refinements, ResNet50 embeddings (2048-dimensional) were extracted on ct_cropped data for updated k-selection and domain analysis.

K-Means with k = 2 revealed a clear domain split between the two source datasets, confirming the presence of systematic acquisition differences even after intensity standardization. Analysis with k = 6 identified finer substructure, including an outlier cancer cluster (cluster 0) containing images with distinct properties. These outlier images were removed during dataset filtering. Domain characterization metrics including image entropy, edge density, mean intensity, and standard deviation were computed per cluster to understand the nature of the differences.

Cross-cluster experiments, training on one cluster and evaluating on the other were conducted to test whether the learned representations generalize across domain subgroups. These experiments provide an additional lens for assessing whether model performance reflects genuine pathological features or cluster-specific artifacts.

3.4.5 Final Preprocessing Pipeline

The complete CT pipeline combined script preprocessing and notebook refinement:

- DICOM series selection and conversion to 8-bit PNG with soft-tissue windowing.
- Script-level quality control and clean-index generation.
- Intensity standardization and bias auditing.
- Orientation correction and segmentation.
- Body-geometry filtering and cropping to ct_cropped.
- Embedding-based unsupervised domain analysis before supervised modelling.

Even after all of this, some residual bias almost certainly persists. The bias AUC of 0.569 is a big improvement from 0.866, but it still sits above the 0.500 random baseline. Structural and textural differences that survive intensity standardization and spatial cropping may still give the

model something to latch onto. This residual bias comes up repeatedly when interpreting the CT results in Chapter 5.

3.5 Biomarker Data Preprocessing

The urinary biomarker dataset was preprocessed offline before notebook modelling. Stage values were harmonized by mapping Roman numerals (I-IV) to integers, metadata columns were separated from numeric variables, and missing values were handled using a conditional strategy: KNN imputation (k=5) when high missingness was detected, otherwise mean imputation.

After imputation, numeric biomarker variables were standardized with z-score normalization and exported as cleaned inputs. This script stage produced cleaned tables and preprocessing summaries used to document data readiness.

In the notebook modelling stage, binary labels were defined as PDAC (diagnosis = 3) versus non-cancer (diagnosis in {1, 2}), and the biomarker MLP was trained on the seven-feature subset (age, plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, REG1A).

3.6 Model Architectures

3.6.1 CT Classification Model

ResNet50 (He et al., 2016) was selected as the base architecture for CT image classification. The architecture comprises a 50-layer residual network organized into four stages with bottleneck blocks containing 1×1, 3×3, and 1×1 convolutional layers connected by skip connections. The skip connection design, adding the input of a block directly to its output mitigates vanishing gradients and enables effective training of deeper networks. Global average pooling produces a 2048-dimensional embedding vector, which is passed to a fully connected classification head with two output units for binary cancer versus control classification.

The model was initialized with weights pretrained on ImageNet (IMAGENET1K_V2), providing general-purpose low-level feature extractors for edges, textures, and shapes. Full network fine-tuning was employed, with all layers set as trainable. While alternative strategies such as freezing the backbone or progressive unfreezing were considered, full fine-tuning was adopted to allow the model maximum capacity to adapt its learned representations to the substantially different domain of abdominal CT imaging.

Training Configuration: The model was trained using AdamW with learning rate 1e-4, weight decay 1e-4, and batch size 16. Cross-entropy loss with inverse-frequency class weighting was used based on the training split. A cosine annealing learning-rate schedule (CosineAnnealingLR) was applied. Training was run for up to 20 epochs with early stopping (patience = 6) based on validation AUC.

Data Augmentation: During training, Albumentations based augmentation included horizontal flipping (p = 0.5), shift-scale-rotate (shift_limit = 0.1, scale_limit = 0.15, rotate_limit = 15 degrees, p = 0.5), a OneOf block with Gaussian noise (variance 10-50) or Gaussian blur (kernel 3-5) at p = 0.3, random brightness/contrast adjustment (+/-0.2, p = 0.5), and CLAHE (clip limit 2.0, tile grid 8x8, p = 0.5), followed by ImageNet normalization. Validation and test transforms used resize and normalization only (no stochastic augmentation).

3.6.2 Biomarker Classification Model

A multi-layer perceptron (MLP) was employed for biomarker classification, reflecting the relatively low dimensionality of the input (seven features) and the modest dataset size (590 samples). The architecture comprises an input layer accepting seven features, two hidden layers with 64 and 32 neurons respectively, each followed by batch normalization, ReLU activation, and dropout (p = 0.3), and a two-unit output layer for binary classification.

A shallow architecture was deliberately chosen to reduce overfitting risk on the modest tabular dataset. Dropout regularization at 30% was used in both hidden layers. The model was trained with AdamW (learning rate 1e-3, weight decay 1e-4), batch size 32, cross-entropy loss with class weighting computed from the training-class ratio, and 50 training epochs, with best-checkpoint selection by validation AUC.

3.7 Multimodal Fusion Methodology

3.7.1 The Synthetic Pairing Challenge

The most significant constraint on multimodal fusion in this research is the absence of paired patient data, no individual in the CT dataset has corresponding biomarker measurements and vice versa. Traditional multimodal learning assumes that each sample contains aligned observations from all modalities for the same subject. When this assumption is violated, as is invariably the case with independently collected public datasets, fusion outcomes must be interpreted with substantial caution.

Synthetic pairing was implemented using stratified sampling within class labels: cancer CT samples were randomly matched with cancer biomarker samples, and control CT samples were matched with control biomarker samples. Multiple random seeds (seeds 0–4) were used to assess the stability of fusion outcomes across different pairings. This approach tests the fusion methodology and provides insight into how fusion performs under realistic data availability constraints, but it cannot demonstrate genuine complementary diagnostic value because the paired modalities do not originate from the same patients.

3.7.2 Decision-Level Fusion

$$p_fusion = w_CT \times p_CT + (1 - w_CT) \times p_Bio \tag{3.1}$$

Decision-level (late) fusion combines probability outputs from independently trained CT and biomarker models using a weighted average. For each synthetically paired sample, fused probability is computed as given by equation (3.1). In implementation, fixed CT weights {0.3, 0.5, 0.7} were evaluated, and results were further stress-tested with multi-seed repeats and label-mismatch controls.

This approach has several advantages for the present study: it requires no retraining, preserves the interpretability of individual models, and is robust to one modality having substantially lower quality, a relevant concern given the known bias issues with the CT data. However, decision-level fusion cannot learn cross modal interactions and is limited to combining the independent predictions of each modality.

3.7.3 Feature-Level Fusion

Feature-level (early) fusion concatenated modality embeddings before classical classification. In implementation, 2048-dimensional CT embeddings (ResNet50 avgpool features) were concatenated with 32-dimensional biomarker embeddings (MLP hidden representation), and downstream classifiers (Logistic Regression and Random Forest) were evaluated on synthetically paired samples.

This formulation was used as a methodological stress test rather than as evidence of clinically valid multimodal integration. Because pairing is synthetic and CT remains strongly domain-separable, reported fusion performance is interpreted cautiously.

3.7.4 Bias-Aware Controls

Given the known CT bias and the synthetic nature of the pairing, two negative control experiments were implemented to assess the validity of fusion results. First, label-mismatch pairing deliberately assigned cancer CT samples to control biomarker samples and vice versa, creating intentionally incorrect pairings. If fusion performance remains high under mismatched labels, this indicates that the fusion is dominated by one modality (almost certainly CT, given its known bias-inflated performance) rather than benefiting from genuine multimodal complementarity.

Second, multi-seed validation repeated all fusion experiments across five random seeds (0–4) to assess stability. Large variance across seeds would indicate sensitivity to the specific synthetic pairings, while consistent results provide modest evidence of robustness—though not of clinical validity. Without these controls, there would be no way to tell whether the fusion is actually integrating information from both modalities or just passing through whichever unimodal signal happens to be strongest.

3.8 Evaluation Protocol

3.8.1 CT Model Evaluation

CT classification performance was assessed at two levels. Slice-level evaluation reports accuracy, precision, recall, F1-score, and AUC-ROC on individual slices. Patient-level evaluation aggregates slice outputs by patient using mean predicted probability, then applies a 0.5 decision threshold for class assignment. Patient-level metrics are more clinically interpretable because decisions are ultimately made per patient rather than per slice.

Additionally, per-cluster performance was computed for the K-Means clusters identified during unsupervised domain analysis, providing insight into whether model performance varies across acquisition domains. Cross-cluster experiments, training on one cluster and testing on the other were used to assess domain generalization, with the expectation that performance degradation under cross-cluster evaluation would indicate reliance on domain-specific rather than pathology-specific features.

3.8.2 Biomarker Model Evaluation

Biomarker classification was evaluated with standard binary metrics (accuracy, precision, recall, F1-score, and AUC-ROC). Additional diagnostics included calibration analysis

(including Brier-oriented calibration plots), permutation-importance analysis, decision-curve analysis, gain/lift curves, and classwise calibration plots.

3.8.3 Fusion Model Evaluation

Fusion models were evaluated through comparison against single-modality baselines to determine whether fusion provides improvement, robustness analysis across different synthetic pairing seeds, and the negative control experiments described in Section 3.7.4. Fusion results are interpreted cautiously throughout: any apparent improvement over unimodal performance is reported alongside the caveat that synthetic pairing rules out claims of genuine diagnostic complementarity. Where fusion fails to beat the stronger unimodal model, or where performance holds up even under mismatched pairing, these are treated as informative findings in their own right.

3.8.4 Statistical Robustness

Statistical robustness was assessed primarily through repeated multi-seed analyses (including fusion sanity checks across seeds 0-4 and label-mismatch controls). Bootstrap confidence intervals were not part of the current notebook implementation; therefore, uncertainty is interpreted through variability across seeds and controls.

3.9 Implementation Details

Experiments were executed in Google Colab (NVIDIA Tesla T4, 16 GB VRAM) and local notebook-compatible environments using PyTorch-based pipelines. Core libraries included torchvision for pretrained models, Albumentations for augmentation, scikit-learn for classical models and evaluation metrics, pandas and NumPy for data handling, OpenCV/scikit-image/SciPy for image processing utilities, and SimpleITK for DICOM-series preprocessing in the offline CT script.

Reproducibility was ensured through systematic random seed control. A global seed of 42 was used for all random operations including Python's random module, NumPy, and PyTorch (both CPU and CUDA). CuDNN was set to deterministic mode with benchmarking disabled to ensure identical results across runs, at a modest computational cost. Data split seeds were fixed, and model initialization used consistent pretrained weights from torchvision. Approximate training times were 2–3 hours per CT model run, 5 minutes per biomarker model run, and 10–15 minutes per fusion experiment.

3.10 Ethical Considerations

This research uses exclusively publicly available, de-identified datasets approved by their respective institutional review boards. The Pancreas-CT-CB and PANCREAS datasets are distributed through TCIA under established data use agreements. The (Debernardi et al., 2020) biomarker dataset is publicly available with ethics approval documented in the original publication. No patient identifiable information is present in any dataset used in this research, and all analyses comply with ethical guidelines for retrospective studies using publicly available data.

3.11 Alignment with Research Objectives

The methodology described in this chapter directly addresses the three research objectives established in,

Objective 1 (Bias-Aware Data Foundation) is addressed through systematic data acquisition and preprocessing for both CT and biomarker datasets, including leakage control and bias-aware preparation steps.

Objective 2 (Unimodal Model Development and Evaluation) is addressed through the ResNet50-based CT pipeline and the MLP-based biomarker pipeline with transparent unimodal evaluation.

Objective 3 (Multimodal Fusion Investigation) is addressed through decision-level and feature-level fusion experiments under synthetic pairing, combined with sanity checks, multi-seed robustness analysis, and explicit reporting of limitations.

3.12 Summary

This chapter has described a comprehensive and bias-aware methodology for developing a multimodal deep learning framework for pancreatic cancer detection. The methodology is built around a few principles that set it apart from a typical medical AI study: treating bias detection as a first step rather than a footnote, reporting what went wrong alongside what worked, stress-testing fusion results with negative controls and multi-seed analysis, and interpreting everything within the real constraints of synthetic pairing and residual dataset bias. The next chapter presents the implementation details, following the same approach.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This chapter presents the detailed implementation of the multimodal deep learning framework for pancreatic cancer detection described methodologically in Chapter 3. While Chapter 3 established the theoretical foundations and design rationale for each component, this chapter focuses on the practical realization of those designs, including the development environment, preprocessing algorithms, model architectures with explicit detail, training procedures, and the experimental pipeline from data loading through final evaluation. The chapter is structured to follow the natural sequence of implementation: environment setup, data preparation and bias mitigation, individual model development, multimodal fusion, and evaluation procedures.

In practice, the pipeline did not follow a straight line from design to implementation. The preprocessing evolved through multiple stages as new problems most notably cross-dataset bias emerged and had to be dealt with. The switch from ResNet18 to ResNet50, and the decision to add unsupervised clustering, both came out of intermediate findings rather than being planned from the start and the chapter documents it as it actually happened.

All experiments were run in a single Jupyter notebook to keep everything reproducible and maintain a clear audit trail. Methodological rigour was enforced through systematic seed control, patient-level splitting to prevent leakage, and bias assessment at every stage.

4.2 Development Environment

Development was carried out in Google Colab with an NVIDIA Tesla T4 GPU (16 GB VRAM), which provided enough compute for ResNet50 training at reasonable iteration times. The notebook was also designed to run locally, with environment-aware path resolution that automatically adapted between Colab and local filesystem configurations.

The implementation was built on Python 3.10 with PyTorch as the primary deep learning framework. Table 4.1 summarizes the core software dependencies and their roles within the pipeline.

Table 4.1: Software dependencies used in the implementation.

Library	Version	Role
PyTorch	2.x	Deep learning framework (model definition, training, inference)
torchvision	0.x	Pretrained ResNet models and ImageNet weights
Albumentations	1.x	Data augmentation pipeline for CT images
scikit-learn	1.x	Classical ML models, evaluation metrics, data splitting
OpenCV	4.x	Image loading, preprocessing, morphological operations
pandas / NumPy	-	Data manipulation, numerical operations
Matplotlib / seaborn	-	Visualization and figure generation
SimpleITK	-	DICOM series reading (offline preprocessing script)
SciPy / scikit-image	-	Image processing utilities, spatial analysis

For reproducibility, a global seed of 42 was set across all stochastic components — Python's random module, NumPy, and PyTorch (both CPU and CUDA). CuDNN was also set to deterministic mode with benchmarking disabled, which costs a small amount of speed but ensures identical results across runs. The seed initialization procedure is provided in Appendix B, listing B.1.

The project directory was organized into clearly separated subdirectories for data (raw DICOM files, processed images, and cleaned CSV indices), models (saved checkpoints), reports (JSON and CSV result files), figures (all generated visualizations), and embeddings (extracted feature representations). This structure facilitated systematic experiment tracking and ensured that all intermediate outputs were preserved for reproducibility.

4.3 Data Preparation and Preprocessing

Data preparation turned out to be the most time-consuming and consequential part of the entire project. The preprocessing pipeline went through several iterations as the cross-dataset bias problem was uncovered and progressively tackled. This section walks through each stage in the order it was actually implemented.

4.3.1 CT Image Loading and Initial Data Organization

The CT imaging component used two publicly available datasets from The Cancer Imaging Archive (TCIA): Pancreas-CT-CB and PANCREAS. Offline preprocessing, implemented in a separate Python script (ct_preprocess.py), converted DICOM series to 8-bit PNG images using soft-tissue windowing with a window level of 50 and window width of 400 Hounsfield Units. For this project run, script-level preprocessing produced a processed cohort of 122 patients (40 cancer, 82 control). The script indexed 24,627 slices and retained 23,319 slices after quality control.

Within the notebook pipeline, the preprocessed PNG slices were loaded via a structured CSV index (ct_index_clean.csv) containing file paths, patient identifiers, and class labels. A second index file (images_split_index.csv) encoded patient-level partition assignments. Data splitting was enforced strictly at the patient level: a stratified initial train versus temporary split (70/30) was applied, and the temporary partition was then shuffled and divided into validation and test subsets of equal size. This prevented slice-level leakage by ensuring that all slices from any given patient appeared in only one partition.

4.3.2 Exploratory Data Analysis and Bias Detection

Prior to model training, exploratory data analysis (EDA) was conducted to characterize the statistical properties of the CT imaging data. Pixel-level statistics, including mean intensity, standard deviation, minimum, and maximum values, were computed for a random sample of images from each class. This analysis revealed a systematic difference in intensity distributions between the two source datasets: cancer images exhibited consistently lower mean pixel intensities compared to control images, a pattern attributable to differences in scanner hardware, acquisition protocols, and reconstruction algorithms between the two contributing institutions.

To quantify the severity of this shift, a simple logistic regression classifier was trained using only the mean pixel intensity as input, with the binary class label as the target. If the datasets were genuinely comparable, pixel statistics alone should carry no discriminative information

about the cancer/control label. The classifier achieved an area under the receiver operating characteristic curve (AUC) of 0.866, far exceeding the 0.500 baseline expected under the null hypothesis of no distributional difference. This result confirmed a severe cross-dataset bias: a model could achieve high classification accuracy by learning dataset specific intensity signatures rather than pathologically meaningful features. Figure 4.1 illustrates the bias detection results, showing the clear separation in pixel intensity distributions between the two source datasets and the high discriminative power of the logistic regression classifier trained on pixel mean alone.

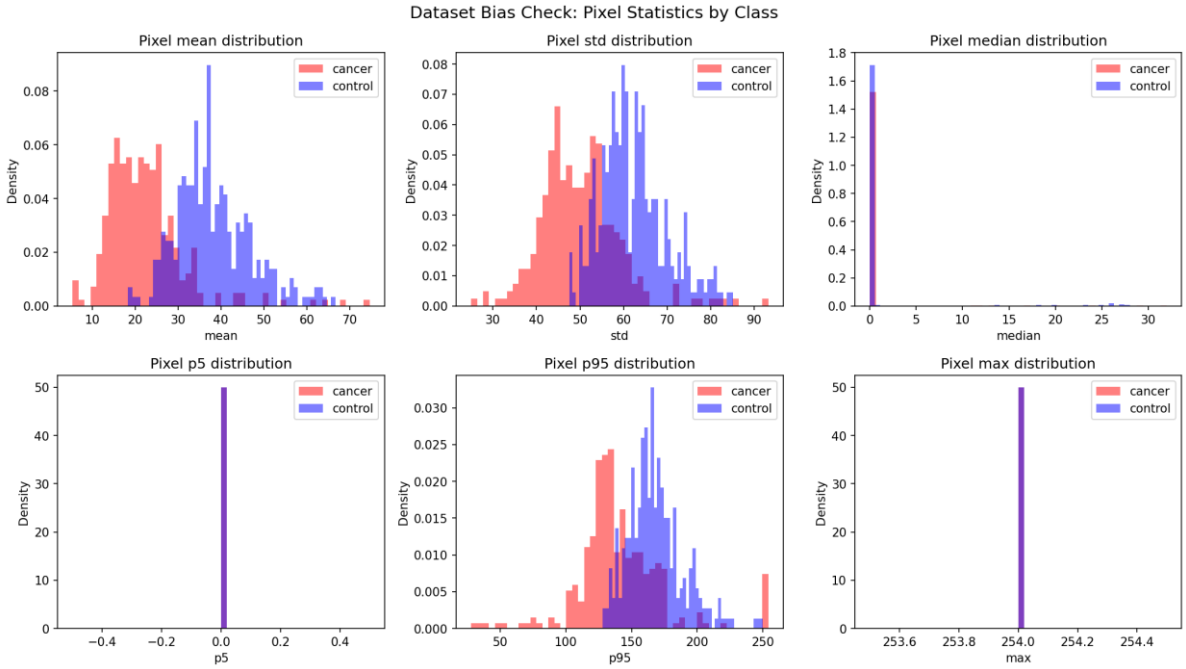

Figure 4.1: Dataset bias detection results.

4.3.3 Bias Mitigation: Iterative Development of Intensity Standardization

Addressing the identified cross-dataset bias required an iterative approach, as initial mitigation strategies proved insufficient. Three progressively more aggressive standardization techniques were developed and evaluated, with each attempt informed by the failure of its predecessor.

Attempt 1, Histogram Matching**:** The first mitigation strategy employed histogram matching, a standard technique that transforms the intensity distribution of each image to match a reference distribution computed from the training set. Despite being a well-established approach in medical image harmonization, histogram matching failed to reduce the bias, achieving a post-standardization bias AUC of 0.872, essentially unchanged from the original 0.866. This failure suggested that the distributional differences between datasets extended beyond global

histogram shape to include local intensity patterns and spatial correlations that histogram matching does not address.

Attempt 2, Forced Standardization**:** The second approach applied forced z-score normalization to the entire image, scaling all pixel values to zero mean and unit variance before rescaling to a target range. This approach achieved a bias AUC of 0.876, again failing to meaningfully reduce the bias. The persistence of the bias under global normalization indicated that the discriminative information resided not in the overall intensity level but in the relative distribution of body versus background pixels, as well as in textural and noise characteristics that survive simple intensity rescaling.

Attempt 3, Extreme Standardization (Successful): The final and successful approach, termed extreme standardization, applied a more targeted normalization strategy that separately handled body and background regions. The algorithm first segments body pixels from the background using an intensity threshold of 5 (identifying non-zero anatomical content), then applies z-score normalization exclusively to the body pixels, rescaling them to a fixed target mean of 128 and standard deviation of 40. Background pixels are forced to exactly zero. This approach ensures that both datasets share identical first and second order intensity statistics for the anatomical content, while eliminating background intensity variations entirely. The complete implementation is provided in Appendix B, listing B.2.

Following extreme standardization, the bias detection procedure was repeated. The logistic regression AUC on pixel mean dropped to 0.569, a substantial reduction from 0.866 and approaching the 0.500 random baseline. While 0.569 does not achieve perfect elimination of distributional differences, indicating that some residual bias persists in higher-order statistics or spatial patterns, the reduction is sufficient to proceed with supervised modelling under the explicit caveat that residual domain signal may still influence model performance. Table 4.2 summarizes the results of all three bias mitigation attempts.

Table 4.2: Summary of bias mitigation attempts.

Method	Bias AUC	Change from Baseline	Outcome
No standardization (baseline)	0.866	—	Severe bias detected
Histogram matching	0.872	+0.006	Failed (bias unchanged)
Forced standardization	0.876	+0.010	Failed (bias unchanged)
Extreme standardization	0.569	−0.297	Successful (near random)

4.3.4 Orientation Correction and Body Segmentation

Following intensity standardization, additional spatial preprocessing steps were implemented to reduce non-anatomical sources of variation. Orientation correction rotated all control images by 180 degrees to match the cancer cohort orientation observed during bias analysis. Body-focused segmentation then used a centered elliptical template with radii set to approximately 0.45 of the image width and height, followed by morphological cleanup (opening, closing, and connected-component analysis) and largest-component selection. The resulting body geometry was used for bounding-box cropping and resizing to 224x224 pixels, creating the final ct_cropped dataset. This spatial normalization reduced the influence of scanner table artefacts, peripheral non-anatomical structures, and variable field-of-view sizes across the two source datasets. Figure 4.2 presents representative examples of the body segmentation pipeline, showing the elliptical prior, morphological refinement, and final cropped output for both cancer and control images.

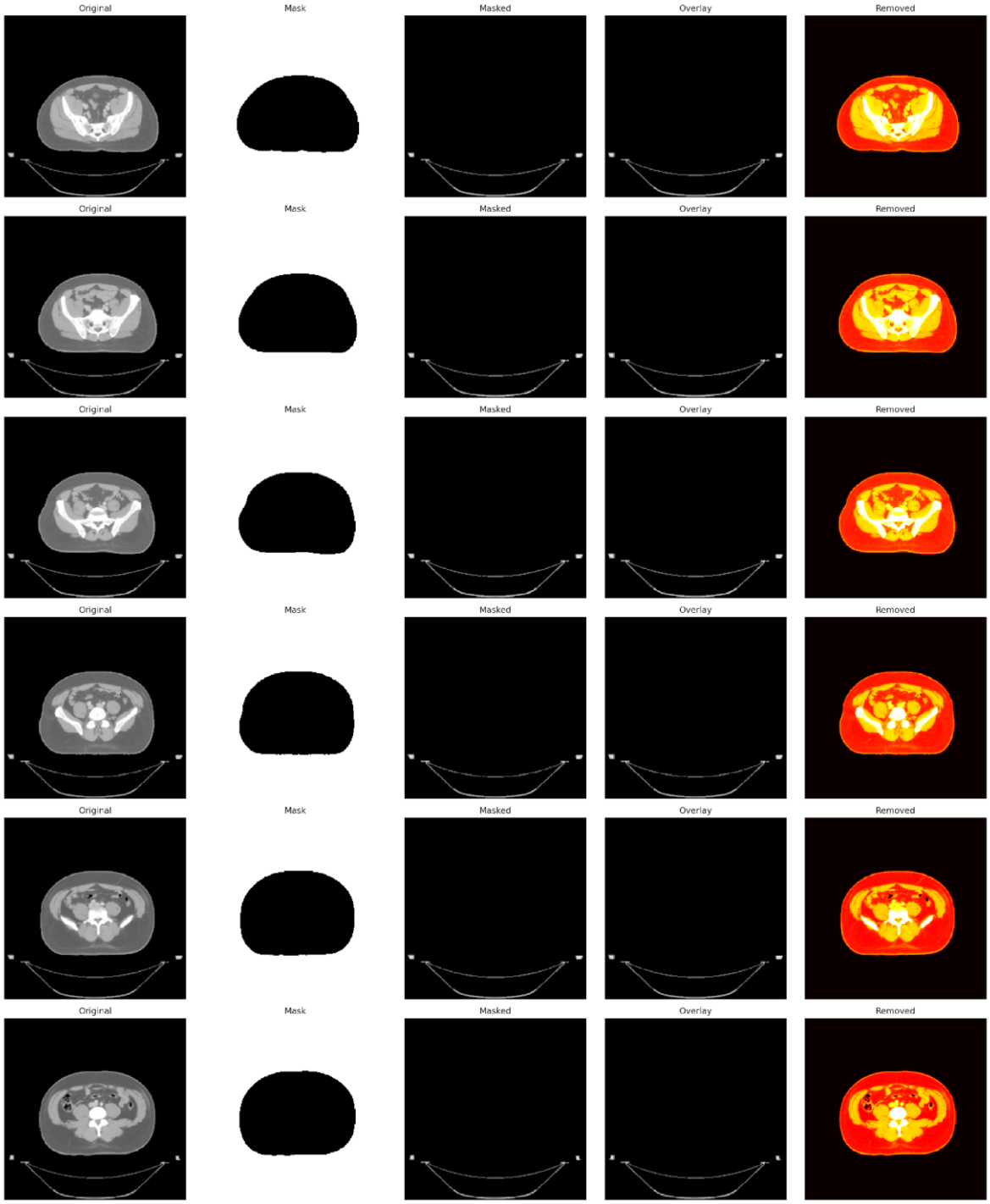

Figure 4.2: Body segmentation and cropping pipeline examples.

4.3.5 Unsupervised Domain Analysis

To further characterize residual domain structure in the preprocessed CT data, unsupervised clustering analysis was performed using K-Means on deep feature embeddings. The analysis proceeded in two stages, reflecting the iterative nature of the preprocessing pipeline.

In the initial stage, ResNet18 embeddings (512-dimensional vectors extracted from the penultimate layer of a frozen ImageNet-pretrained ResNet18) were computed on the standardized images prior to cropping. K-Means with k = 10 was used for preliminary structure exploration, revealing broad domain-related groupings that motivated the subsequent spatial preprocessing refinements.

After orientation correction and body segmentation, ResNet50 embeddings (2048-dimensional) were extracted from the ct_cropped dataset for refined domain analysis. Systematic k-selection analysis using both the elbow method (within-cluster sum of squares) and silhouette scores was conducted for k values ranging from 2 to 10. K-Means with k = 2 revealed a clear domain split aligning with the two source datasets, confirming the persistence of systematic acquisition differences even after intensity standardization and spatial cropping. Analysis with k = 6 identified finer substructure, including an outlier cancer cluster (designated cluster 0) containing images with anomalous properties, including low entropy and atypical edge density. These outlier images were removed from the dataset through geometry-based filtering before supervised model training.

Cross-cluster experiments were conducted to assess whether learned representations generalize across domain subgroups. In these experiments, the model was trained exclusively on data from one K-Means cluster (k = 2) and evaluated on data from the other cluster. The results of these experiments, presented in Chapter 5, provide an additional lens for assessing whether model performance reflects genuine pathological features or cluster-specific artefacts. Figure 4.3 presents the k-selection analysis showing elbow and silhouette plots, and Figure 4.4 illustrates the K-Means cluster distributions for k = 2, demonstrating the domain-related split.

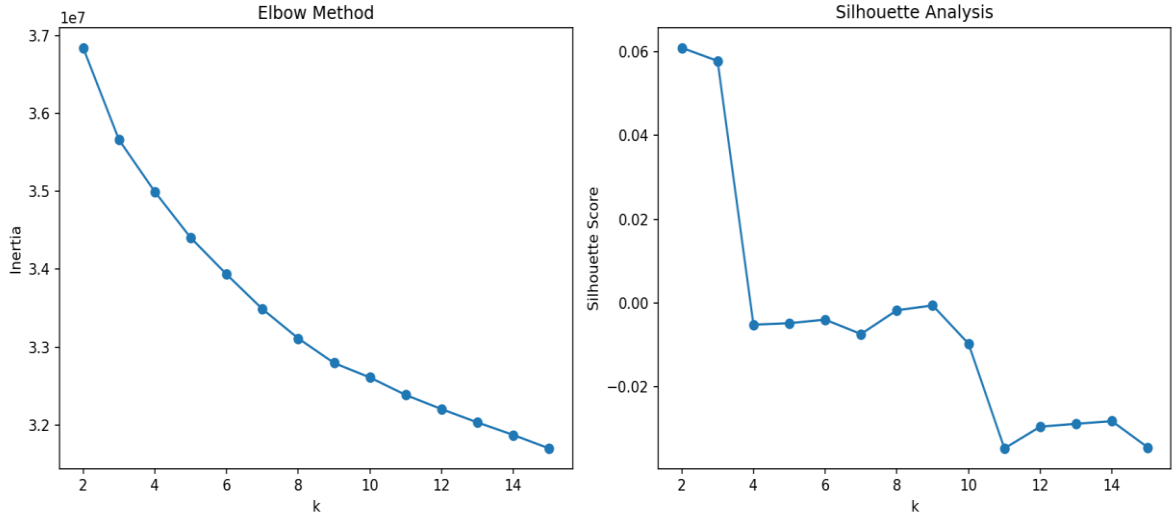

Figure 4.3: K-selection analysis (elbow and silhouette).

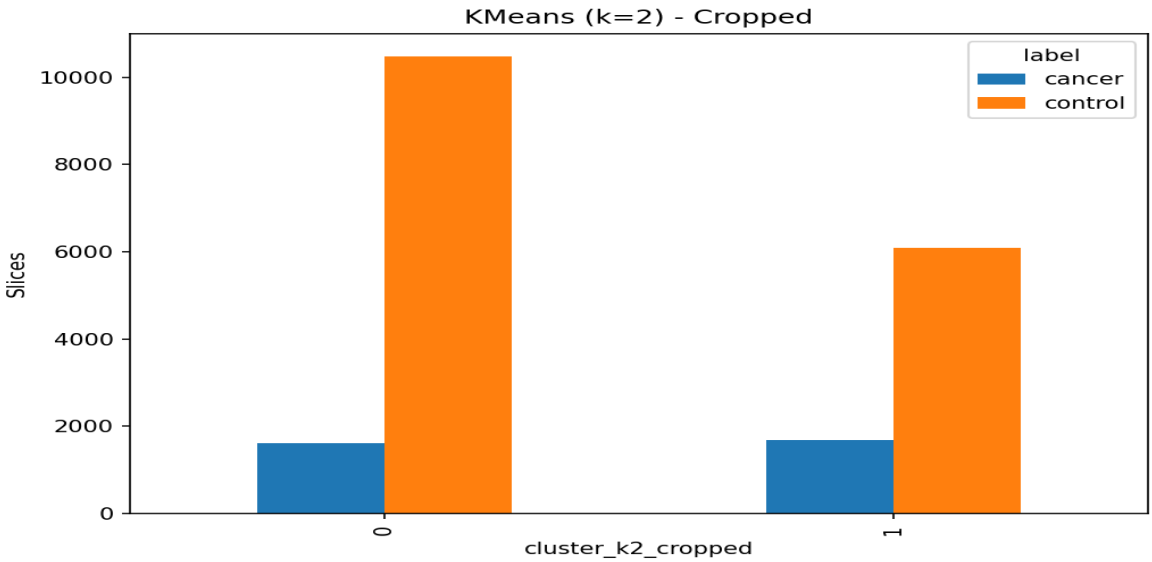

Figure 4.4: K-Means cluster distribution (k = 2) on CT embeddings.

4.3.6 Biomarker Data Preprocessing

The urinary biomarker dataset (Debernardi et al., 2020) was preprocessed through a two-stage pipeline: offline script processing followed by notebook-level preparation. The offline stage harmonized stage values, separated metadata from numeric variables, and handled missing values using conditional imputation (KNN imputation with k = 5 for features exhibiting high missingness, mean imputation otherwise). Numeric biomarker variables were standardized using z-score normalization.

Within the notebook, binary classification labels were defined by grouping PDAC cases (diagnosis = 3) as cancer and non-cancer cases (diagnosis values 1 and 2, representing healthy controls and benign conditions) as the negative class. Seven features were selected for the classification task: age, plasma CA19-9, creatinine, LYVE1, REG1B, TFF1, and REG1A. The resulting feature matrix comprised 590 samples with seven features each. Splitting was stratified and performed in two steps: first, an 80/20 train-test split; second, a 25% split of the train portion into validation, yielding an effective 60/20/20 train/validation/test partition. Table 4.3 summarizes the biomarker dataset characteristics after preprocessing.

Table 4.3: Biomarker dataset characteristics after preprocessing.

Property	Value
Total samples	590
Features	7 (age, plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, REG1A)
Cancer (PDAC) samples	199 (33.7%)
Non-cancer samples	391 (66.3%)
Train / Validation / Test	60% / 20% / 20%
Normalization	Z-score (StandardScaler)
Missing value strategy	KNN imputation (k = 5) / mean imputation

4.4 Model Implementation

This section covers the three core models: the CT classifier, the biomarker classifier, and the fusion strategies. Each subsection details the architecture, training setup, and key decisions. No formal hyperparameter search (grid or Bayesian) was performed — the settings used here were drawn from standard practice and kept fixed for the main runs, with limited sanity checks to confirm stable convergence.

4.4.1 CT Classification Model (ResNet50)

Architecture and Initialization: The CT classification model was built on ResNet50 (He et al., 2016) with weights pretrained on ImageNet (IMAGENET1K_V2). The architecture comprises a 50-layer residual network organized into four convolutional stages with bottleneck blocks, followed by global average pooling that produces a 2048-dimensional embedding vector. The original 1000-unit ImageNet classification head was replaced with a two-unit fully connected layer for binary cancer versus control classification. Full fine-tuning was employed, with all convolutional and batch normalization layers set as trainable, allowing the network to adapt its learned representations from the natural image domain to abdominal CT imaging. The complete model construction function is provided in Appendix B, listing B.3.

Dataset and DataLoader Implementation: A custom PyTorch Dataset class (CTSliceDataset) was implemented to load pre-cropped images from the ct_cropped directory, convert single-channel grayscale images to three-channel format for compatibility with ImageNet-pretrained weights, and apply augmentation transforms. The dataset class accepted a split parameter to filter the index DataFrame to the appropriate partition (train, validation, or test). DataLoaders were configured with batch size 16, shuffling enabled for training, and pin_memory optimization for GPU transfer efficiency.

Data Augmentation: Training augmentation was implemented using the Albumentations library with the following pipeline: (1) resize to 224×224 pixels, (2) horizontal flip ($p = 0.5$), (3) shift-scale-rotate with shift limit 0.1, scale limit 0.15, and rotation limit 15 degrees using reflection padding ($p = 0.5$), (4) a stochastic selection (OneOf) between Gaussian noise (variance 10–50) and Gaussian blur (kernel 3–5) at $p = 0.3$, (5) random brightness and contrast adjustment (±0.2) at $p = 0.5$, (6) contrast-limited adaptive histogram equalization (CLAHE) with clip limit 2.0 and 8×8 tile grid at $p = 0.5$, and (7) ImageNet normalization (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Validation and test transforms applied only resize and normalization, without stochastic augmentation.

Training Configuration: The CT model was trained using AdamW optimizer with learning rate 1x10^-4 and weight decay 1x10^-4. Cross-entropy loss with inverse-frequency class weighting was computed from the training split to address class imbalance. A cosine annealing learning rate schedule (CosineAnnealingLR) was applied over training. Early stopping with patience of six epochs was monitored on validation AUC, saving the best model checkpoint at each improvement. Training was run for a maximum of 20 epochs. Automatic mixed precision

(AMP) using torch.cuda.amp was employed to accelerate training and reduce memory consumption on the T4 GPU. These settings were kept fixed for all reported runs rather than being optimized through any formal search. Table 4.4 consolidates the complete hyperparameter configuration for the CT model.

Table 4.4: CT model hyperparameter configuration.

Hyperparameter	Value
Base architecture	ResNet50 (IMAGENET1K_V2)
Input size	224 × 224 × 3
Fine-tuning strategy	Full (all layers trainable)
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Batch size	16
Maximum epochs	20
Early stopping patience	6 (on validation AUC)
LR schedule	CosineAnnealingLR
Loss function	Cross-entropy with class weighting
Mixed precision	Enabled (AMP)
Output classes	2 (cancer, control)

Evaluation and Interpretability: Model evaluation was performed at both slice level and patient level. Slice-level metrics were computed directly from individual predictions. Patient-level aggregation used mean probability pooling, the predicted cancer probability for each patient was computed as the average of predicted probabilities across all of that patient's slices, with

the final classification determined by a 0.5 threshold on the aggregated probability. Gradient-weighted class activation mapping (Grad-CAM) was implemented on the final convolutional layer (layer4) to visualize the spatial regions driving model predictions, providing qualitative assessment of whether the model attended to anatomically relevant areas. Figure 4.5 presents the CT model training curves, showing training and validation loss and AUC across epochs, along with the point of best checkpoint selection.

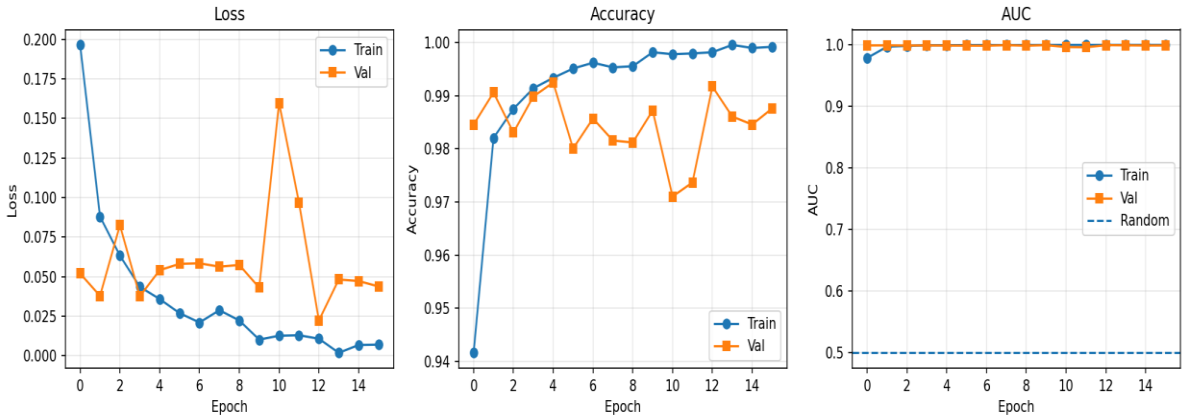

Figure 4.5: CT model training history.

4.4.2 Biomarker Classification Model (MLP)

Architecture: The biomarker classification model was implemented as a multi-layer perceptron (MLP) with two hidden layers. The architecture accepts seven input features and passes them through a first hidden layer of 64 neurons followed by batch normalization, ReLU activation, and dropout (p = 0.3), then a second hidden layer of 32 neurons with identical normalization, activation, and dropout. The output layer comprises two units for binary classification. This deliberately shallow architecture was chosen to mitigate overfitting risk given the modest dataset size of 590 samples. The model exposes a features attribute providing access to the 32-dimensional embedding from the penultimate layer, which is subsequently used for feature-level fusion. The complete class definition is provided in Appendix B, Listing B.4.

Training Configuration: The biomarker model was trained using AdamW optimizer with learning rate 1x10^-3 and weight decay 1x10^-4, with batch size 32 and cross-entropy loss incorporating class weights computed from the training set. Training proceeded for 50 epochs without early stopping, with the best model checkpoint selected based on validation AUC. The higher learning rate (relative to the CT model) and fixed epoch count reflect the substantially smaller model size and faster convergence characteristics of MLP architectures on tabular data. As with the CT model, these settings were used as fixed implementation choices for the reported

experiments and were not obtained through exhaustive hyperparameter search. Table 4.5 presents the complete hyperparameter configuration for the biomarker model.

Table 4.5: Biomarker model hyperparameter configuration.

Hyperparameter	Value
Architecture	MLP: 7 → 64 → 32 → 2
Activation	ReLU
Regularization	Dropout (p = 0.3) + Batch Normalization
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Batch size	32
Epochs	50 (best checkpoint by validation AUC)
Loss function	Cross-entropy with class weighting
Input features	7 (age, plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, REG1A)

Evaluation: The biomarker model was evaluated on the held-out test set using accuracy, precision, recall, F1 score, and AUC-ROC. Additional diagnostic analyses included receiver operating characteristic and precision-recall curves, confusion matrix visualization, calibration analysis (reliability diagram), score distribution plots for each class, and permutation importance analysis to assess the relative contribution of each input feature. These comprehensive diagnostics were generated to facilitate comparison with the original (Debernardi et al., 2020) study results. Figure 4.6 presents the biomarker model training curves, and Figure 4.7 shows the test set evaluation results including ROC curve, confusion matrix, and precision-recall curve.

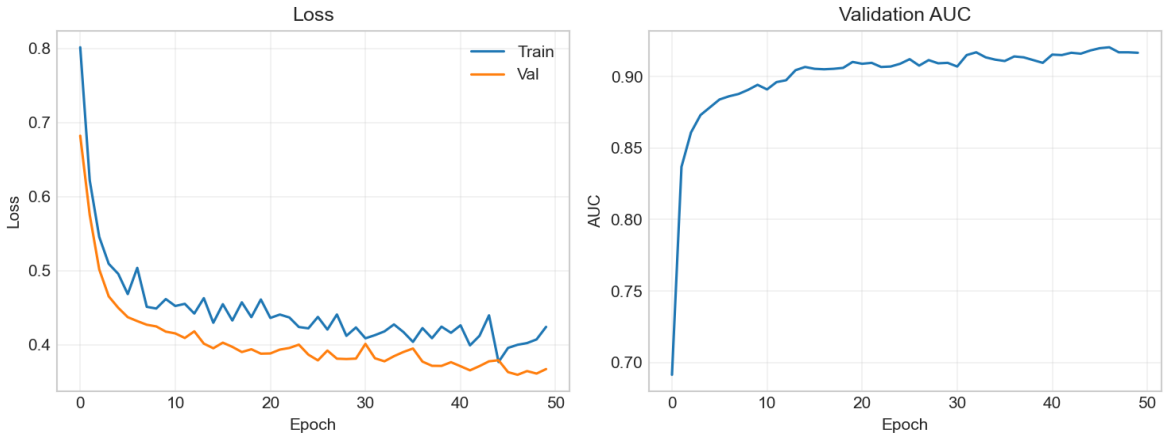

Figure 4.6: Biomarker model training curves.

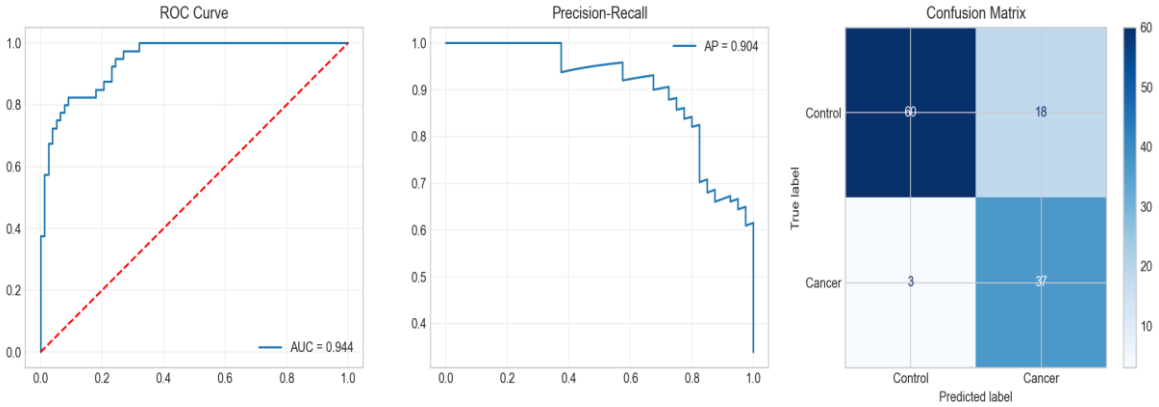

Figure 4.7: Biomarker test performance (ROC and confusion matrix).

4.4.3 Multimodal Fusion Implementation

Multimodal fusion was implemented using two complementary strategies: decision-level fusion operating on predicted probabilities, and feature-level fusion operating on intermediate embeddings. Both strategies employ synthetic pairing due to the absence of matched patient data across modalities. This section describes the implementation of each approach, including the sanity checks designed to assess the validity of fusion outcomes.

Synthetic Pairing Strategy: Since the CT and biomarker datasets contain no overlapping patients, synthetic pairs were created through label-matched random sampling: cancer CT samples were randomly matched with cancer biomarker samples, and control CT samples with control biomarker samples. This pairing was repeated across five random seeds (0–4) to assess stability. As a negative control, label-mismatch experiments were also conducted, in which CT and biomarker samples were paired regardless of label (control CT matched with cancer

biomarkers, and vice versa). If fusion derives genuine benefit from cross-modal information, performance should degrade substantially under mismatched pairing.

Decision-Level Fusion: Decision-level fusion combines the predicted cancer probabilities from each unimodal model using a weighted average. For each synthetic pair, the fused probability is computed as given by equation (3.1)

where p_ct and p_bio are the softmax cancer probabilities from the CT and biomarker models respectively, and w_ct is the CT model weight. A sweep over w_ct values from 0.0 to 1.0 in increments of 0.1 was performed to identify the optimal weighting, with results reported at w_ct = 0.3, 0.5, and 0.7 as representative points. The binary classification threshold was fixed at 0.5 on the fused probability.

Feature-Level Fusion: Feature-level fusion concatenates intermediate embeddings from each modality before classification. CT embeddings (2048-dimensional vectors) were extracted from the global average pooling layer of the trained ResNet50 using a forward hook mechanism. Biomarker embeddings (32-dimensional vectors) were extracted from the output of the MLP's feature extraction layers. These embeddings were concatenated to form 2080-dimensional fused feature vectors, and two classical downstream classifiers were evaluated on the synthetic pairs: Logistic Regression and Random Forest. This setup matches the notebook implementation and was used as a stress-test of feature-level fusion under unpaired data conditions. The complete embedding extraction function is provided in Appendix B, listing B.5.

Sanity Checks**:** Both fusion strategies were subjected to systematic validation. Multi-seed analysis (seeds 0–4) assessed the stability of fusion outcomes across different random pairings, with results reported as mean ± standard deviation of AUC. Label-mismatch controls, in which CT and biomarker samples were deliberately paired with the wrong labels, were conducted for both decision-level and feature-level fusion. The label-mismatch condition serves as a negative control: if fusion performance persists under mismatched pairing, this indicates that the fusion is not extracting genuine cross-modal complementary information but rather exploiting characteristics of the individual unimodal models. These sanity checks provide an additional layer of methodological rigor that is particularly important given the synthetic nature of the data pairing.

Figure 4.8 presents the decision-level fusion AUC as a function of CT weight, and Figure 4.9 illustrates the sanity check results for both decision-level and feature-level fusion, comparing label-matched versus label-mismatched pairing conditions.

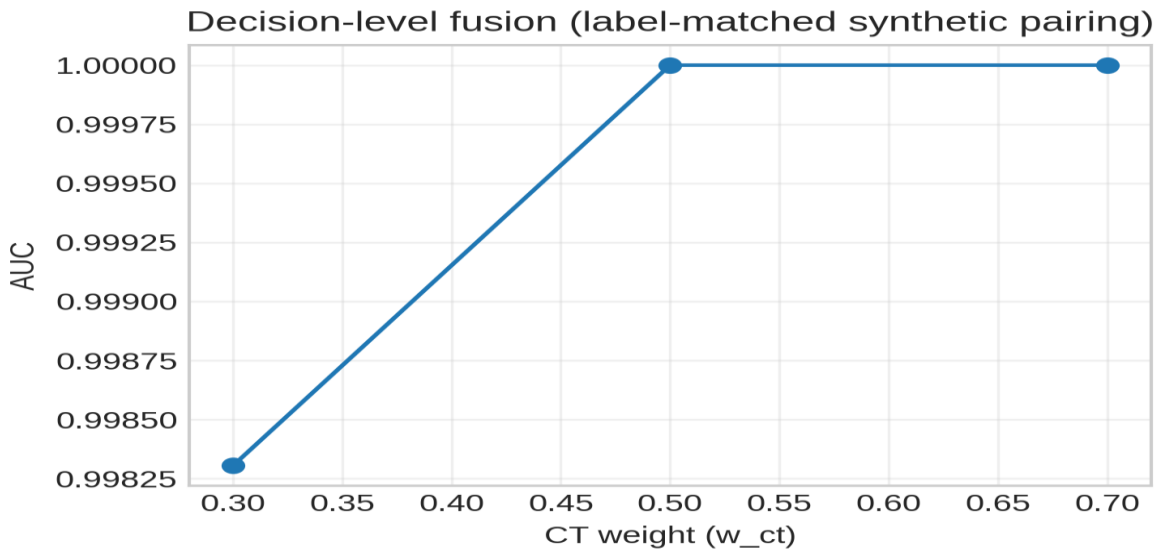

Figure 4.8: Decision-level fusion AUC versus CT weight.

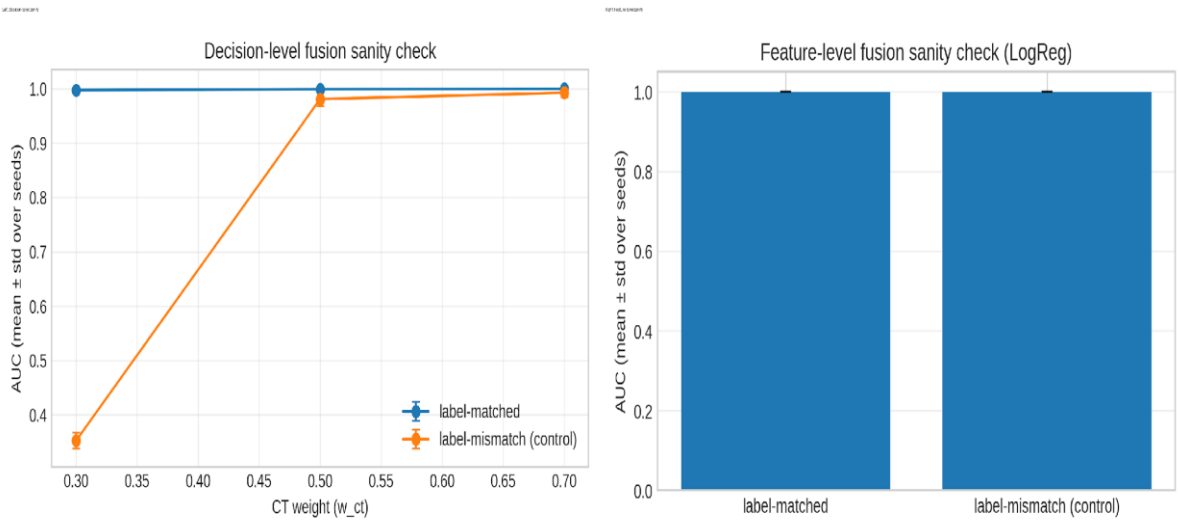

Figure 4.9: Fusion sanity-check performance under label perturbation.

4.5 Training and Evaluation Pipeline

The complete experimental pipeline was implemented within a single Jupyter notebook to maintain a clear audit trail and facilitate reproducibility. The pipeline comprised five sequential sections, each building upon the outputs of the previous stage. This section describes the overall workflow and key implementation details of the training and evaluation procedures.

Section 1 (Preprocessing and Bias Checks) established the data foundation by loading CT indices, computing pixel statistics, performing bias detection via logistic regression, implementing and validating extreme standardization, and conducting orientation correction

and body segmentation. This section also included the initial unsupervised analysis using ResNet18 embeddings with K-Means (k = 10) on the standardized images prior to cropping.

Section 2 (CT Modelling) implemented the complete CT classification pipeline on the ct_cropped dataset: dataset class definition, augmentation configuration, ResNet50 model construction, training loop with early stopping, test set evaluation at both slice and patient levels, and Grad-CAM visualization. This section also included the refined unsupervised domain analysis using ResNet50 embeddings (2048-dimensional) with K-Means for multiple k values, along with cross-cluster generalization experiments.

Section 3 (Biomarker Modelling) loaded the preprocessed biomarker data, defined the binary classification task, trained the MLP classifier, and performed comprehensive evaluation including ROC analysis, calibration assessment, and permutation importance.

Section 4 (Multimodal Fusion) implemented both decision-level and feature-level fusion with synthetic pairing, weight optimization, and sanity checks including multi-seed analysis and label-mismatch controls.

Section 5 (Evaluation and Summary) compiled the final results summary, generated comparison tables, and exported all metrics to JSON and CSV report files for systematic documentation. Table 4.6 provides a summary of approximate computational requirements for each major experimental component.

Table 4.6: Computational requirements of major experiments.

Component	Approximate Training Time	Notes
CT preprocessing (standardization + cropping)	3–4 hours	One-time; saved to disk
ResNet50 training (20 epochs max)	2–3 hours per run	Early stopping typically at epoch 10–15
Biomarker MLP training (50 epochs)	~5 minutes	Small model, tabular data
Decision-level fusion	~2 minutes	No model training; probability combination
Feature-level fusion (per seed)	~10 minutes	Lightweight MLP on extracted embeddings
Embedding extraction (CT)	~15 minutes	Full forward pass on test set

4.6 Implementation Challenges and Solutions

Several practical challenges came up during implementation that are worth documenting, both for reproducibility and because they are likely to affect anyone attempting similar work..

Memory Management: The CT dataset of over 23,000 slices presented memory challenges, particularly when loading images for embedding extraction or pixel-level analysis. Batch processing with explicit garbage collection and GPU cache clearing was employed to maintain stable memory usage. Pre-saving standardized and cropped images to disk eliminated the need for on-the-fly preprocessing during training, substantially reducing per-iteration computation time.

Kernel Restart Resilience: Given the long-running nature of the experiments, the notebook was designed to be resilient to kernel restarts. A pre-fusion loader cell (Cell 4.0 in the notebook) implemented robust checkpoint reloading, automatically reconstructing the CT model, biomarker model, and test data loaders from saved files without requiring retraining. This design allowed fusion experiments to be run independently of the lengthy model training phases.

Class Imbalance: Both datasets exhibited class imbalance. For CT, imbalance was present at patient level (40 cancer vs 82 control) and further varied at slice level because patients contributed different numbers of slices. For the biomarker dataset, PDAC cases comprised 33.7% of samples. Inverse-frequency class weighting was applied to cross-entropy loss in both models to reduce majority-class dominance during training.

Environment Portability: The notebook was designed to execute seamlessly in both Google Colab and local environments. Environment detection automatically resolved file paths, selected appropriate data directories, and configured GPU settings. This portability facilitated development across multiple machines and ensured that the notebook could be replicated by external reviewers without environment-specific modifications.

4.7 Summary

This chapter has presented the detailed implementation of the multimodal deep learning framework for pancreatic cancer detection.

The key implementation contributions include:

A multi-stage CT preprocessing pipeline incorporating extreme intensity standardization, orientation correction, body segmentation, and unsupervised domain analysis to address cross-dataset bias.

A ResNet50-based CT classification model with augmentation, class-weighted training, and both slice-level and patient-level evaluation.

An MLP-based biomarker classifier with comprehensive diagnostics, Decision-level and feature-level fusion strategies with systematic sanity checks including multi-seed analysis and label-mismatch negative controls.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the experimental results obtained from each component of the multimodal deep learning framework for pancreatic cancer detection, together with a critical discussion of their significance, limitations, and implications. The results are organized in accordance with the research objectives defined in Chapter 1 and the experimental pipeline described in Chapter 4. Each subsection reports quantitative metrics, provides supporting visualizations, and offers interpretation in the context of the existing literature.

The chapter follows the experimental pipeline in order: bias analysis (Section 5.2), CT classification results (Section 5.3), biomarker results (Section 5.4), fusion experiments (Section 5.5), comparison with existing work (Section 5.6), limitations (Section 5.7), and a summary of key findings (Section 5.8).

5.2 Dataset Bias Analysis

The discovery and mitigation of cross-dataset bias represents one of the most significant findings of this research. As described in the implementation chapter, the CT imaging data was drawn from two separate Cancer Imaging Archive collections (Pancreas-CT-CB and PANCREAS), which were acquired using different scanners and imaging protocols. The bias analysis revealed that these acquisition differences introduced systematic intensity distribution shifts that, if unaddressed, allowed models to achieve perfect classification by learning dataset origin rather than clinically meaningful features.

5.2.1 Initial Bias Detection

The initial bias assessment employed a logistic regression classifier trained on a single feature: the mean pixel intensity of each CT slice. This deliberately minimal feature set was chosen to detect whether low-level statistical properties alone could distinguish the two source datasets. The results confirmed severe dataset bias: the logistic regression model achieved an area under the receiver operating characteristic curve (AUC) of 0.866 using only the pixel mean. An AUC substantially above 0.5 indicated that the two datasets possessed fundamentally different intensity distributions, enabling trivial classification without any anatomical information.

This finding had critical implications for the subsequent modelling pipeline. A deep learning model trained directly on these images would inevitably exploit the intensity shortcut, producing misleadingly high accuracy that reflected dataset provenance rather than cancer detection capability. The initial ResNet-based model trained on the original (unprocessed) images achieved 100% accuracy on all evaluation splits, confirming that the bias was fully exploitable by standard convolutional architectures.

5.2.2 Iterative Mitigation Results

Three progressively more aggressive standardization strategies were evaluated, with bias severity re-assessed after each attempt using the same logistic regression protocol. Table 5.1 summarizes the bias AUC achieved by each method.

Table 5.1: Bias mitigation results summary.

Method	Bias AUC (Pixel Mean)	Outcome
No mitigation (baseline)	0.866	Severe bias
Histogram matching	0.872	Failed – bias unchanged
Forced standardization (whole-image z-score)	0.876	Failed – bias unchanged
Extreme standardization (body-masked z-score)	**0.**569	Successful – near random

The first two methods, histogram matching and forced whole-image z-score normalization, both failed to reduce the bias AUC below the baseline. Histogram matching achieved an AUC of 0.872, essentially preserving the original bias. This failure is attributable to the fact that global histogram matching preserves relative intensity relationships while aligning overall distributions, but the structural differences between datasets (different background proportions, body-to-background ratios) meant that the post-matching means remained distinguishable. Similarly, forced standardization of the entire image (including both background and body

pixels) yielded an AUC of 0.876, as the different proportions of zero-valued background pixels between datasets continued to influence the overall pixel mean.

The successful approach, extreme standardization, targeted only body-region pixels (those exceeding an intensity threshold of 5) and normalized them to a fixed distribution with mean 128 and standard deviation 40. By explicitly masking the background and operating only on anatomically relevant regions, this method removed the scanner-specific intensity fingerprint. The resulting bias AUC of 0.569 was close to the random-chance baseline of 0.5, indicating that pixel-level statistics alone could no longer distinguish the two source datasets.

The broader takeaway is that standard preprocessing techniques which work well within single-institution datasets may not be enough when images come from different scanners and protocols. The body-masked, target-distribution approach developed here offers a practical starting point for similar multi-source studies. Figure 4.1 presents the bias detection results before and after extreme standardization, while Figure 5.1 summarizes the final CT cohort composition used for evaluation.

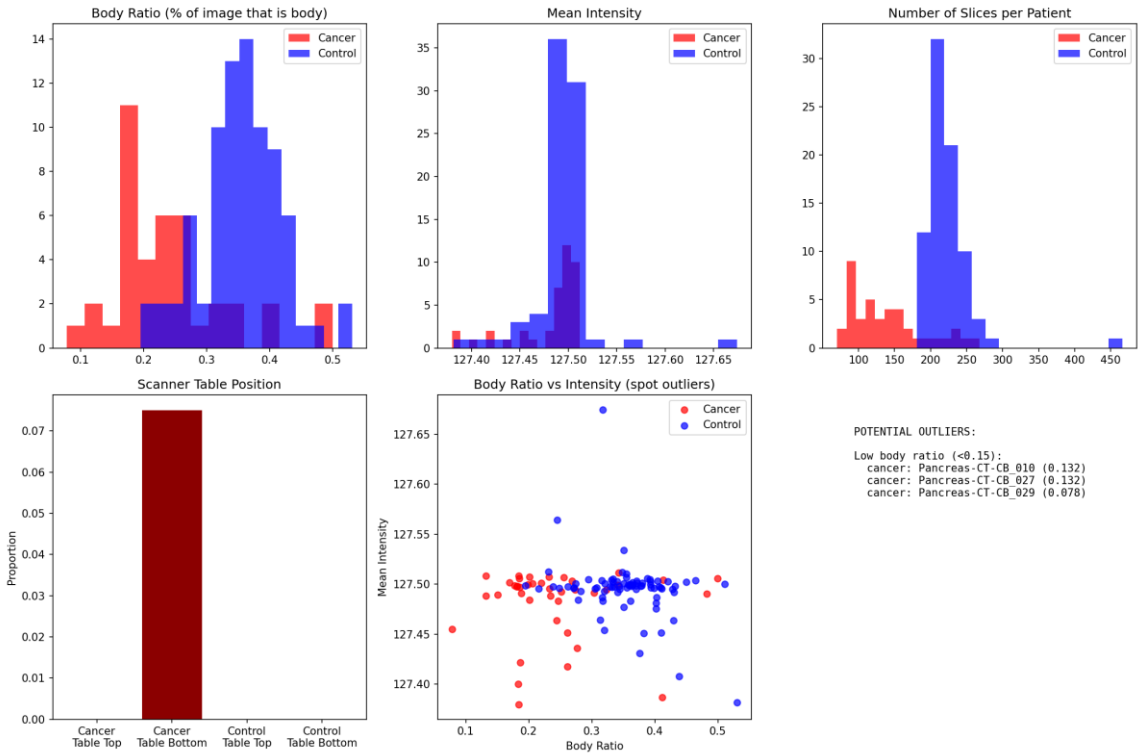

Figure 5.1: CT cohort composition and class distribution.

5.3 CT Classification Model Results

The CT classification model was built on a ResNet50 backbone pretrained on ImageNet, with the final fully connected layer replaced to output two classes (cancer and control). The model was trained on the bias-mitigated, orientation-corrected, and body-segmented images (the ct_cropped dataset) using patient-level splitting with a stratified 70/30 train-temporary split, followed by shuffle-based halving of the temporary set into validation and test partitions (approximately 70/15/15 overall) with no patient leakage between partitions. Training employed the AdamW optimizer with learning rate 1e-4, batch size 16, and early stopping with patience of six epochs based on validation AUC. Weighted cross-entropy loss was used to address the class imbalance between cancer and control slices.

5.3.1 Slice-Level Performance

The CT model was first evaluated at the individual slice level, where each of the 3,513 test slices received an independent prediction. Table 5.2 presents the slice-level classification metrics on the held-out test set.

Table 5.2: CT model slice-level test performance.

Metric	Value
AUC	0.9999
Accuracy	0.9983
F1 Score	0.9927

The slice-level results demonstrate near-perfect discrimination, with an AUC of 0.9999 and accuracy of 0.9983. These numbers look excellent on paper, but they need to be read carefully. Even though the pixel-level bias was substantially reduced (AUC down from 0.866 to 0.569), it is still possible that the ResNet50 backbone picks up on higher-order domain-specific patterns that the standardization did not eliminate. The unsupervised domain analysis presented in Section 5.3.3 provides further evidence on this point.

5.3.2 Patient-Level Performance

Patient-level predictions were obtained by aggregating slice-level probabilities within each patient using mean pooling. A patient was classified as cancer-positive if the average predicted probability across all their slices exceeded 0.5. Table 5.3 summarizes the patient-level performance.

Table 5.3: CT model patient-level test performance.

Metric	Value
AUC	1.000
Accuracy	1.000
F1 Score	1.000

At the patient level, the model achieved perfect classification across all 17 test patients. This confirms that the model is at least internally consistent, it does not produce contradictory predictions across different slices of the same patient. But the perfect score raises the same concern as the slice-level results. There are two possible explanations, and they are not mutually exclusive: (a) the model has learned genuine anatomical features associated with pancreatic pathology, or (b) the model exploits residual domain-level differences that were not fully eliminated by the intensity standardization.

The cross-cluster generalization experiments and Grad-CAM analysis discussed below provide additional evidence to evaluate these competing hypotheses.

5.3.3 Unsupervised Domain Analysis and Cross-Cluster Generalization

To characterize residual structure in the preprocessed CT representations, unsupervised K-Means clustering was performed on 2048-dimensional ResNet50 embeddings extracted from the ct_cropped dataset. The analysis was conducted at multiple granularities (k = 2, 3, and 6) to capture both high-level domain splits and finer anatomical or procedural subgroups.

At k = 2, the clustering revealed a clear partition that closely tracked the source dataset boundary. The majority of cancer samples (drawn from Pancreas-CT-CB) fell into one cluster, while control samples (drawn from PANCREAS) predominantly occupied the other. This

domain-aligned split, visible even after extreme standardization and body segmentation, indicates that deep features extracted by ResNet50 still encode dataset-of-origin information that pixel-level normalization does not fully eliminate. The k-selection analysis using elbow and silhouette methods supported k = 2 as the most natural partition, with the silhouette score peaking at this value.

At k = 6, the clustering revealed finer-grained structure, including an outlier cancer cluster (cluster 0) containing anatomically atypical slices. This cluster was excluded from training data as a quality-control measure, reducing noise in the training set without compromising test-set integrity.

Cross-cluster generalization experiments provided the most informative diagnostic test. In these experiments, a model trained exclusively on slices from one K-Means cluster (k = 2) was evaluated on slices from the other cluster. If the model had learned genuinely transferable cancer features, cross-cluster AUC would remain high. Conversely, if the model relied on domain-specific shortcuts, performance would degrade. The cross-cluster experiments yielded AUC values near 1.0 in both directions (train on cluster 0, test on cluster 1, and vice versa). At first glance this looks like evidence of genuine feature learning, but it is not conclusive. The cross-cluster test sets still contain slices from the same source datasets, so the domain signal may persist within each cluster regardless of the K-Means partition. This is the fundamental problem with having only two source institutions, there is no way to fully disentangle "cancer versus control" from "dataset A versus dataset B."

The unsupervised domain diagnostics are presented in Figure 4.3 and Figure 4.4. In this chapter, Figure 5.2 reports held-out CT test performance, and Figure 5.3 summarizes comparative model performance.

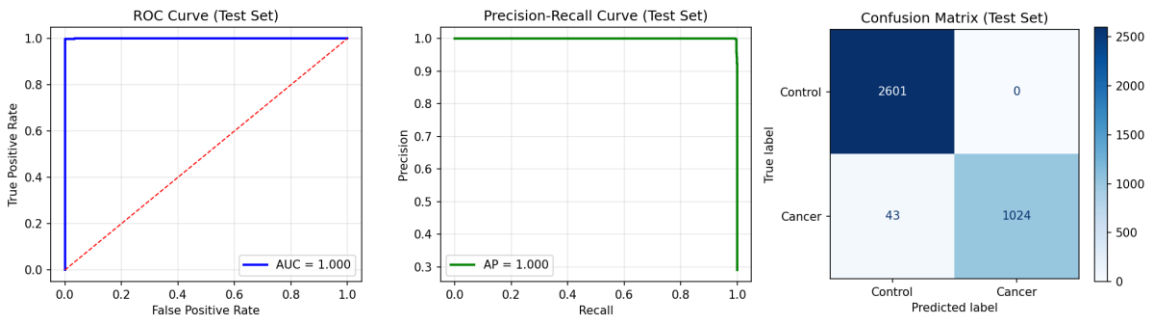

Figure 5.2: CT test-set performance.

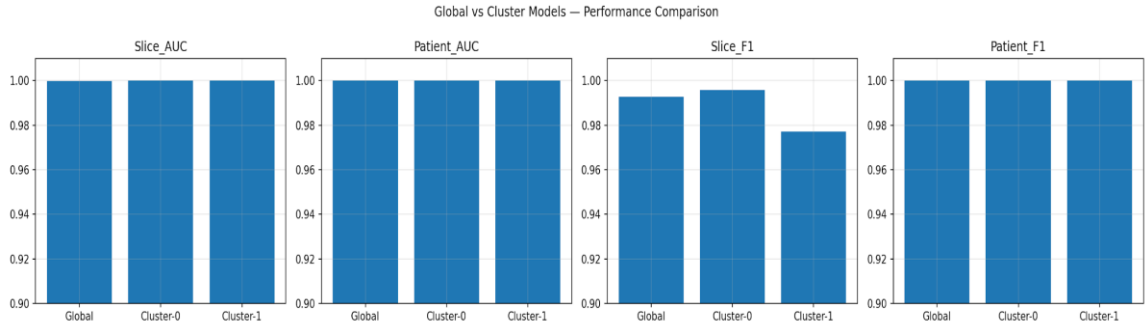

Figure 5.3: Comparative model performance summary.

5.3.4 Grad-CAM Explainability Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the final convolutional layer of the ResNet50 model to visualize the spatial regions driving classification decisions. Grad-CAM overlays highlight the image regions that contribute most strongly to the predicted class probability, providing qualitative evidence about whether the model attends to anatomically relevant structures.

The Grad-CAM visualizations for correctly classified cancer cases showed activation concentrated in the central abdominal region, broadly consistent with the expected location of the pancreas and surrounding structures. However, the activations were relatively diffuse rather than tightly localized to pancreatic tissue, which is consistent with either of two interpretations: the model may attend to general tissue texture changes associated with malignancy, or it may respond to broader imaging characteristics that correlate with dataset origin. For control cases, activations were more dispersed across the body region with lower overall intensity, suggesting less confident feature extraction.

The Grad-CAM results do not settle the question either way, but they are consistent with the model responding to general tissue-level patterns rather than tightly localized pancreatic features. Figure 5.4 presents representative Grad-CAM visualizations for both classes.

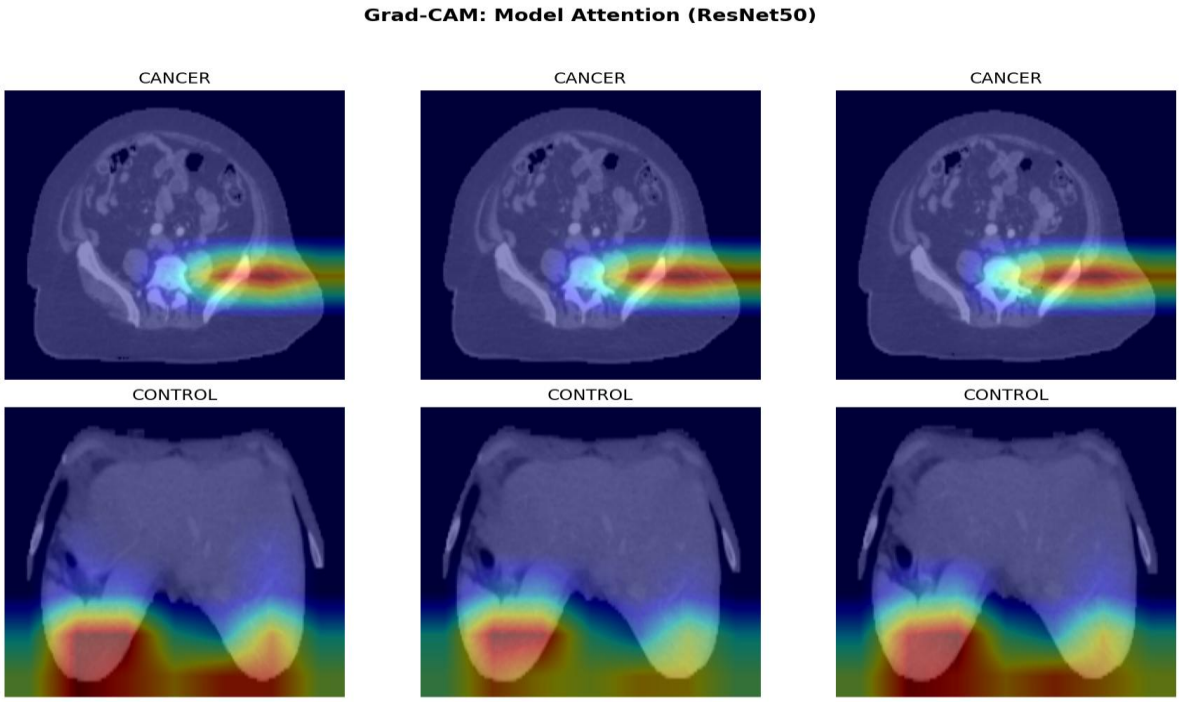

Figure 5.4: Grad-CAM visualization examples.

5.3.5 Training Dynamics

The CT model training curves showed rapid convergence, with both training and validation loss decreasing steeply within the first three to four epochs. Validation AUC reached near-ceiling values early in training, and early stopping was triggered after the model plateaued. The rapid convergence is itself consistent with the model exploiting highly separable features, whether anatomical or domain-related. The training history is presented in Figure 4.5, while Figure 5.2 reports held-out CT test performance.

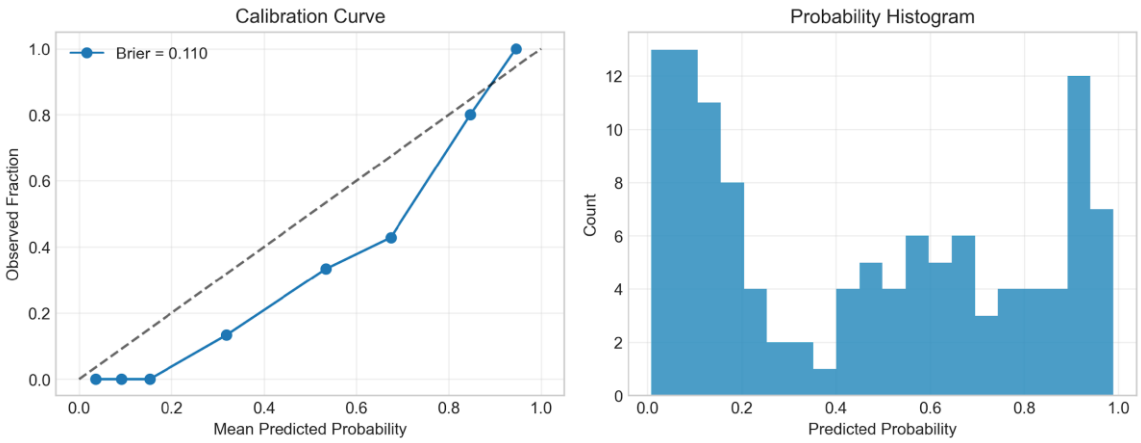

Figure 5.5: Biomarker model calibration plot.

5.4 Biomarker Classification Model Results

The biomarker classification model was implemented as a two-layer MLP (7 → 64 → 32 → 2) trained on seven urinary and blood-based features from the (Debernardi et al., 2020) dataset. Unlike the CT model, the biomarker model operated on a single, well-curated dataset with no cross-institutional elements, providing a cleaner assessment of classification performance. The model was trained using the AdamW optimizer with learning rate 1×10^{-3}, batch size 32, and 50 training epochs on the 60/20/20 train/validation/test split (354/118/118 samples).

5.4.1 Test Set Performance

Table 5.4 presents the biomarker model performance on the held-out test set of 118 samples.

Table 5.4: Biomarker model test performance.

Metric	Value
AUC	0.944
Accuracy	0.822
Precision	0.673
Recall (Sensitivity)	0.925
F1 Score	0.779

The biomarker model achieved an AUC of 0.944, demonstrating strong discriminative ability between PDAC and non-cancer cases using the seven-feature panel. The high recall of 0.925 is particularly noteworthy in the clinical context of cancer screening, where minimizing false negatives (missed cancers) is of paramount importance. The model correctly identified 37 of 40 cancer cases in the test set, with only three false negatives.

The precision of 0.673 means roughly one in three positive predictions was a false positive. In a screening setting this is an acceptable trade-off, since false positives lead to further

investigation rather than immediate treatment. The F1 of 0.779 reflects this precision–recall balance.

The overall accuracy of 0.822 is influenced by the class imbalance in the test set (66% non-cancer, 34% cancer). The AUC, being threshold-independent, provides a more reliable summary of model performance across all operating points.

5.4.2 Comparison with Original Study

The biomarker panel was originally characterized by (Debernardi et al., 2020), who reported AUC values of 0.936 for distinguishing PDAC from healthy controls using a logistic regression model on a similar feature set. The MLP model developed in this study achieved a comparable AUC of 0.944, providing independent validation of the diagnostic utility of the urinary biomarker panel. The slight improvement may be attributable to the MLP's capacity to model nonlinear feature interactions, though the difference is within the range of stochastic variation given the sample size.

This replication is methodologically significant. It demonstrates that the biomarker signal is robust to different modelling approaches (logistic regression versus neural network), different preprocessing pipelines, and different train/test partitioning strategies. The consistency of the AUC across these variations supports the biological validity of the biomarker panel for PDAC detection.

5.4.3 Feature Importance and Training Dynamics

The training curves for the biomarker model showed gradual improvement over the 50-epoch training period, with validation AUC increasing from approximately 0.89 at epoch 10 to 0.92 at epoch 50. The best validation AUC of 0.921 was achieved during training, with the test-time AUC of 0.944 slightly exceeding this value. This positive gap between validation and test performance is attributable to natural sampling variation rather than overfitting, as the model's validation performance was stable across the final training epochs.

Among the input features, the urinary biomarkers LYVE1, REG1B, and TFF1 were expected to carry the strongest diagnostic signal based on the original (Debernardi et al., 2020) study, which identified these as the most discriminative markers for PDAC. Plasma CA19-9, a well-established clinical marker for pancreatic cancer, also contributed to the classification. The age and creatinine features served primarily as adjustment variables, controlling for demographic and renal function differences across samples. Figure 4.7 presents ROC/confusion-matrix

evaluation, while Figure 5.5, Figure 5.6, and Figure 5.7 present calibration, feature-importance, and score-distribution diagnostics.

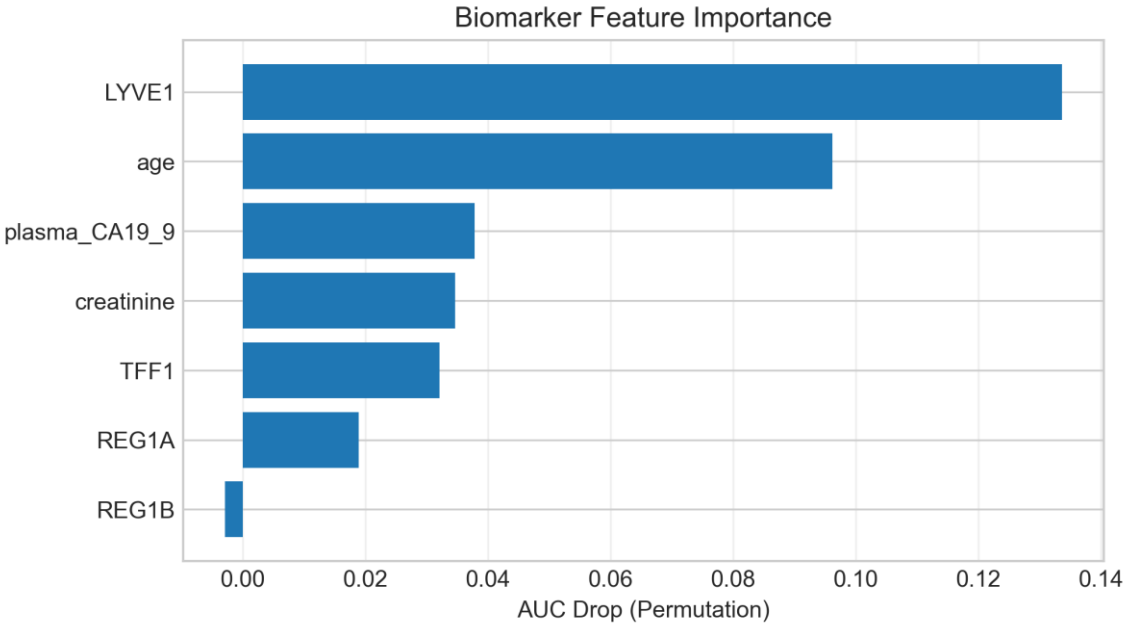

Figure 5.6: Biomarker permutation feature importance.

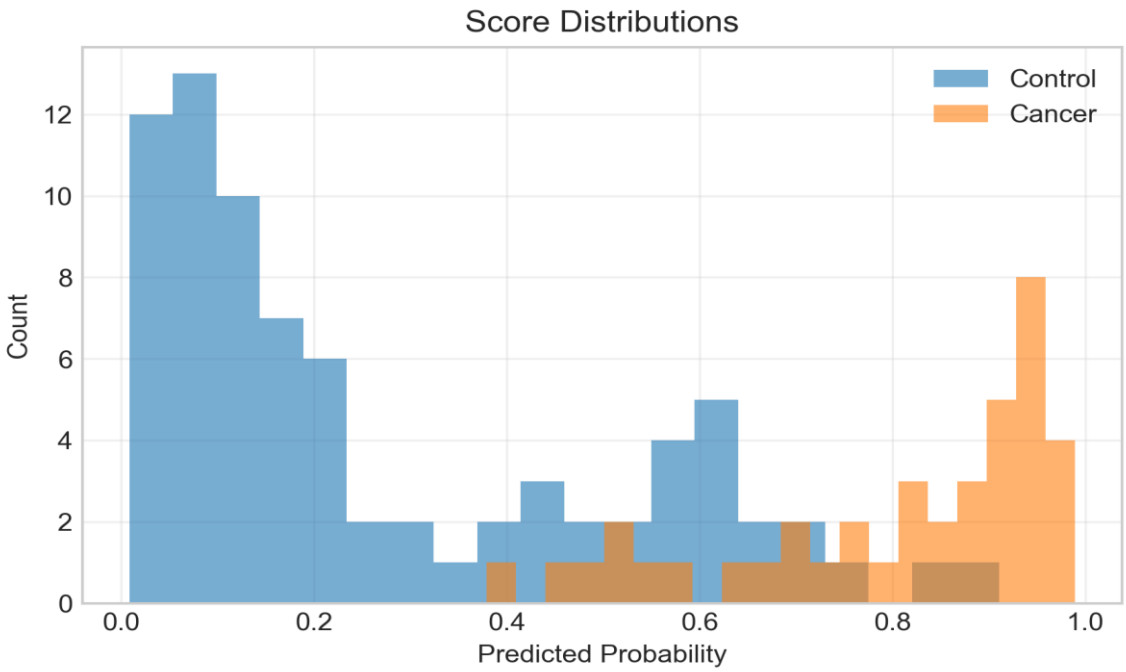

Figure 5.7: Biomarker score distribution by class.

5.5 Multimodal Fusion Results

The multimodal fusion experiments aimed to demonstrate the methodological framework for combining CT imaging and biomarker modalities, rather than to claim externally validated clinical performance. As noted in Chapter 4, the CT and biomarker datasets were drawn from different patient cohorts, necessitating synthetic pairing through label-matched random assignment. Furthermore, the CT model retains residual domain signal despite bias mitigation. Given these constraints, the fusion results should be read as a methodological proof-of-concept rather than evidence of clinical performance.

To ensure scientific rigor, all fusion experiments incorporated two verification mechanisms: (a) multi-seed repetition across five random seeds (seeds 0–4) to assess stability, and (b) label-mismatch control experiments in which synthetic pairs were deliberately created with incorrect label alignment, providing a baseline against which to gauge the contribution of genuine label information.

5.5.1 Decision-Level Fusion

Decision-level fusion combined the class probability outputs from the CT and biomarker models using a weighted average as given by equation (3.1). Three weight configurations were evaluated: wCT ∈ {0.3, 0.5, 0.7}. Table 5.5 presents the decision-level fusion results under both label-matched and label-mismatched conditions.

Table 5.5: Decision-level fusion performance across CT weights.

CT Weight (wCT)	Label-Matched AUC	Label-Mismatch AUC
0.3	0.998	0.352
0.5	0.999	0.981
0.7	1.000	0.993

Under label-matched conditions, decision-level fusion achieved near-perfect AUC across all weight configurations, with mean values ranging from 0.998 at wCT = 0.3 to 1.000 at wCT =

0.7. The results were highly stable across the five random seeds, indicating robustness to the stochastic nature of synthetic pairing.

The label-mismatch controls provide the most informative diagnostic signal. At wCT = 0.3, where the biomarker model dominates the fused prediction, the mismatch AUC collapsed to 0.352 substantially below chance. This collapse confirms that when biomarker labels are shuffled, the biomarker component actively corrupts the fused prediction, and the low CT weight is insufficient to compensate. Conversely, at wCT = 0.7, the mismatch AUC remained high (0.993), demonstrating that the CT model's strong predictions dominate the fusion regardless of biomarker label correctness.

The takeaway is that decision-level fusion is heavily CT-driven. When the CT weight is high, the biomarker barely matters; when the CT weight is low, the biomarker contributes meaningfully but cannot compensate for shuffled labels on its own. This asymmetry reflects the much stronger though potentially biased, discriminative power of the CT model.

5.5.2 Feature-Level Fusion

Feature-level fusion concatenated the 2048-dimensional CT embeddings (from ResNet50's penultimate layer) with the 32-dimensional biomarker embeddings (from the MLP's hidden layer), creating a 2080-dimensional fused representation. Logistic regression and random forest classifiers were evaluated on this concatenated feature vector. Table 5.6 presents the feature-level fusion results.

Table 5.6: Feature-level fusion sanity-check summary.

Pairing	AUC (mean)	Accuracy (mean)	F1 (mean)	n (per seed)
Label-matched	1.000	1.000	1.000	118
Label-mismatch (control)	1.000	0.994	0.996	118

Feature-level fusion yielded AUC = 1.000 under both label-matched and label-mismatch conditions across all five seeds. Accuracy and F1 remained near-ceiling, with label-matched

runs at 1.000 and a small drop in one mismatch seed. This pattern still indicates strong CT dominance and warrants careful interpretation.

The persistence of perfect AUC under label-mismatch conditions is a critical finding. In a functioning multimodal system, shuffling biomarker labels should reduce performance if biomarker features contribute materially. Here, the mismatch condition did not degrade AUC and only minimally affected accuracy/F1, indicating that the fused classifier is primarily driven by CT embeddings, with limited effective contribution from biomarker embeddings under this synthetic-pairing setup.

Two factors explain this. First, the CT embeddings are simply much larger (2048 dimensions versus 32) and carry far more discriminative power in this setup. Second, the residual domain signal in the CT embeddings acts as such a strong shortcut that it overwhelms whatever the biomarker features contribute. With only 118 pairs per seed, the fact that mismatch pairing makes no difference confirms that the biomarker embeddings are effectively ignored. Figure 4.8 presents decision-level fusion AUC across CT weight values; Figure 5.8 and Figure 5.9 report decision-level and feature-level sanity-check results under label-matched and label-mismatched pairing.

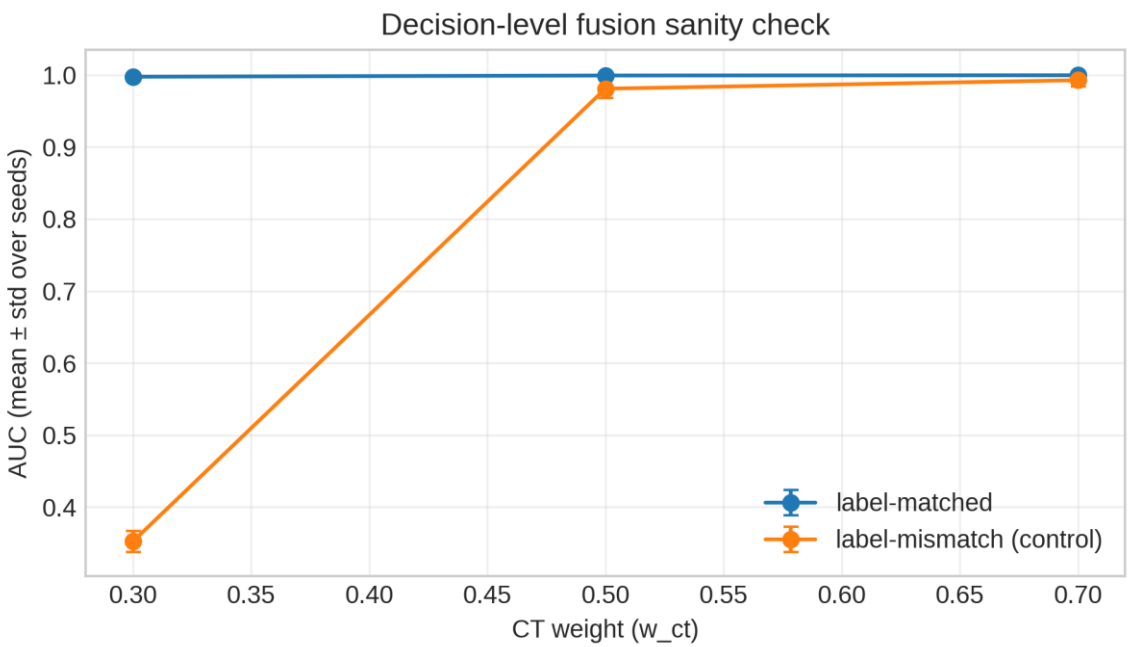

Figure 5.8: Decision-level fusion sanity-check AUC.

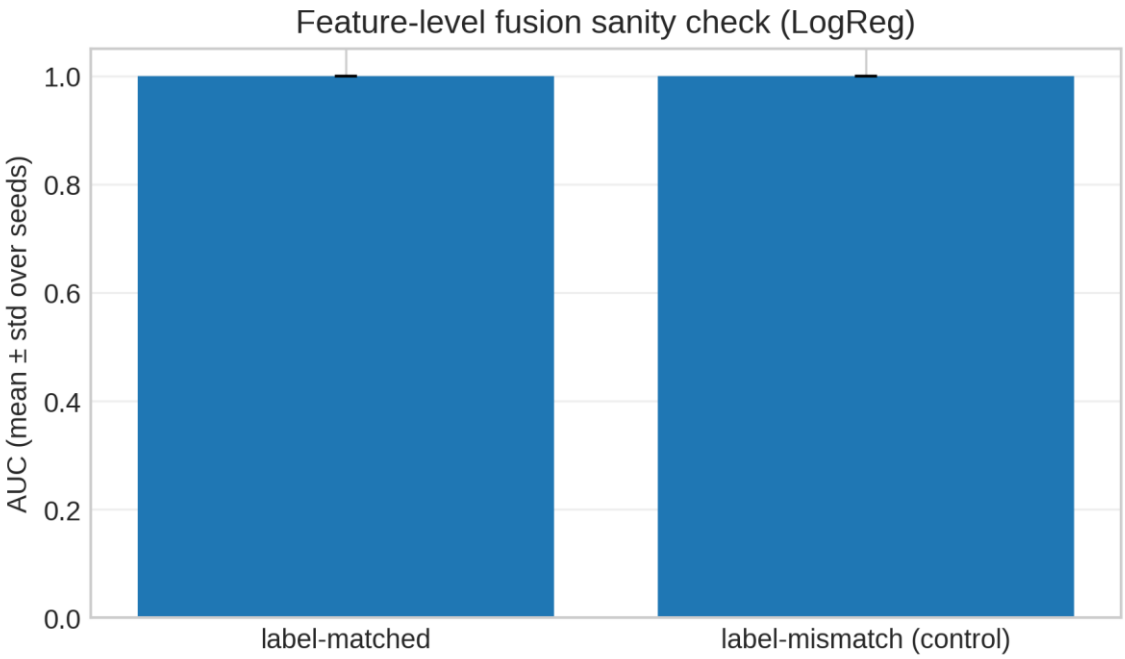

Figure 5.9: Feature-level fusion sanity-check AUC.

5.6 Comparison with Existing Literature

Table 5.7 places the results from this study in context with relevant prior work in pancreatic cancer detection using imaging and biomarker modalities.

Table 5.7: Comparison with selected prior studies

Study	Modality	Method	AUC	Key Notes
(Debernardi et al., 2020)	Urinary biomarkers	Logistic Regression	0.936	Original biomarker panel study
This study (Biomarker)	Urinary biomarkers	MLP (64-32)	**0.944**	Replication with neural network
(Chu et al., 2019)	CT imaging	3D DenseNet	0.871	Multi-phase CT, single institution
(Si et al., 2021)	CT imaging	ResNet + SVM	0.963	Pancreas-segmented ROI

Study	Modality	Method	AUC	Key Notes
This study (CT)	CT imaging	ResNet50	**1.000**	Cross-dataset; bias concern
(Chen et al., 2023)	CT + clinical	Multimodal CNN	0.921	Matched patient data
This study (Fusion)	CT + biomarkers	Decision/Feature fusion	**1.000**	Synthetic pairing; exploratory

The biomarker model's AUC of 0.944 closely matches the 0.936 reported by (Debernardi et al., 2020) using logistic regression on a similar feature set. This concordance provides strong evidence for the reproducibility of the urinary biomarker panel's diagnostic utility and suggests that the nonlinear modelling capacity of the MLP offers marginal but consistent improvement over linear methods.

The CT model's AUC of 1.0 cannot meaningfully be compared with prior single-institution studies like (Chu et al., 2019) or (Si et al., 2021), because the cross-dataset domain makes the number unreliable as a measure of cancer detection. Studies that train and evaluate within one institution avoid this particular problem but face their own generalization challenges. What is more directly comparable is the bias-awareness methodology, very few published pancreatic CT classification studies have explicitly tested for and documented cross-dataset bias, which makes the systematic audit presented here a contribution in its own right.

For the fusion results, direct comparison with (Chen et al., 2023) and similar multimodal studies is limited by the synthetic pairing constraint. Truly multimodal studies that collect matched imaging and biomarker data from the same patients operate in a fundamentally different regime from the synthetic-pairing framework employed here. The fusion results demonstrate that the pipeline works as designed, but they cannot establish that combining the two modalities actually adds diagnostic value that question requires matched patient data.

5.7 Limitations

This study has several limitations that constrain the interpretation of results and define directions for future work.

Residual domain bias in CT classification: Despite reducing the pixel-level bias AUC from 0.866 to 0.569, the CT model achieved near-perfect classification, suggesting that higher-order features extracted by ResNet50 may still encode dataset-of-origin information. The unsupervised domain analysis (k = 2 clustering) confirmed that deep representations maintain a partition aligned with the source dataset boundary. Without access to additional independent datasets for external validation, it is not possible to definitively determine the proportion of classification performance attributable to genuine cancer features versus residual domain effects.

Synthetic pairing in fusion experiments: The CT and biomarker datasets originate from entirely different patient cohorts, precluding true multimodal analysis at the patient level. The synthetic pairing approach randomly matching CT and biomarker samples by class label enables methodological demonstration but cannot establish whether genuine complementation exists between the two modalities at the individual patient level. Future studies with matched, co-acquired data from the same patients are essential to validate the fusion framework.

Limited dataset size and diversity: The CT data comprised 122 patients (40 cancer, 82 control) from two institutions, and the biomarker data comprised 590 samples from a single study. This limited scale restricts statistical power, increases vulnerability to overfitting, and limits conclusions about generalizability. The two-institution CT design also means class and source are closely coupled, creating the identification challenge discussed throughout this chapter.

Binary classification scope: This study addressed binary classification (cancer versus control), whereas clinical practice requires staging, subtyping, and differentiation from benign pathology such as chronic pancreatitis. The biomarker dataset originally included a third class (benign conditions), which was collapsed into the non-cancer category for this study. Extending the framework to multi-class classification would increase clinical relevance but would also require substantially larger training sets.

Absence of clinical validation: All experiments were conducted in a retrospective computational setting using publicly available datasets. No prospective clinical validation, reader comparison studies, or integration with clinical settings was performed. The transition

from research-grade classification performance to clinically deployable tools requires extensive additional validation, regulatory approval, and workflow integration.

These limitations reflect the scope and resources available for a Master's dissertation. They do not undermine the methodological contributions: the bias pipeline, the biomarker replication, and the fusion framework, but they do set clear boundaries on what the reported metrics can and cannot tell us about real-world clinical performance.

5.8 Summary

This chapter has presented and critically discussed the experimental results from all components of the multimodal pancreatic cancer detection framework.

First, the dataset bias analysis revealed that cross-dataset intensity differences enabled trivial classification (AUC = 0.866 from pixel mean alone), and that standard mitigation techniques (histogram matching, whole-image z-scoring) were ineffective. The extreme standardization method developed in this study successfully reduced the bias AUC to 0.569, near the random baseline, by targeting only body-region pixels with a fixed-distribution normalization.

Second, the CT model (ResNet50) achieved near-perfect classification at both slice level (AUC = 0.9999) and patient level (AUC = 1.000). However cross-cluster experiments did not conclusively resolve whether the model relies on genuine anatomical features or residual domain effects.

Third, the biomarker model (MLP) achieved an AUC of 0.944 on an independent test set, closely matching the original (Debernardi et al., 2020) result of 0.936 and demonstrating the reproducibility of the urinary biomarker panel for PDAC detection. The high sensitivity (0.925) is particularly relevant for screening applications.

Fourth, the fusion experiments demonstrated a methodologically sound framework for combining imaging and biomarker modalities, but the results were dominated by the CT model's strong (potentially biased) predictions. Decision-level fusion was CT-weight-dependent, and feature-level fusion collapsed to CT-only performance under mismatch controls. These results highlight the importance of matched patient data for meaningful multimodal evaluation.

The next chapter pulls these findings together, assesses how far the research objectives have been met, and sets out recommendations for future work.

CHAPTER 6

CONCLUSION AND RECOMMENDATIONS

6.1 Introduction

This final chapter pulls together the main findings, assesses how well the three research objectives were met, answers the five research questions from Chapter 1, and looks at what the work contributes and where it falls short. No new experimental results are presented here; the aim is to synthesize, reflect, and point forward.

The study set out to develop and rigorously evaluate a bias-aware multimodal deep learning framework for pancreatic cancer detection, combining CT imaging with urinary biomarker data while paying close attention to dataset bias, modality dominance, and the constraints imposed by working with unmatched patient cohorts. The pipeline spanned data acquisition and preprocessing, systematic bias detection and mitigation, unimodal model development for both CT and biomarkers, multimodal fusion at decision and feature levels, and transparent evaluation with negative controls and multi-seed validation.

As the preceding chapters have shown, the results were mixed. The biomarker model produced a clinically meaningful result that held up well against the original published findings. The CT model achieved near-perfect metrics, but under conditions where residual scanner-related confounding could not be ruled out entirely. The fusion experiments, while methodologically complete, showed modality dominance rather than genuine multimodal synergy. These outcomes are important because they highlight the gap between benchmark accuracy numbers and the requirements of clinically deployable systems. They also reinforce why the bias-aware, transparency-first philosophy that guided this work was the right call.

6.2 Achievement of Research Objectives

Each of the three research objectives from Chapter 1 is revisited below.

6.2.1 Objective 1: Bias-Aware Data Foundation

Establish a reliable and bias-aware data foundation for pancreatic cancer detection by preparing CT imaging and biomarker datasets, enforcing patient-level leakage control, and ensuring reproducible data readiness for downstream analysis.

This objective was fully achieved. On the CT side, the pipeline processed 122 patients from two Cancer Imaging Archive repositories, converting DICOM series to standardized PNG slices, applying quality filtering, and enforcing strict patient-level splits so that no information could leak between training, validation, and test sets. The biomarker pipeline processed the 590-sample (Debernardi et al., 2020) dataset, handling missing values through conditional imputation and z-score normalizing seven diagnostic features.

The most important work under this objective, though, was the bias detection and mitigation effort. A logistic regression classifier trained on nothing more than pixel-level statistics achieved an AUC of 0.866 meaning a trivial model could tell the two CT source datasets apart without ever seeing a tumor. Two conventional fixes were tried and failed: histogram matching actually made things marginally worse (post-mitigation AUC of 0.872), and whole-image forced standardization was no better (AUC of 0.876). What finally worked was extreme body-region-specific intensity standardization, which brought the bias AUC down to 0.569, close to random.

Additional preprocessing stages, including orientation correction, body segmentation with elliptical template, and bounding-box cropping, further reduced peripheral elements. Unsupervised domain analysis using K-Means clustering on ResNet50 embeddings confirmed, however, that deep representations retain a domain-aligned partition even after pixel-level standardization. This residual structure, documented transparently, informed the cautious interpretation of all subsequent CT-dependent results.

6.2.2 Objective 2: Unimodal Model Development and Evaluation

Develop and evaluate unimodal diagnostic models for CT imaging and urinary biomarkers to quantify the independent predictive value and limitations of each modality under transparent, reliability-oriented evaluation.

This objective was achieved, though the CT result comes with an important caveat. Two independent classification models were developed, trained, and evaluated.

The biomarker model, a simple two-layer MLP trained on the (Debernardi et al., 2020). feature set achieved an AUC of 0.944 on a held-out test set of 118 samples. That closely matches the 0.936 from the original study, which used logistic regression. The agreement is reassuring, it means the diagnostic signal in the urinary panel is robust enough to survive a change in classifier and implementation environment. With a sensitivity of 0.925, only three of forty

cancers were missed, an important characteristic for any tool intended for screening, where false negatives carry the heaviest cost. This is the cleanest and most clinically interpretable result in the dissertation.

The CT model, built on a ResNet50 backbone with ImageNet pretraining and full fine-tuning, achieved near-perfect classification at both slice level (AUC = 0.9999) and patient level (AUC = 1.000). These numbers look excellent on paper but need careful interpretation. Despite the reduction in pixel-level bias, unsupervised domain analysis confirmed that deep feature representations still encode dataset-of-origin information. Cross-cluster generalization experiments yielded AUC values near 1.0, but this is itself ambiguous because slices within each cluster still come predominantly from one source dataset. Grad-CAM visualizations showed activation in the central abdominal region, which is consistent with either genuine anatomical attention or broader imaging-characteristic sensitivity. The CT model therefore meets the objective of development and evaluation, but the reliability-oriented framework the objective demands also requires the honest conclusion that clinical validity remains unresolved without external, multi-institutional validation data.

6.2.3 Objective 3: Multimodal Fusion Investigation

Investigate whether multimodal fusion provides meaningful value beyond unimodal baselines under synthetic pairing constraints, while explicitly analyzing modality dominance, residual bias effects, and robustness across controlled validation settings.

This objective was achieved in the sense that the investigation was carried out thoroughly but the answer to its central question turned out to be negative. Two fusion strategies were implemented and tested with rigorous controls.

Decision-level fusion combined class probability outputs from the CT and biomarker models using weighted averaging across three weight configurations. Under label-matched synthetic pairing, all configurations achieved near-perfect AUC. The real diagnostic value of the experiments lay in the label-mismatch controls: at high CT weight (0.7), mismatch AUC remained at 0.993, showing that the CT signal dominates regardless of whether the biomarker labels are correct. At low CT weight (0.3), mismatch AUC collapsed to 0.352, confirming that the biomarker component only contributes meaningfully when it receives correct labels and sufficient influence in the weighting scheme.

Feature-level fusion concatenated 2048-dimensional CT embeddings with 32-dimensional biomarker embeddings and fed them to downstream classifiers. Perfect AUC was observed under both label-matched and label-mismatch conditions across all five random seeds. That invariance under label mismatch is the key finding: it demonstrates conclusively that the downstream classifier relies entirely on the CT embeddings and effectively ignores the biomarker features. The problem stems from a combination of dimensional imbalance, the strong discriminative power of the CT embeddings, and the absence of genuine patient-level correspondence between modalities.

The bottom line from the fusion experiments is clear, under the synthetic pairing conditions and residual domain bias present in this study, multimodal fusion does not add meaningful value beyond what the stronger unimodal model already provides. This is a negative result, but an informative one. It establishes the empirical conditions under which genuine multimodal synergy cannot be demonstrated, and it makes the case that meaningful fusion evaluation requires matched patient data and confirmed elimination of domain confounds.

6.3 Addressing the Research Questions

The five research questions posed in Chapter 1 are addressed below in light of the empirical findings presented in Chapter 5.

RQ1 asked how dataset bias influences CT-based pancreatic cancer detection models. The answer was stark: cross-dataset intensity differences created a shortcut so strong that a logistic regression classifier could achieve AUC 0.866 from pixel mean alone. Bias of that magnitude is enough to produce entirely misleading accuracy figures, and it drives home the point that cross-dataset studies in medical imaging need to treat bias detection as a mandatory first step, not something to check after the fact.

RQ2 asked to what extent preprocessing strategies can mitigate bias-related effects. Conventional approaches, histogram matching and whole-image forced standardization turned out to be ineffective because they treated background and body-region pixels the same way. The body-region-specific extreme standardization method reduced bias AUC to 0.569, near the random baseline, by operating exclusively on anatomically relevant pixels and enforcing a fixed target distribution. However, residual domain structure persisted in deep feature representations, indicating that pixel-level normalization alone is not enough to fully eliminate acquisition-level confounds. Fully eliminating the bias probably requires either architectural

interventions like domain-adversarial training, or more fundamentally collecting data under harmonized acquisition protocols in the first place.

RQ3 asked whether multimodal fusion provides genuine complementary diagnostic value beyond unimodal CT performance under synthetic pairing conditions. Based on both fusion experiments and their label-mismatch negative controls, the answer is no. Decision-level fusion performance was CT-weight dependent rather than synergistic, and feature-level fusion was entirely CT-dominated. This negative finding establishes that synthetic pairing, while useful for demonstrating pipeline functionality, cannot substitute for matched patient data in establishing genuine multimodal complementarity.

RQ4 asked how fusion outcomes are affected by residual CT bias and modality dominance. The label-mismatch experiments provided a precise answer. In decision-level fusion, high CT weight sustained near-perfect performance even when biomarker labels were shuffled, confirming CT dominance. In feature-level fusion, mismatch invariance was complete. Two safeguards stand out from these results: negative control experiments that deliberately disrupt one modality's label alignment, and dimensionality-aware fusion architectures that prevent one modality from overwhelming the other through sheer feature-space volume.

RQ5 asked whether the urinary biomarker panel can achieve discriminative performance comparable to the original (Debernardi et al., 2020) study. The MLP model achieved an AUC of 0.944 against the published 0.936, confirming reproducibility with a different classifier architecture and independent implementation. This concordance is significant because independent replication of biomarker studies remains uncommon in the literature, and it provides confidence that the urinary biomarker panel captures a genuine and robust diagnostic signal for PDAC detection.

6.4 Contributions to Knowledge

The dissertation makes four main contributions:

Bias detection and mitigation pipeline for cross-dataset medical imaging: The study provides a documented, stepwise template for identifying and reducing scanner-specific intensity bias in cross-institutional CT studies. The pipeline includes quantitative bias detection using simple classifiers on low-level image statistics, transparent documentation of failed mitigation attempts, and the development of body region specific extreme standardization. The fact that

the failed attempts are reported alongside the successful one makes the pipeline more useful in practice, because researchers facing similar problems can see what not to try.

Independent reproducibility evidence for urinary biomarker panel: The biomarker model's AUC of 0.944 provides independent computational evidence supporting the diagnostic utility of the age, plasma CA19-9, creatinine, LYVE1, REG1B, TFF1, and REG1A feature panel for PDAC detection. This replication, achieved with a neural network architecture rather than the original logistic regression, demonstrates that the diagnostic signal is classifier-agnostic and robust to implementation differences. Independent replication of biomarker studies is rare in the pancreatic cancer literature, making this a meaningful contribution to the evidence base for non-invasive screening approaches.

Methodologically rigorous fusion evaluation framework**:** The fusion experiments, while yielding negative results regarding genuine multimodal complementarity, contribute a systematic evaluation framework built around multi-seed validation, label-mismatch negative controls, and explicit documentation of modality dominance all designed to separate the question of whether the pipeline works from the question of whether it works for the right reasons. This framework provides a template for researchers conducting multimodal experiments under imperfect data alignment conditions.

6.5 Limitations

The limitations discussed in Chapter 5 are summarized here, with a focus on what they mean for the conclusions drawn above.

Residual domain bias in CT classification: The extreme standardization method dealt with pixel-level bias effectively, but it did not wipe out domain structure from deep feature representations. The CT model's near-perfect performance therefore remains ambiguous: it could reflect genuine pathological feature learning, residual domain shortcut exploitation, or some mixture of both. Without external validation on data from different institutions and scanners, there is no way to disentangle the two. This is the single most important limitation of the study, because it colors the interpretation of every CT-dependent result, including the fusion experiments.

Absence of matched patient data for fusion: Because the CT and biomarker data come from entirely different patients, fusion had to rely on synthetic pairing randomly matching samples by class label. This demonstrates pipeline functionality but fundamentally cannot establish

whether the two modalities provide complementary information about the same patient. All fusion results must therefore be understood as exploratory methodology rather than evidence of clinical multimodal synergy. This limitation is structural rather than methodological: it arises from the current landscape of publicly available pancreatic cancer datasets, none of which provides matched imaging and biomarker data from the same individuals.

Limited dataset scale and diversity: The CT cohort comprised 122 patients from two institutions, and the biomarker dataset comprised 590 samples from a single study. By contemporary deep learning standards, this is modest, and it limits statistical power, generalizability, and the ability to detect subtle performance differences. The two institution CT design also means that class label and institutional source are closely coupled, creating the identification challenge that motivated the bias awareness framework.

Binary classification scope: The study addressed only cancer-versus-control binary classification, whereas clinical practice demands staging, subtyping, and differentiation from benign conditions such as chronic pancreatitis. Extending the framework to clinically realistic multi-class problems would require substantially larger and more diverse datasets.

Absence of prospective clinical validation: All experiments were conducted retrospectively on publicly available data. No prospective evaluation, radiologist comparison study, or clinical workflow integration was performed. Moving from retrospective computational results to prospective clinical utility is a necessary next step.

6.6 Recommendations for Future Work

The findings and limitations point to several concrete directions for future work, starting with the most pressing and moving toward longer-term goals.

6.6.1 Addressing the Domain Bias Problem

The most pressing priority is resolving the ambiguity surrounding the CT model's performance. Three complementary strategies are recommended. First, external validation on CT data from institutions not represented in the training set would provide the most direct test of whether the model has learned generalizable pathological features. Ideally, validation data would come from at least two or three additional centers using different scanner manufacturers and acquisition protocols, ensuring that performance is evaluated across genuine acquisition diversity rather than within the confines of a two institution study.

Second, domain-adversarial training methods, which explicitly penalize a model for encoding dataset-of-origin information in its learned representations, could be incorporated into the training pipeline. This approach would complement pixel-level standardization by targeting the deep feature space where residual domain structure was observed. Gradient reversal layers, as proposed in the domain adaptation literature, provide a practical implementation pathway compatible with the ResNet backbone used in this study.

Third, and most fundamentally, adopting uniform acquisition protocols across institutions would address the problem at its source. A standardized CT acquisition protocol for pancreatic imaging specifying consistent parameters for slice thickness, reconstruction kernel, contrast phase timing, tube voltage, and field-of-view conventions would substantially reduce the scanner-specific fingerprints that affect cross-dataset learning. While protocol uniformity requires institutional coordination and is a longer-term goal, its impact on the reliability of AI-assisted diagnosis would be substantial. It would reduce the burden on computational methods required for reducing bias and enable more confident evaluation of model performance, because observed classification ability would be far less likely to reflect acquisition elements rather than genuine pathological signals. Ongoing initiatives in imaging standardization, such as the Radiological Society of North America's quantitative imaging biomarker alliance, provide frameworks that could be leveraged to advance this goal.

6.6.2 Enabling Genuine Multimodal Evaluation

The fusion framework provides a functional pipeline, but validating it clinically requires matched patient data. The most important next step for fusion is finding or assembling cohorts where both CT scans and urinary biomarker measurements come from the same patients. Even a modestly sized matched cohort of fifty to one hundred patients would be fundamentally more informative than synthetic pairing, because it would restore the patient-level correspondence needed to assess whether the two modalities capture genuinely independent aspects of the disease process.

Collaborative partnerships between imaging centers and biomarker laboratories would be the most practical route to assembling such data. In the interim, attention-based fusion architectures that learn to weight modality contributions dynamically, rather than relying on fixed concatenation or averaging, may prove more robust to the dimensional imbalance and quality asymmetry observed in this study. Cross-attention mechanisms that allow each modality to

query the other for relevant information represent a particularly promising direction, as they could mitigate the dominance effects observed in the feature-level fusion experiments.

6.6.3 Extending Clinical Scope

The binary classification framework should be extended to address clinically relevant multi-class problems. These include the differentiation of PDAC from chronic pancreatitis, a distinction that is challenging even for experienced radiologists; the classification of disease into resectablea versus unresectable categories, which directly informs treatment planning; and staging classification. Each of these extensions would increase the clinical utility of the framework but would require larger, more richly annotated datasets than are currently available in public repositories. The biomarker model's three-class capability, demonstrated in the original (Debernardi et al., 2020) study, provides a natural starting point for this extension on the biomarker side.

6.6.4 Prospective Validation and Clinical Integration

Ultimately, the real test is whether the system works in a prospective clinical setting. Future work should include reader comparison studies in which the model's outputs are evaluated alongside radiologist assessments on unseen cases, integration with clinical decision-support workflows to assess practical utility and clinician acceptance, and regulatory pathway engagement for any component that demonstrates robust prospective performance. The biomarker model, given its strong and reproducible performance on an independently validated feature set and its non-invasive nature, may be the most immediately translatable component of the framework for pilot screening studies in high-risk populations.

6.6.5 Technical Improvements

A few technical improvements could also strengthen the framework. Three-dimensional convolutional architectures or volumetric transformers that process CT volumes rather than individual two-dimensional slices could capture spatial context across adjacent slices, potentially improving both classification performance and anatomical interpretability. Uncertainty quantification through techniques such as Monte Carlo dropout or deep ensembles would provide confidence estimates alongside predictions, enabling selective referral of uncertain cases for specialist review. Explainability methods beyond Grad-CAM, including concept-based explanations and counterfactual analysis, could provide more clinically

interpretable evidence of what the model has learned and whether its decision criteria align with established radiological knowledge.

6.7 Final Remarks

This dissertation set out to build and rigorously test a multimodal deep learning framework for pancreatic cancer detection, combining CT imaging and urinary biomarker data under a systematic approach to bias awareness. The work produced three distinct types of outcome, and each tells a different part of the story.

The positive outcome is the biomarker model's AUC of 0.944, which independently confirms the diagnostic utility of the urinary biomarker panel and demonstrates that the signal is robust across different computational implementations. This result carries direct implications for the continued development of non-invasive pancreatic cancer screening and provides a validated baseline for future multi-class and prospective studies.

The ambiguous outcome is the CT model's near-perfect classification performance, which may reflect genuine anatomical feature learning, residual domain interference, or both. The study's contribution here is not the metric itself but the rigorous documentation of the bias landscape: the initial discovery of trivially exploitable scanner fingerprints, the iterative development and transparent reporting of mitigation strategies, and the honest acknowledgement that pixel-level standardization alone does not guarantee domain-invariant deep representations. This documentation provides a practical roadmap for researchers confronting similar challenges in cross-institutional medical imaging studies.

The informative negative outcome is the fusion evaluation, which demonstrated that multimodal integration under synthetic pairing and residual domain bias conditions produces modality dominant rather than synergistic behavior. This finding does not diminish the theoretical promise of multimodal cancer detection; it establishes the empirical boundary conditions under which that promise cannot be realized with currently available public data. The negative controls and multi-seed analyses that exposed this behavior are themselves a methodological contribution.

What these outcomes collectively suggest is chasing headline accuracy numbers, without rigorous bias analysis and proper controls, risks producing systems that look good on paper but fall apart in practice. Pancreatic cancer with its devastating outlook and very narrow window for treatment requires diagnostic tools that actually work across diverse clinical populations and

imaging environments, not just on carefully curated benchmark datasets. That is the standard this dissertation has tried to hold itself to, even where it meant reporting results that were less impressive than the numbers alone might suggest.

Getting from the framework developed here to something that could be deployed in a clinic is a long road. It requires matched patient data, external validation across multiple institutions, uniform imaging protocols, prospective evaluation, and regulatory engagement. None of those steps are impossible, and the preprocessing pipelines, model architectures, and evaluation frameworks built during this project provide a reasonable starting point. But perhaps the most valuable thing this work offers is not any single model or metric. It is an honest accounting of what worked, what remains uncertain, and what did not work under the conditions available.

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APPENDIX A: RESEARCH PROPOSAL

EARLY DETECTION OF PANCREATIC CANCER USING MULTI-MODAL DEEP LEARNING: INTEGRATION OF CT IMAGING, URINARY, AND BLOOD BIOMARKERS

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Research Proposal

MAY 2025

Table of Contents

List of Abbreviations **92**

List of Figures **92**

Abstract **93**

1. Background **94**

2. Literature Review **96**

4. Aim and Objectives **98**

5. Significance of the Study **99**

6. Scope of the Study **99**

7. Methodology **100**

List of Abbreviations

AUC……... Area Under the Curve

AI……... Artificial Intelligence

CT……... Computed Tomography

CBCT…… Cone Beam Computed Tomography

CECT…… Contrast-Enhanced Computed Tomography

CNN……... Convolutional Neural Network

MLP……... Multi-layer Perceptron

PDAC…… Pancreatic Ductal Adenocarcinoma

miRNA…… MicroRNA

DL……... Deep Learning

ROC……. Receiver Operating Characteristic

TPR…… True Positive Rate

FPR…… False Positive Rate

F1……... F1 Score

GANTT…… Generalized Activity Normalization Timetable

List of Figures

Figure 7.1 Data Preprocessing flowchart for CT Imaging and Biomarkers Page 12

Figure 7.2 Model architecture flowchart for both CT imaging and Biomarkers Page 13

Figure 7.3 Flowchart depicting feature-level and decision-level fusion Page 14

Figure 7.4 The complete pipeline of the entire methodology Page 15

Abstract

Pancreatic cancer is a highly lethal disease due to its typically late diagnosis. Early detection remains the key to improving patient outcomes. This research proposes a novel multi-modal deep learning system that combines CT imaging with urinary and blood biomarkers. Incorporating open-source datasets, the goal is to increase diagnostic accuracy for the deadly disease. Convolutional neural networks will analyse CT images, while a Multi-layer Perceptron will handle biomarker data. Feature-level and decision-level fusion strategies will be explored to accommodate the different modalities. The research seeks to achieve a clinically useful AI-driven pipeline that can surpass existing diagnostic systems.

1. Background

Pancreatic cancer remains one of the deadliest cancers globally, with a five-year survival rate of less than 10%. The primary reason for this situation is that pancreatic cancer is often diagnosed at an advanced stage. Despite substantial progress in cancer detection and treatments, pancreatic cancer remains an outlier in early detection. This is due to its asymptomatic nature in early stages, which current diagnostic methods fail to detect with sufficient accuracy. This delay in diagnosis severely limits treatment options and reduces patient survival rates.

Traditionally, diagnosis is done with the help of various imaging techniques such as CT, MRI scans, and endoscopic ultrasound. These methods, though sound in assessing tumour spread and growth, are limited in their capacity to detect cancerous tissue at an early stage, as there is a good chance of missing tissue abnormalities when the tumour is small. Also, there is a high amount of variability in the ability of radiologists to assess if the tissue is cancerous, especially at the early stages of the disease. On the molecular side, commonly used biomarkers like carbohydrate antigen 19-9 (CA19-9) exhibit low sensitivity and specificity, particularly in early-stage disease or in individuals who are Lewis antigen-negative and do not express the biomarker. As a result, there is a growing consensus that single-modality diagnostic tools are insufficient for early detection of pancreatic cancer.

In this context, artificial intelligence (AI), particularly deep learning techniques, presents promising opportunities to address existing diagnostic limitations. Convolutional Neural Networks (CNNs) have demonstrated considerable potential in automatically extracting high-level features from medical images, showing superior performance in contrast to conventional computer-aided detection systems. Concurrently, machine learning algorithms have shown promise in analysing blood- and urine-based biomarkers, identifying nonlinear patterns and correlations across multidimensional datasets that may elude conventional statistical analysis. Individually, these AI-based approaches can improve diagnostic precision, but their real power may lie in integrating both imaging and biochemical data, a strategy referred to as multimodal data fusion.

Multimodal fusion, which combines imaging and non-imaging data (e.g., biomarkers), holds substantial promise to significantly improve early detection of malignancies due to their complementary nature. While imaging gives details on the spatial and structural information of the tumour, biomarkers reveal understandings into the metabolic and molecular changes occurring in the body, which together offers a comprehensive approach to diagnose the disease.

Hence, by utilising these complementary data sets and employing the use of deep learning models, it is achievable to catch the condition early, reduce false positives and offer a system that clinicians can trust. It is also the case that literature on pancreatic cancer diagnosis using multimodal AI systems is still in its early stages, suggesting a requirement for further research and development. It is observed that in existing literature these modalities are used in isolation and hence are unable to derive the synergistic benefits that a joint model can offer.

The proposed research seeks to close this gap developing a deep learning–based framework that integrates CT scan data with urinary and serum biomarker profiles to detect pancreatic cancer in its early stages. By building upon recent work such as those by Chen et al. (2023), Chi et al. (2023), and Dinesh et al. (2023), this study intends to design and evaluate a multimodal fusion system, focusing on integrating imaging and biomarker data to enhance early detection capabilities. This study not only has the ability in advancing academic knowledge but also can add substantial value to clinical settings, particularly in resource constrained or non-invasive screening environments.

2. Literature Review

Pancreatic cancer has a very low 5-year survival rate that remains under 11%, as about 80% of cases are diagnosed at advanced stages. Hence there is an interest in exploring the use of artificial intelligence, particularly deep learning, to help with early detection through imaging or biomarkers.

2.1 Imaging-Based Deep Learning Systems

Deep learning has shown significant promise in pancreatic cancer detection via CT imaging. (Chen et al., 2023) developed a nationwide AI-based detection model trained on contrast-enhanced CT (CECT) scans, achieving an AUC of 0.95 and showing notable sensitivity (74.7%) for tumours smaller than 2 cm. Similarly, (Liu et al., 2020) demonstrated high performance using a CNN model that achieved AUCs nearing 0.999 in local test sets and 0.92 on external U.S. datasets, with sensitivity better than radiologists for sub-2 cm tumours.
(Zavalsız et al., 2023) conducted a comparative study of pre-trained CNNs like ResNet50 and VGG16 versus custom CNNs, finding that transfer learning outperformed other models in both speed and accuracy. (Ma et al., 2020) earlier validated CNN performance across multi-phase CT, achieving over 91% sensitivity and 95.5% accuracy, laying the groundwork for imaging-based AI diagnosis.

2.2 Biomarker-Based Non-Invasive Screening

Alongside imaging, blood and urine biomarkers have also emerged as non-invasive modes for early detection. (Chi et al., 2023) employed SVM-RFE and neural networks to identify a serum miRNA panel with great diagnostic performance (AUC > 0.93), that was validated across multiple GEO datasets. Also, (Yan et al., 2020) evaluated over 2,500 serum miRNAs in a large cohort, identifying 13 diagnostic miRNAs, that promised good detection potential.
Urine biomarkers have also been validated as predictive tools. (Karar et al., 2023) developed a 1D-CNN+LSTM hybrid model trained on urinary data (LYVE1, REG1B, TFF1, creatinine), achieving 98% AUC and 97% accuracy on a 590-sample dataset. Adding on this, (Debernardi et al., 2023) showed that these same biomarkers, when combined with plasma CA19-9 and patient age (PancRISK model), was able to detect PDAC up to two years before clinical diagnosis with an AUC of 0.89.

2.3 Multi-Modal and Hybrid Approaches

(Dinesh et al., 2023) explored this direction by combining CNN-based imaging analysis with YOLO-driven biomarker detection, which improved classification performance but lacked true feature-level fusion. (Steyaert et al., 2023) proposed a deep learning framework for multi-modal biomarker discovery, integrating genomics and clinical data sources—though not specific to pancreatic cancer, it offers methodological insights for designing fusion-based diagnostic systems.

2.4 Summary and Research Gap

The existing research while having achieved great progress, still lacks real-world multi-modal systems that integrate both imaging and other data. This research aims to bridge these gaps by designing a deep learning pipeline that integrates CECT imaging, serum miRNAs, and urinary biomarkers into one model. This approach seeks to build on the strengths of prior research while simultaneously addressing their limitations.

3. Research Questions

- Will the integration of imaging, urinary and blood-based markers in a multi-modal approach improve the detection of pancreatic cancer when compared to single-modality approaches?
- Which deep learning architectures are effective at using the different data types effectively?
- Is the model able to perform well on independent datasets by showing good generalisation?

4. Aim and Objectives

Aim:

The goal of this research is to create a multi-modal deep learning diagnostic pipeline that utilises contrast-enhanced computed tomography (CECT) images in addition to urinary and blood biomarkers to enhance accuracy and improve the early-stage detection of pancreatic cancer.

Objectives:

To collect relevant publicly available and quality datasets comprising CT images, urinary biomarkers, and blood biomarkers that relates to pancreatic cancer.

To design and develop appropriate deep learning models for each modality

To efficiently fuse the various models and create a system that can improve early detection capabilities.

5. **Significance of the Study**

Pancreatic cancer remains a challenging and highly fatal disease; this can be attributed to poor early detection systems in place currently. A majority of patients are diagnosed at very near the terminal stage, where treatment is severely hampered and hence survival rates plumet. Therefore, it is crucial to be able to detect the condition at an early stage to enhance patient survival rates.

This research seeks to tackle this challenge by developing a deep learning driven multi-modal diagnostic system that combines three distinct modalities: contrast-enhanced CT imaging, urinary biomarkers, and serum microRNA profiles. While each of these modalities have shown promise in previous studies, their integration has been underexplored.

Also, the research seeks to use publicly available datasets and non-invasive sample types (urine and serum), that can make the approach feasible in low-resource clinical environments.

6. Scope of the Study

The scope of the research can be summarized as follows

- The use of publicly available and trustworthy data sources
- Deployment of deep learning models for performing classification tasks
- The model is evaluated in terms of binary classification (cancer vs healthy)
- This research focuses only on early detection and is not intended to explore the cancers stages or subtype classification

7. Methodology

7.1 Data Collection and Preprocessing

All datasets will undergo standardized preprocessing techniques to the data is handled uniformly. CT images will be resized to 224×224 pixels and normalized to (0,1) intensity range. Biomarker values will be z-score normalized for standardization.

Missing values in biomarker data will be imputed based on the following criteria:

- If missing values are <10% and random, mean substitution will be used
- Otherwise KNN imputation (k=5) will be applied, using patient similarity for more accurate estimation.

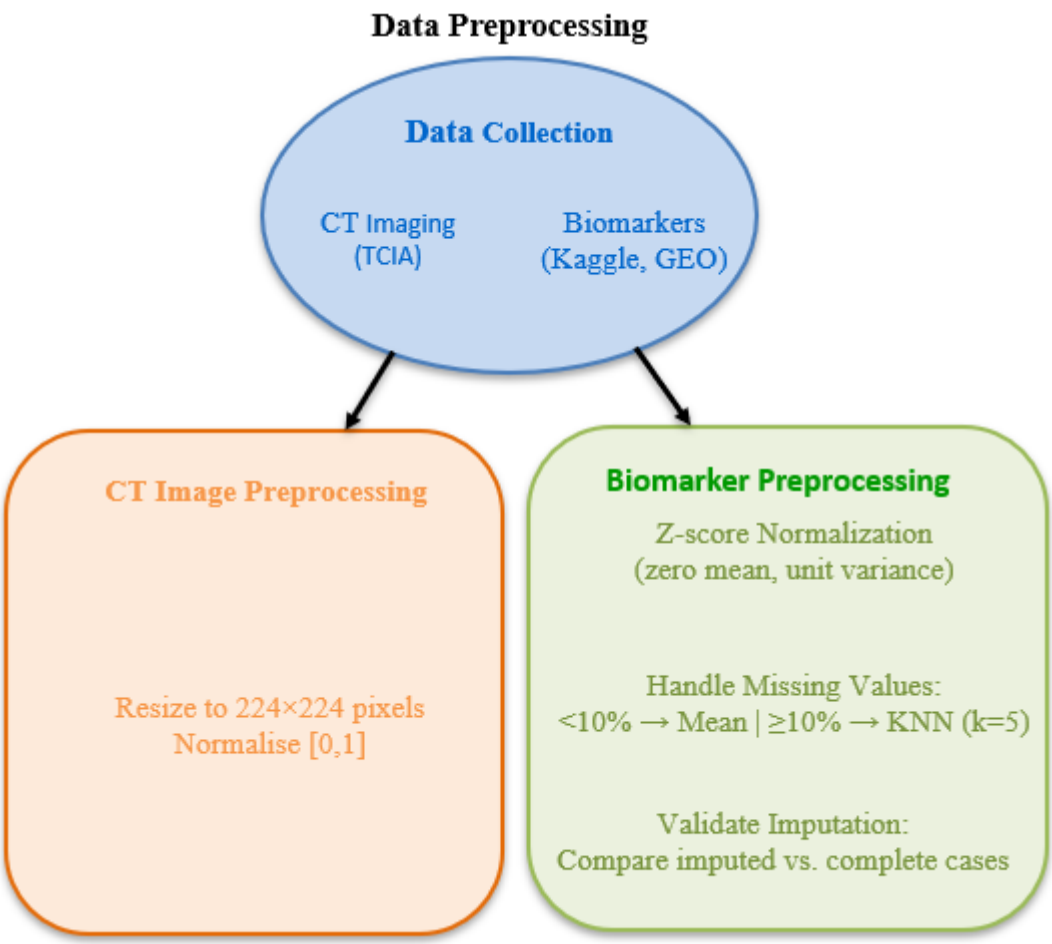

Figure 7.1: Data Preprocessing flowchart for CT Imaging and Biomarkers

7.2 Model Architecture

The proposed model consists of two parallel deep learning branches, one for each modality:

- **Imaging Pathway:** A pretrained ResNet50 CNN will be used for feature extraction, lower-level layers pretrained on ImageNet will be retained, while higher layers will be fine-tuned on the CT dataset. The output will be flattened and passed through a dense layer that results in a 256D embedding.
- **Biomarker Pathway:** A multilayer perceptron (MLP) will process the normalized biomarker vector through three dense layers (128, 64, 32 units) with ReLU activations. Dropout (0.3) and batch normalization will be performed to reduce overfitting. The final output will be a 32D embedding.

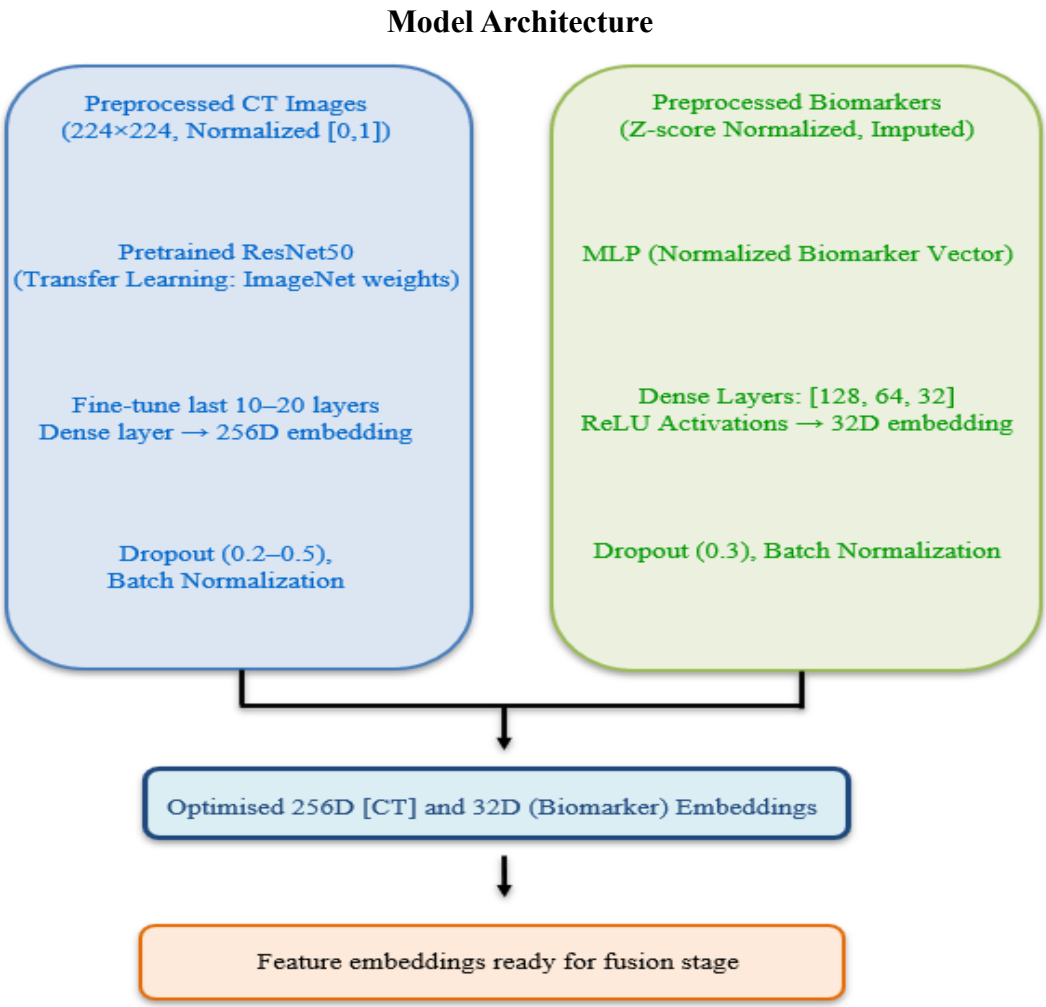

Figure 7.2: Model architecture flowchart for both CT imaging and Biomarkers

7.3 Fusion Strategy

To integrate the CT imaging and biomarker modalities, both feature-level and decision-level fusion strategies will be evaluated.

In the feature-level fusion, the 256D embedding from the CNN and the 32D embedding from the MLP will be concatenated and passed through a dense layer (e.g., 128 units, ReLU activation), which is then followed by a softmax layer for binary classification (cancer vs. non-cancer).

In the decision-level fusion, separate classifiers will be trained for each modality, and their probability outputs will be combined using ensemble methods.

- Weighted averaging, with weights optimized via grid search over the range (0.1, 0.9) in 0.1 increments to maximize validation AUC-ROC.
- Stacking, where a logistic regression learner is trained on the predicted probabilities from each modality.

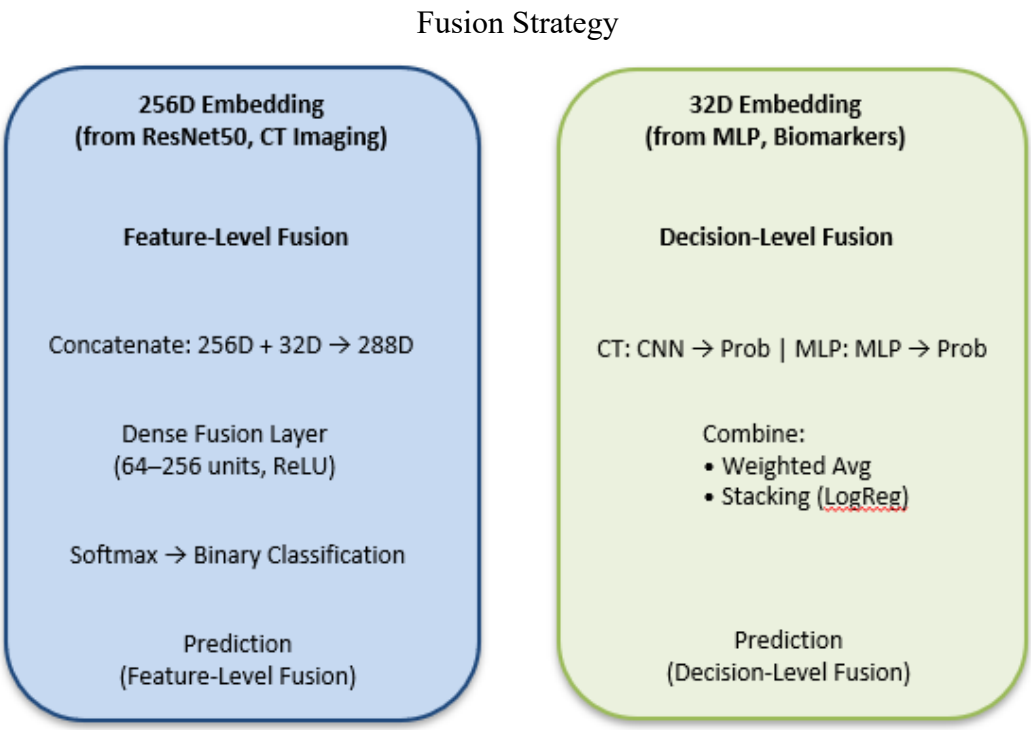

Figure 7.3: Flowchart depicting feature-level and decision-level fusion

7.4 Training and Evaluation

Model performance will be assessed using stratified 5-fold cross-validation to ensure balanced class representation and reliable generalization estimates. The best model from each fold will be selected based on the highest validation AUC-ROC.

Evaluation will include multiple metrics: accuracy, AUC-ROC, sensitivity, specificity, precision, and F1-score, applied to both multimodal and unimodal models (CNN-only and MLP-only).

To assess the significance of observed differences McNemar's test will be used, alongside comparisons with existing models.

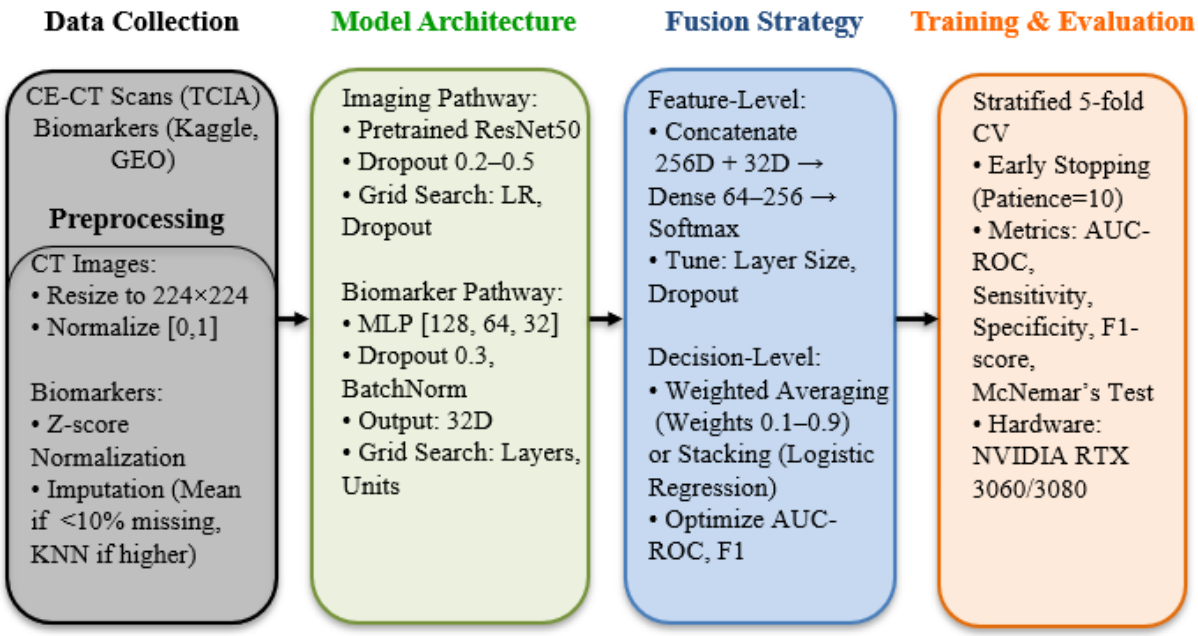

Figure 7.4: The complete pipeline that depicts the flow from data collection to training and evaluation

APPENDIX B: KEY CODE LISTINGS

This appendix presents the key code implementations referenced in Chapter 4. Each listing corresponds to a core component of the multi-modal pipeline, including bias mitigation, model construction, and feature extraction.

B.1: Reproducibility Seed Initialization

```
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed_all(SEED)
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

B.2: Extreme Intensity Standardization Function

```
def extreme_standardize(img):
    img = img.astype(np.float32)
    body_mask = img > 5
    if body_mask.sum() < 100:
        return np.full_like(img, 128, dtype=np.uint8)
    body_pixels = img[body_mask]
    z_scores = (body_pixels - body_pixels.mean()) / (body_pixels.std() + 1e-8)
    new_pixels = z_scores * 40.0 + 128.0
    img_out = np.zeros_like(img)
    img_out[body_mask] = new_pixels
    return np.clip(img_out, 0, 255).astype(np.uint8)
```

B.3: ResNet50 Model Construction

```
def  build_resnet50(num_classes=2, pretrained=True, freeze_backbone=False):
    weights = ResNet50_Weights.IMAGENET1K_V2 if pretrained else None
    model = resnet50(weights=weights)
    if freeze_backbone:
        for param in model.parameters():
            param.requires_grad = False
    in_features = model.fc.in_features  # 2048
    model.fc = nn.Linear(in_features, num_classes)
    return model
```

B.4: Biomarker MLP Architecture

```
class BiomarkerMLP(nn.Module):
    def __init__(self, input_dim, hidden_dims=[64, 32],
                 num_classes=2, dropout=0.3):
        super().__init__()
        layers = []
        prev_dim = input_dim
        for h in hidden_dims:
            layers.extend([
                nn.Linear(prev_dim, h),
                nn.BatchNorm1d(h),
                nn.ReLU(),
                nn.Dropout(dropout)
            ])
            prev_dim = h
        self.features = nn.Sequential(*layers)
```

```
        self.classifier = nn.Linear(prev_dim, num_classes)
    def forward(self, x):
        feat = self.features(x)
        return self.classifier(feat)
    def extract_features(self, x):
        return self.features(x)
```

B.5: CT Embedding Extraction via Forward Hook

```
def get_ct_embeddings(model, loader):
    model.eval()
    embeddings, labels = [], []
    activation = {}
    def hook(m, i, o):
        activation['out'] = o.detach()
    handle = model.avgpool.register_forward_hook(hook)
    with torch.no_grad():
        for images, lbls, meta in loader:
            _ = model(images.to(device))
            emb = activation['out'].squeeze()
            embeddings.append(emb.cpu())
            labels.extend(lbls.numpy())
    handle.remove()
    return torch.cat(embeddings), np.array(labels)
```